GENERAL LINE CONES AND THE LEADING-FORM MAP

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Abstract. Let Z be a general set of r points in projective three-space and $C(Z) \subset \mathbb{A}^4$ the affine cone over its standard homogeneous coordinate ring, a reduced curve consisting of r lines through the origin. Write $r = \binom{d+2}{3} + s$ with $0 \le s < \binom{d+2}{2}$. We prove that all cotangent groups $T^1_j(C(Z))$ of negative degree vanish for every fixed pair with $d \ge 84$ and $d+3 \le s \le \binom{d+2}{2} - 7d - 427$. Equivalently in degree -1, the differential of the map sending a configuration of affine points to the leading forms of its degree-d equations has the maximal rank $3r-4$ allowed by translations and dilation, as predicted by Douvropoulos, Jelisiejew, Nødland and Teitler and by Stevens. Consequently these cones admit neither an algebraic nor an analytic smoothing, and the set of branch numbers r covered has natural density one. The lower end is sharp: vanishing holds at $s = d+3$ for every $d \ge 27$, the first position permitted by the dimension count, whereas at $s = d+2$ the group T^1_{-1} of a general cone has dimension $4(d+1)$ for every $d \ge 4$. The proof combines Fourier analysis on degenerate cyclic plane arrangements, finite characteristic-zero certificates, Horace-type degenerations to plane point sets, and Pinkham's equivariant versal deformation.

1. Introduction

A reduced curve singularity consisting of r lines through the origin of $\mathbb{A}^n$ is the affine cone over a set of r points of $\mathbb{P}^{n-1}$. Such cones are among the oldest sources of non-smoothable curve singularities; the examples obtained by dimension counts, going back to Pinkham [7], are surveyed and extended by Stevens [9, Sections 3.1–3.2], and the subject has recently been revisited also by Bozlee [1]. A second, more rigid mechanism is negative grading. The cotangent space T^1 of a cone is graded, and if all its pieces of negative degree vanish, Pinkham's equivariant versal deformation has only parameters of non-positive weight; the vertex then persists in every deformation, and no smoothing exists. For finite schemes the analogous statement is part of Jelisiejew's theory of trivial negative tangents [4, Theorem 1.2].

For lines in $\mathbb{A}^4$ this paper concerns the second mechanism. Work over $\mathbb{C}$. For an ordered set Z of r distinct points of $\mathbb{P}^3$ let $C(Z) \subset \mathbb{A}^4$ be the affine cone defined by the homogeneous ideal of the points, that is, the spectrum of the standard graded coordinate ring of the given embedding. Put

$$L_d = \binom{d+2}{3}, \qquad M_d = \binom{d+2}{2}, \qquad r = L_d + s, \qquad m = M_d - s. \tag{1.1}$$

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Every $r \geq 1$ is uniquely of this form with $0 \leq s < M_d$; for general Z the ideal then has no equations of degree less than d and an m-dimensional space of equations of degree d. By "general" we always mean: in a nonempty Zariski open subset of the space of ordered configurations, for the fixed pair (d, s) under consideration; we do not claim a common open set for infinitely many pairs.

Theorem 1.1. *Let $d \geq 84$ and $d+3 \leq s \leq M_d - 7d - 427$. For general Z,*

$$T^1_j(C(Z)) = 0 \qquad \text{for every } j < 0,$$

and the reduced curve $C(Z)$ admits neither an algebraic nor an analytic smoothing. The set of branch numbers $r = L_d + s$ covered by these pairs has natural density one.

The lower end of the range is sharp, and it is reached in every degree from $d = 27$ on.

Theorem 1.2. *For general Z, $T^1_j(C(Z)) = 0$ for every $j < 0$, and $C(Z)$ admits neither an algebraic nor an analytic smoothing, in each of the following cases:*

(a) *$d \geq 27$ and $s = d + 3$;*
(b) *$d \geq 27$ and $d + 3 \leq s \leq 2d - 24$;*
(c) *$d \geq 28$ and $d + 3 \leq s \leq 3d - 52$.*

On the other hand, for $d \geq 4$ and $s = d + 2$ the group $T^1_{-1}(C(Z))$ of a general cone has dimension $4(d + 1)$.

The last assertion only locates the onset of vanishing; it makes no statement about smoothability at $s = d + 2$.

The degree -1 part has a concrete meaning. Choose the chart $X_0 \neq 0$ and write the points as $(1 : p_i)$ with $P = (p_1, \ldots, p_r) \in (\mathbb{A}^3)^r$. Let F_j be the polynomials of degree at most j in three variables and S_d the forms of degree d. For general P, evaluation is injective on F_{d-1} and surjective on F_d; the equations $K(P) = \ker(\mathrm{ev}_P\,|_{F_d})$ form an m-dimensional space whose leading forms span

$$\Phi(P) = \{f_d : f \in K(P)\} \in \mathrm{Gr}(m, S_d).$$

The ideal $\Phi(P) \oplus S_{\geq d+1}$ is the limit of the homothetic configurations tP as $t \to 0$, so Φ is the homothety-limit map to the Grassmannian of very compressed algebras studied in [3, Section 5] and [5, Section 2.3]. Translations and the dilation of $\mathbb{A}^3$ do not change Φ, so its differential has rank at most $3r - 4$. We show (Proposition 2.3) that $T^1_{-1}(C(Z))$ is the kernel of this differential modulo these four motions, and we prove the corresponding maximal-rank statement.

Theorem 1.3. *In all cases of Theorems 1.1 and 1.2*(a)–(c)*, the differential of Φ at a general configuration has rank $3r - 4$; equivalently, the image of Φ has dimension $3r - 4$, and a general fibre has dimension four. For $d \geq 4$ and $0 \leq s \leq d + 2$ the differential at a general configuration is surjective.*

The statement about the fibres is local: it does not assert that a general fibre is a single orbit, nor any global injectivity. A dimension count (Section 2) shows that the rank $3r - 4$ is possible only if

$$F_d(s) = s(M_d - s) - 3(L_d + s) + 4 \geq 0,$$

and for $d \geq 11$ this holds exactly for $d + 3 \leq s \leq M_d - d - 6$; moreover $F_d(d + 2) = -4(d + 1)$ and $F_d(d + 3) = (d - 11)(d + 2)/2$. Thus Theorem 1.2(a) is the first

dimensionally possible position in each degree, and Theorems 1.1 and 1.2 leave $6d + 421$ positions open at the upper end of each layer $d \geq 84$. In the language of [9], whose Proposition 3.5 gives the lower bound $\dim T^1_{-1} \geq \max(0, -F_d(s))$ for general lines in $\mathbb{A}^4$, our results establish the equality predicted in [9, Conjecture 3.6] for all $s \leq d + 2$ ($d \geq 4$) and throughout the ranges above, where both sides vanish. They also establish, in these ranges, the maximal rank of the homothety-limit map conjectured in [3, Remark 5.4], where it was verified by computer for configurations of at most 95 points. Existence results for non-smoothable reduced curve singularities with prescribed genus and number of branches are proved in [1, Theorem 1.5]; the present results concern the specific family of general line cones in $\mathbb{A}^4$ and the vanishing of all their negative cotangent groups. To our knowledge, this vanishing had previously been established for line cones in $\mathbb{A}^4$ only for finitely many values of r, by direct computation [9, Section 3.2].

Outline of the proof. The identification of T^1_{-1} with the kernel of the moving differential J_P uses the description of T^1 of a curve through its normalization (Section 2). Vanishing in degree -1 propagates to all negative degrees by an explicit obstruction map whose injectivity is witnessed by a lattice configuration (Section 3), and non-smoothability follows from Pinkham's construction and a Fitting-ideal argument (Section 4).

The rank statements are proved by exhibiting, for each pair (d, s), one configuration of maximal rank. For the lower boundary $s = d + 3$ we use the $L_d = \binom{n}{3}$ triple points of $n = d+2$ planes, n further points, one on each plane, and the origin. Products of the plane equations interpolate the triple points, and the velocities of the triple points can be eliminated exactly (Section 5). We specialize to a cyclic arrangement, which is not a genuine arrangement but on which the cyclic group of order n acts; Fourier decomposition reduces the remaining equations, for each character, to a system with six or seven unknowns, whose kernel (the *special kernel*) has total dimension $2n - 2$ (Section 6). Moving the auxiliary points off the planes by a homothety, we show that the first-order perturbation has rank $2n-6$ on the special kernel; the four remaining directions are the motions (Section 7). For $n \geq 89$ this is proved uniformly by a three-column detector and exact estimates for three Fourier moments; for $29 \leq n \leq 88$ it is verified by exact modular computations lifted to characteristic zero. Further positions are reached by coupled Horace transitions from degree e to $e + 1$, which add $b = M_e + c$ points on a plane (Section 8); their two auxiliary rank conditions are proved with canonical modules and syzygies of general plane points. The bands of Theorem 1.2(b),(c) are obtained from the lower boundary by sparse transitions with $c = 2, 3$, and the broad range of Theorem 1.1 by a finite propagation from an explicit rational configuration in degree 11, followed by an analytic induction (Sections 9 and 10).

Dependencies. Besides standard facts, the proof uses the following results that we do not reprove: Pinkham's construction of an equivariant formal versal deformation [7]; the cotangent sequence and the residue description of the canonical module of a curve of lines as presented in [9]; Sauer's count of minimal generators for general plane points [8, Proposition 1.7]; and the Buchsbaum–Rim exactness criterion [6]. It also uses finite exact computations: modular certificates for $29 \leq n \leq 88$, a rational configuration in degree 11, a finite list of transitions, and a fixed finite set of explicit symbolic identities and nonvanishing certificates; their scope is described in

Appendices A–E. The statements for infinitely many degrees are proved by uniform arguments and inductions which use these fixed certificates as input (for instance as the base of an induction, or as identities of rational functions in a parameter); they are never inferred from tests of finitely many degrees.

Organization. Section 2 sets up the cone ring and the moving differential. Sections 3 and 4 treat lower negative degrees and non-smoothability. Section 5 introduces plane arrangements and proves the statement at $s \le d+2$. Sections 6 and 7 prove the lower boundary. Section 8 proves the Horace transitions, Section 9 the seed and the propagation, and Section 10 completes the proofs. Appendix A describes the modular certificates, Appendix B lists the leading coefficients and interval bounds for the boundary characters, Appendices C and D give the seed and the finite propagation, and Appendix E describes the supplementary programs and data.

2. The cone ring and the moving differential

2.1. **Conventions.** Throughout, $d \ge 2$ and $0 \le s < M_d$ are integers, and L_d, M_d, r and m are as in (1.1); thus $m \ge 1$, and the pair (d,s) is the unique representative of r. We write $F_j \subset \mathbb{C}[x,y,z]$ for the space of polynomials of degree at most j (with $F_j = 0$ for $j < 0$) and S_j for the space of forms of degree j; thus $\dim F_{d-1} = L_d$, $\dim S_d = M_d$ and $\dim F_d = L_d + M_d$. For $f \in F_j$ we write $f = \sum_{k \le j} f_k$ with $f_k \in S_k$, and $D_b f$ denotes the derivative of f in the direction $b \in \mathbb{C}^3$.

Let $\mathrm{Conf}_r \subset (\mathbb{P}^3)^r$ be the space of ordered r-tuples of distinct points; it is irreducible. For $Z = (a_1, \dots, a_r) \in \mathrm{Conf}_r$ let $I_Z \subset S = \mathbb{C}[X_0, \dots, X_3]$ be the homogeneous ideal of the reduced set $\{a_1, \dots, a_r\}$, let $R = R(Z) = S/I_Z$ be its standard graded coordinate ring, and let $C(Z) = \operatorname{Spec} R \subset \mathbb{A}^4$ be the affine cone; it is the union of the r lines through the origin corresponding to the points. We always use this ring, which is the coordinate ring of the specified embedding; it differs in general from the full section ring $\bigoplus_j H^0(\mathcal{O}_Z(j))$. The cotangent groups $T^1_j = T^1(C(Z))_j$ are taken with respect to the grading of R, so that a derivation of degree j raises degrees by j.

Most computations take place in the affine chart $X_0 \ne 0$. For $P = (p_1, \dots, p_r) \in (\mathbb{A}^3)^r$ with distinct entries put $a_i = (1 : p_i)$, and let

$$\mathrm{ev}_P \colon F_j \to \mathbb{C}^r, \qquad f \mapsto (f(p_1), \dots, f(p_r)), \qquad V_j(P) = \mathrm{ev}_P(F_j).$$

The *evaluation open* $U^{\mathrm{ev}}_{d,s} \subset (\mathbb{A}^3)^r$ is the set of configurations of distinct points for which $\mathrm{ev}_P|_{F_{d-1}}$ is injective and $\mathrm{ev}_P|_{F_d}$ is surjective. On $U^{\mathrm{ev}}_{d,s}$ we put

$$K(P) = \ker(\mathrm{ev}_P|_{F_d}), \qquad Q(P) = \mathbb{C}^r / V_{d-1}(P), \tag{2.1}$$

so that $\dim K(P) = m$ and $\dim Q(P) = s$. Every nonzero $f \in K(P)$ has degree exactly d, since $K(P) \cap F_{d-1} = 0$; hence the leading-form map $f \mapsto f_d$ is injective on $K(P)$, and

$$\Phi(P) = \{ f_d : f \in K(P) \} \in \mathrm{Gr}(m, S_d)$$

is well defined. The *moving differential* is the linear map

$$J_P \colon (\mathbb{C}^3)^r \to \mathrm{Hom}(K(P), Q(P)), \qquad J_P(v)(f) = \big[(D_{v_i} f(p_i))_{i=1}^r \big]. \tag{2.2}$$

Its source has dimension $3r$ and its target dimension sm. The subspace

$$W_P = \{ (b + a\, p_i)_{i=1}^r : b \in \mathbb{C}^3,\ a \in \mathbb{C} \} \tag{2.3}$$

consists of the velocities of translations and of the dilation of $\mathbb{A}^3$.

Lemma 2.1. *Let $\Lambda \subset \mathbb{Z}^3_{\geq 0}$ be a finite set that is closed under passing to coordinatewise smaller multi-indices, and let $P_\Lambda = (\gamma)_{\gamma\in\Lambda}$, regarded as a configuration of $|\Lambda|$ distinct points of $\mathbb{A}^3$. For $\alpha \in \mathbb{Z}^3_{\geq 0}$ put*

$$N_\alpha(x_1, x_2, x_3) = \prod_{j=1}^{3} \prod_{i=0}^{\alpha_j - 1} (x_j - i).$$

Then $(N_\alpha)_{\alpha\in\Lambda}$ restricts to a basis of functions on P_Λ, and N_α vanishes on P_Λ whenever $\alpha \notin \Lambda$.

Proof. $N_\alpha(\gamma) = 0$ unless $\alpha \leq \gamma$ coordinatewise, and $N_\gamma(\gamma) = \gamma_1!\gamma_2!\gamma_3! \neq 0$. Ordering Λ by a linear extension of the coordinatewise order makes the matrix $(N_\alpha(\gamma))_{\alpha,\gamma\in\Lambda}$ triangular with nonzero diagonal. If $\alpha \notin \Lambda$ and $\gamma \in \Lambda$, then $\alpha \leq \gamma$ would give $\alpha \in \Lambda$; hence $N_\alpha(\gamma) = 0$. □

Corollary 2.2. *For $d \geq 2$ and $0 \leq s < M_d$ the evaluation open $U^{\mathrm{ev}}_{d,s}$ contains configurations whose points affinely span $\mathbb{A}^3$; in particular it is nonempty.*

Proof. Let Λ consist of all multi-indices of degree at most $d-1$ and of s multi-indices of degree d. By Lemma 2.1 the polynomials N_α, $\alpha \in \Lambda$, which lie in F_d, give a basis of $\mathbb{C}^r$ after evaluation, and those with $|\alpha| \leq d-1$ form a basis of F_{d-1} that evaluates to independent vectors. Since $d \geq 2$, Λ contains $0, e_1, e_2, e_3$. Both conditions defining $U^{\mathrm{ev}}_{d,s}$ are given by the nonvanishing of minors and are therefore open. □

2.2. **The graded pieces of the cone ring.** Let $P \in U^{\mathrm{ev}}_{d,s}$ with $r \geq 4$, and suppose that the points affinely span $\mathbb{A}^3$. Parametrize the i-th line of $C(Z)$ by $t_i \mapsto t_i a_i$. Restriction to the lines embeds R into the graded ring

$$K = \prod_{i=1}^{r} \mathbb{C}[t_i, t_i^{-1}], \qquad K_j = \{(c_i t_i^j)_i\} \cong \mathbb{C}^r,$$

which is the ring of homogeneous fractions of R. In degree $j \geq 0$ the image of R_j is $V_j(P)$, since a form G of degree j restricts to $(G(1, p_i)t_i^j)_i$. Hence $R_j = K_j$ for $j \geq d$, $\dim R_j = \dim F_j$ for $j \leq d-1$, and I_Z has no nonzero elements of degree less than d. The dehomogenizations $f(x) = G(1, x)$ of the forms $G \in (I_Z)_d$ are exactly the elements of $K(P)$. Each line is smooth away from the vertex and two lines meet only at the vertex, so $C(Z)$ is a reduced curve with an isolated singular point; since $d \geq 2$ it has embedding dimension four. In particular all T^1_j are finite dimensional and vanish for all but finitely many j.

Proposition 2.3. *Let P be as above and $Z = ((1 : p_i))_i$.*

(a) *There is a natural isomorphism $T^1_{-1} \cong \ker J_P/W_P$, and $\dim W_P = 4$.*

(b) *Let $j \leq -2$. Every class in T^1_j has a unique representative given by a vector $v = (v_i) \in (\mathbb{C}^3)^r$, and the class is zero if and only if $v = 0$. If $j = -2$, the representative satisfies*

$$(D_{v_i} f(p_i))_{i=1}^r \in V_{d-2}(P) \qquad \text{for every } f \in K(P). \tag{2.4}$$

(c) *For $j \leq -2$, multiplication by X_0 maps $T^1_j \to T^1_{j+1}$ and preserves the representing vectors v.*

Proof. We use the exact sequence obtained from $0 \to R \to K \to K/R \to 0$; see [9, Section 1.4, (1)]. Since all points lie in the chart $X_0 \neq 0$, the vertex is the only point of $C(Z)$ with $X_0 = 0$, and $K = R[X_0^{-1}]$ is the coordinate ring of the disjoint union of the r punctured lines; it is smooth over $\mathbb{C}$. Cotangent cohomology is compatible with localization, so $T^1(R, K) = T^1(R[X_0^{-1}]/\mathbb{C}, K) = 0$, and $T^1(R)$ is the cokernel of the map $\mathrm{Der}(R, K) \to \mathrm{Der}(R, K/R)$. A derivation into K/R is represented by a vector $\xi \in K^4$, determined modulo R^4, such that $\xi \cdot \nabla G \in R$ for every $G \in I_Z$; it is enough to impose this for a set of generators of I_Z. On the i-th line the Jacobian matrix of I_Z has rank three over $\mathbb{C}(t_i)$, with kernel spanned by the radial vector a_i, so the image of $\mathrm{Der}(R, K)$ consists of the vectors whose i-th component is a multiple of a_i. In degree j the i-th component of ξ is $\xi_i t_i^{j+1}$ with $\xi_i \in \mathbb{C}^4$; subtracting the radial multiple $(\xi_i)_0\, a_i t_i^{j+1}$ we may assume that $\xi_i = (0, v_i)$, and this normalization is unique.

Let $G \in (I_Z)_q$ and $f = G(1, x)$. For $k = 1, 2, 3$ we have $\partial_k G(t, tp) = t^{q-1} \partial_k f(p)$, so the i-th component of $\xi \cdot \nabla G$ is $t_i^{q+j} D_{v_i} f(p_i)$. The condition is therefore $(D_{v_i} f(p_i))_i \in V_{q+j}(P)$. For $j = -1$ and $q \geq d+1$ it is empty, because $V_{q-1}(P) = \mathbb{C}^r$; for $q = d$ it is the condition $J_P(v)(f) = 0$. Thus the space of normalized representatives in degree -1 is $\ker J_P$. The indeterminacy R_0^4 consists of the constant vectors $e = (e_0, e') \in \mathbb{C}^4$; after normalization they become $v_i = e' - e_0 p_i$, which is W_P. Since the points affinely span $\mathbb{A}^3$, the map $(b, a) \mapsto (b + a p_i)_i$ is injective. This proves (a).

If $j \leq -2$, then $R_{j+1}^4 = 0$, so there is no indeterminacy, and the condition for $q = d$ is (2.4) when $j = -2$. Multiplication by X_0 multiplies the i-th component by t_i, which in the normalization keeps v_i and raises the degree by one. This proves (b) and (c). □

Lemma 2.4. *For every $P \in U_{d,s}^{\mathrm{ev}}$ we have $W_P \subset \ker J_P$. Consequently* $\operatorname{rank} J_P \leq 3r-4$ *when the points affinely span $\mathbb{A}^3$, and $T^1_{-1} = 0$ if and only if* $\operatorname{rank} J_P = 3r-4$.

Proof. Let $f \in K(P)$. For $v_i = b$ we get $D_b f(p_i)$, and $D_b f \in F_{d-1}$. For $v_i = p_i$, Euler's formula gives $D_p f(p) = \sum_k k f_k(p)$, and since $f(p_i) = 0$ this equals $\big(\sum_k (k-d) f_k\big)(p_i)$, the value of a polynomial in F_{d-1}. In both cases $J_P(v)(f) = 0$. The rest follows from Proposition 2.3(a). □

Comparing dimensions, $\dim T^1_{-1} \geq 3r - 4 - sm = -F_d(s)$, where

$$F_d(s) = s(M_d - s) - 3(L_d + s) + 4. \tag{2.5}$$

Thus $T^1_{-1} = 0$ is possible only if $F_d(s) \geq 0$. A direct computation gives

$$F_d(d+2) = -4(d+1), \qquad F_d(d+3) = \tfrac{1}{2}(d-11)(d+2). \tag{2.6}$$

The function F_d is a concave quadratic in s which is invariant under $s \mapsto M_d - 3 - s$. Hence, for $d \geq 11$, the integers s with $F_d(s) \geq 0$ form exactly the interval

$$d + 3 \leq s \leq M_d - d - 6, \tag{2.7}$$

whose endpoints are exchanged by this symmetry, as are $d+2$ and $M_d - d - 5$.

2.3. **Openness.**

Lemma 2.5. *The set $U_{d,s}^{(-1)}$ of $P \in U_{d,s}^{\mathrm{ev}}$ whose points affinely span $\mathbb{A}^3$ and satisfy* $\operatorname{rank} J_P = 3r-4$ *is Zariski open in $(\mathbb{A}^3)^r$. Its image in* Conf_r *under $P \mapsto ((1 : p_i))_i$ is open, and it consists of configurations Z with $T^1_{-1}(C(Z)) = 0$.*

Proof. Cover $U^{\mathrm{ev}}_{d,s}$ by the open sets on which a fixed $r \times r$ minor of $\mathrm{ev}_P|_{F_d}$ and a fixed $L_d \times L_d$ minor of $\mathrm{ev}_P|_{F_{d-1}}$ do not vanish. On such a set Cramer's rule gives a basis of $K(P)$ and a complement-projection for $V_{d-1}(P)$ whose entries are regular functions, so J_P is represented by a $sm \times 3r$ matrix of regular functions. The condition $\operatorname{rank} J_P \geq 3r-4$ is the nonvanishing of one of its minors of that size, and $\operatorname{rank} J_P \leq 3r-4$ holds everywhere by Lemma 2.4. The map to Conf_r is an open immersion onto the configurations missing the plane $X_0 = 0$, and the last assertion is Lemma 2.4. □

Since $(\mathbb{A}^3)^r$ is irreducible, $U^{(-1)}_{d,s}$ is either empty or dense, and it is nonempty as soon as one configuration in it is exhibited. All of our witnesses are of this kind; some of them are only points of a larger irreducible parameter space mapping to $(\mathbb{A}^3)^r$, and we explain the transfer in each case.

2.4. **The leading-form map.** The map $\Phi\colon U^{\mathrm{ev}}_{d,s} \to \mathrm{Gr}(m, S_d)$ is a morphism: on the charts in the proof of Lemma 2.5 it is given by the leading forms of a regular basis of $K(P)$.

Proposition 2.6. *For $P \in U^{\mathrm{ev}}_{d,s}$ we have $\ker d\Phi_P = \ker J_P$. In particular, in characteristic zero,*

$$\operatorname{rank} J_P = 3r-4 \text{ for general } P \iff \dim \overline{\Phi(U^{\mathrm{ev}}_{d,s})} = 3r-4.$$

Proof. Let $v \in (\mathbb{C}^3)^r$ and let $P_\varepsilon = (p_i + \varepsilon v_i)$ over $\mathbb{C}[\varepsilon]/(\varepsilon^2)$. A basis of $K(P_\varepsilon)$ has the form $f_a + \varepsilon g_a$, where (f_a) is a basis of $K(P)$ and $g_a \in F_d$ satisfies $g_a(p_i) + D_{v_i} f_a(p_i) = 0$ for all i. The tangent vector $d\Phi_P(v)$ is the homomorphism $\Phi(P) \to S_d/\Phi(P)$ sending $(f_a)_d$ to the class of $(g_a)_d$. If $d\Phi_P(v) = 0$, then after subtracting a combination of the f_b from g_a we may assume $(g_a)_d = 0$, so $g_a \in F_{d-1}$ and $J_P(v)(f_a) = 0$. Conversely, if $J_P(v) = 0$, then each g_a may be chosen in F_{d-1}, and $d\Phi_P(v) = 0$. The second assertion follows from generic smoothness of Φ onto its image and from Lemma 2.4. □

Remark 2.7. The ideal $\Phi(P) \oplus S_{\geq d+1} \subset \mathbb{C}[x,y,z]$ is the ideal of the flat limit of the homothetic configurations tP as $t \to 0$: indeed $t^d f(t^{-1}x) \to f_d$ for $f \in K(P)$, and the Hilbert functions agree. Its quotient has Hilbert function $(1, 3, \dots, M_{d-1}, s)$, of the very compressed type discussed in [5, Section 2.3]. Proposition 2.6 therefore identifies $T^1_{-1} = 0$ with the statement that the map $P \mapsto \lim_{t\to 0} tP$ to the Grassmannian $\mathrm{Gr}(m, S_d)$ has image of the maximal dimension $3r-4$ allowed by translations and dilation; this is the map and the maximal-rank question studied in [3, Section 5]. It means that a general fibre has dimension four. It does not assert that a general fibre is a single orbit. In the language of de Boor and Ron, $\Phi(P) \oplus S_{\geq d+1}$ is the space of leading forms of the ideal of P, and [2, Theorem (4.8)] identifies it with the orthogonal complement, for the apolarity pairing, of the least interpolation space of P, which is a correct interpolation space by [2, Theorem (5.8)]; this duality says nothing about the rank of the differential of Φ.

3. Cotangent groups of lower negative degree

In this section we show that vanishing in degree -1 propagates to all negative degrees. For $P \in U^{\mathrm{ev}}_{d,s}$ define

$$\Psi_P\colon \mathbb{C}^3 \oplus \mathbb{C} \to \mathrm{Hom}(K(P), S_{d-1}), \qquad \Psi_P(b,a)(f) = D_b f_d - a\, f_{d-1}. \tag{3.1}$$

On the charts of Lemma 2.5, Ψ_P is a matrix of regular functions, so its injectivity is an open condition on $U^{\mathrm{ev}}_{d,s}$; it does not depend on the choice of basis of $K(P)$.

Proposition 3.1. *Let $d \geq 2$ and $0 \leq s \leq M_d - 4$. Then Ψ_P is injective for a nonempty open set of $P \in U^{\mathrm{ev}}_{d,s}$.*

Proof. Since $m = M_d - s \geq 4$ and $d \geq 2$, the four multi-indices

$$(d,0,0), \quad (0,d,0), \quad (0,0,d), \quad (d-1,1,0)$$

are distinct, and we may choose a set A of m multi-indices of degree d containing them. Let Λ consist of all multi-indices of degree at most $d-1$ together with the multi-indices of degree d not in A, and let $P = P_\Lambda$ be as in Lemma 2.1. Then $|\Lambda| = L_d + M_d - m = r$, the configuration lies in $U^{\mathrm{ev}}_{d,s}$ by the argument of Corollary 2.2, and its points affinely span $\mathbb{A}^3$. By Lemma 2.1 the polynomials N_α, $\alpha \in A$, vanish on P; their leading monomials x^α are distinct, so they form a basis of $K(P)$.

The polynomials attached to the four chosen indices are

$$N_{(d,0,0)} = x^d - \tbinom{d}{2}x^{d-1} + \cdots, \qquad N_{(0,d,0)} = y^d - \tbinom{d}{2}y^{d-1} + \cdots,$$
$$N_{(0,0,d)} = z^d - \tbinom{d}{2}z^{d-1} + \cdots, \qquad N_{(d-1,1,0)} = x^{d-1}y - \tbinom{d-1}{2}x^{d-2}y + \cdots,$$

where the dots denote terms of degree at most $d-2$; the factor y of the last polynomial contributes no term of lower degree. In $\Psi_P(b,a)$ we select the coefficient of x^{d-1}, y^{d-1}, z^{d-1} and $x^{d-2}y$ in the images of these four polynomials, respectively. In the coordinates (b_x, b_y, b_z, a) the resulting 4×4 minor is

$$\det \begin{pmatrix} d & 0 & 0 & \binom{d}{2} \\ 0 & d & 0 & \binom{d}{2} \\ 0 & 0 & d & \binom{d}{2} \\ d-1 & 0 & 0 & \binom{d-1}{2} \end{pmatrix} = d^2\Big(d\tbinom{d-1}{2} - (d-1)\tbinom{d}{2}\Big) = -\frac{d^3(d-1)}{2} \neq 0.$$

For $d = 2$ the last row comes from the coefficient of y in $N_{(1,1,0)} = xy$, and the formula remains valid. Hence Ψ_P is injective at P_Λ. □

Proposition 3.2. *Let $d \geq 2$, $0 \leq s \leq M_d - 4$ and $r \geq 4$. Let $P \in U^{\mathrm{ev}}_{d,s}$ be a configuration whose points affinely span $\mathbb{A}^3$ and for which Ψ_P is injective, and let $Z = ((1 : p_i))_i$. Then multiplication by X_0 induces injections*

$$T^1_j(C(Z)) \hookrightarrow T^1_{-2}(C(Z)) \hookrightarrow T^1_{-1}(C(Z)) \qquad (j \leq -3).$$

In particular, if $T^1_{-1}(C(Z)) = 0$, then $T^1_j(C(Z)) = 0$ for every $j < 0$.

Proof. For $j \leq -3$, multiplication by X_0^{-2-j} preserves the representing vectors of Proposition 2.3(b),(c), and a class in degree at most -2 is zero exactly when its representing vector is zero; hence the first map is injective.

Let v represent a class in T^1_{-2} whose image in T^1_{-1} vanishes. By Proposition 2.3(a),(c) we have $v_i = b + ap_i$ for some $b \in \mathbb{C}^3$ and $a \in \mathbb{C}$. Let $f \in K(P)$. By Euler's formula and $f(p_i) = 0$,

$$D_{v_i} f(p_i) = H_f(p_i), \qquad H_f = D_b f - a\sum_{k=0}^{d-1}(d-k)f_k \in F_{d-1}.$$

By (2.4) there is $g \in F_{d-2}$ with $\mathrm{ev}_P(H_f) = \mathrm{ev}_P(g)$. Since ev_P is injective on F_{d-1}, $H_f = g$, so the degree $d-1$ part of H_f vanishes; this part is $D_b f_d - a f_{d-1} = \Psi_P(b,a)(f)$. As f was arbitrary, $\Psi_P(b,a) = 0$, hence $a = 0$, $b = 0$ and $v = 0$. □

Remark 3.3. Proposition 3.2 uses only necessary conditions coming from the equations of degree d; equations of higher degree could only impose further conditions. It gives no information on T^1_{-1} itself. The Newton configuration P_Λ in the proof of Proposition 3.1 is used only as a point of the open set where Ψ_P is injective; we make no claim about its group T^1_{-1}.

4. Nonsmoothability

The following statement is the form in which negative vanishing is used. For the case of finite schemes compare [4, Theorem 1.2]; the case of curves is analogous, and we give the argument in the generality needed.

Proposition 4.1. *Let $C = \operatorname{Spec} S/I \subset \mathbb{A}^N$ be an affine cone with an isolated singular point at the vertex, where I is a homogeneous ideal generated in degrees at least $d \geq 2$ and $\dim C = 1$. If $T^1_j(C) = 0$ for all $j < 0$, then the germ of C at the vertex admits neither an algebraic nor an analytic smoothing.*

Proof. By Pinkham's construction [7, Proposition 2.3 and (2.6)–(2.10)], the germ has a formal versal deformation on which $\mathbb{G}_m$ acts compatibly with its action on C. Concretely, choose a homogeneous basis of T^1; a basis element of degree e corresponds to a parameter u of weight $-e$, and the defining equations $G^0_1, \dots, G^0_k$ of C, of degrees $q_1, \dots, q_k$, lift to equations $G_1, \dots, G_k \in \mathbb{C}[[u]][X_1, \dots, X_N]$ that are homogeneous of degree q_l for the total weight. By hypothesis $e \geq 0$ for every basis element, so every parameter has weight at most 0. A monomial $u^\alpha X^\beta$ in G_l satisfies $|\beta| + \mathrm{wt}(u^\alpha) = q_l$, hence $|\beta| \geq q_l \geq d$. Thus every G_l lies in $(X_1, \dots, X_N)^d$. In particular $X = 0$ defines a section of the versal family, and along this section all G_l and all their first partial derivatives in X vanish, because $d \geq 2$. The relative continuous cotangent module restricted to the section is therefore free of rank N.

Suppose that a smoothing exists, algebraic or analytic. Restricting it to a curve in its base through the special point which meets the locus of smooth fibres, and passing to the normalization of that curve, we obtain a flat family $\mathcal{X} \to T$ over a smooth curve germ $(T, 0)$ with uniformizer τ, with special fibre the germ of C and with smooth fibres over $T \smallsetminus \{0\}$ near the singular point x. Let $\Omega = \Omega_{\mathcal{X}/T}$. By the Jacobian criterion the relative singular locus of $\mathcal{X} \to T$ is the zero set of $\mathrm{Fitt}_1(\Omega)$; it is contained in the fibre $\tau = 0$, so τ lies in the radical of $\mathrm{Fitt}_1(\Omega)\mathcal{O}_{\mathcal{X},x}$ (by the Nullstellensatz in the algebraic case and by Rückert's Nullstellensatz in the analytic case), and since the local ring is Noetherian, $\tau^n \in \mathrm{Fitt}_1(\Omega)\mathcal{O}_{\mathcal{X},x}$ for some n. Fitting ideals commute with base change and with completion, so $\tau^n \in \mathrm{Fitt}_1(\widehat{\Omega})$, where $\widehat{\Omega}$ is the continuous relative cotangent module of the completed family over $\mathbb{C}[[\tau]]$. By formal versality the completed family is isomorphic, over $\mathbb{C}[[\tau]]$, to the pullback $\mathcal{Y}_\tau$ of the equivariant versal family along a morphism $\mathbb{C}[[u]] \to \mathbb{C}[[\tau]]$; an isomorphism of formal families identifies the continuous cotangent modules and their Fitting ideals, so $\tau^n \in \mathrm{Fitt}_1(\Omega_{\mathcal{Y}_\tau})$. The section $X = 0$ of the versal family pulls back to a section $\iota\colon \mathbb{C}[[\tau]][[X]]/(G) \to \mathbb{C}[[\tau]]$ of $\mathcal{Y}_\tau$, and $\iota^*\Omega_{\mathcal{Y}_\tau}$ is free of rank $N \geq 2$ by the first paragraph. Since Fitting ideals commute with base change, $\iota(\mathrm{Fitt}_1(\Omega_{\mathcal{Y}_\tau})) = \mathrm{Fitt}_1(\iota^*\Omega_{\mathcal{Y}_\tau}) = 0$, whereas $\iota(\tau^n) = \tau^n \neq 0$. This contradiction proves the proposition. □

Remark 4.2. The argument uses only the formal versal family and the formal isomorphism with the pulled-back smoothing; no convergence of an equivariant coordinate change is needed. It applies to every smoothing, equivariant or not. For the cones $C(Z)$ of Section 2 the hypotheses on the singular point and on the degrees of generators were verified before Proposition 2.3, with $N = 4$.

5. Plane arrangements

Throughout this section $n = d + 2 \geq 4$, so that $L_d = \binom{n}{3}$ and $M_d = \binom{n}{2}$.

5.1. **Star interpolation.** Let $\ell_1, \dots, \ell_n$ be affine linear forms $\ell_i(x) = 1 + \nu_i \cdot x$ with $\nu_i \in \mathbb{C}^3$.

Definition 5.1. The arrangement of planes $\{\ell_i = 0\}$ is *genuine* if every three of the normals ν_i are linearly independent and no four of the planes have a common point.

For a 3-subset $T \subset [n]$ let a_T be the common point of the three planes indexed by T; for a 2-subset $\{a, b\}$ (an *edge*) and a 3-subset T put

$$E_{ab} = \prod_{j \notin \{a,b\}} \ell_j \in F_d, \qquad V_T = \prod_{j \notin T} \ell_j \in F_{d-1},$$

and write $V_{abk} = V_{\{a,b,k\}}$.

Lemma 5.2. *Let the arrangement be genuine and let $A = (a_T)_T$. Then the L_d points a_T are distinct, $(V_T)_T$ is a basis of F_{d-1} with $V_T(a_U) = 0$ for $U \neq T$ and $V_T(a_T) \neq 0$, and $(E_{ab})_{ab}$ is a basis of the space $I_A(\leq d)$ of polynomials in F_d vanishing on A. In particular $\dim I_A(\leq d) = M_d$, and the leading forms of the E_{ab} are linearly independent.*

Proof. The point a_T lies on exactly the planes indexed by T, so the points are distinct, $V_T(a_T) \neq 0$ and $V_T(a_U) = 0$ for $U \neq T$. Since $\dim F_{d-1} = L_d$, the V_T form a basis and evaluation on A is an isomorphism on F_{d-1}. Every E_{ab} vanishes on A. If $\sum c_{ab} E_{ab} = 0$, restriction to a general point of the line $\ell_a = \ell_b = 0$ kills every term except $c_{ab} E_{ab}$, so $c_{ab} = 0$. Since A imposes independent conditions on $F_{d-1} \subset F_d$, we have $\dim I_A(\leq d) = \dim F_d - L_d = M_d$. Finally $I_A(\leq d) \cap F_{d-1} = 0$. □

5.2. **The Jacobian in star coordinates.** Let $B = (b_j)_{j \in S}$ be s further points, with $A \cup B$ distinct, and put

$$\mathcal{E} = (E_{ab}(b_j))_{j \in S,\ ab} \qquad (s \times M_d).$$

If $\operatorname{rank} \mathcal{E} = s$, then $P = A \cup B$ lies in $U^{\mathrm{ev}}_{d,s}$ and $K(P) = \{\sum_{ab} c_{ab} E_{ab} :\ c \in \ker \mathcal{E}\}$. At a triple point we use the coordinates $t_{T,k} = \nu_k \cdot w_T$ $(k \in T)$ of a velocity $w_T \in \mathbb{C}^3$; since the three normals in T are independent, this is an invertible change of coordinates.

Proposition 5.3. *Let $P = A \cup B$ be as above with $\operatorname{rank} \mathcal{E} = s$, and let (w, v) be velocities at A and B. For an edge ab and $j \in S$ put*

$$X_{j,ab}(w, v) = D_{v_j} E_{ab}(b_j) - \sum_{k \notin \{a,b\}} t_{abk,k}\, V_{abk}(b_j). \tag{5.1}$$

Then $(w, v) \in \ker J_P$ if and only if $\sum_{ab} c_{ab} X_{j,ab}(w, v) = 0$ for all $c \in \ker \mathcal{E}$ and all $j \in S$.

Proof. Let $f = \sum c_{ab}E_{ab}$ with $c \in \ker \mathcal{E}$. At a_T only the edges contained in T contribute to the gradient, and for $T = \{a, b, k\}$ we get $D_{w_T}E_{ab}(a_T) = V_T(a_T)\, t_{T,k}$. By Lemma 5.2 the unique $g \in F_{d-1}$ with $g(a_T) = D_{w_T}f(a_T)$ for all T is $g = \sum_{ab} c_{ab} \sum_{k \notin \{a,b\}} t_{abk,k} V_{abk}$. The vector $(D_{v_i} f(p_i))_i$ lies in $V_{d-1}(P)$ if and only if it equals $\mathrm{ev}_P(g)$, that is, if and only if $D_{v_j} f(b_j) = g(b_j)$ for all $j \in S$. More precisely, the map $y \mapsto y_B - \mathrm{ev}_B(g_y)$, where $g_y \in F_{d-1}$ interpolates y_A, identifies $Q(P)$ with $\mathbb{C}^S$, and under this identification $J_P(w, v)(f)$ is the vector $\big(\sum_{ab} c_{ab} X_{j,ab}(w, v)\big)_{j \in S}$. □

The condition of Proposition 5.3 says that, for each j, the vector $X_j = (X_{j,ab})_{ab}$ is a linear combination of the rows of $\mathcal{E}$. Introducing the coefficients of these combinations as unknowns, we obtain the *extended system*

$$\Gamma_{j,ab}(w, v, \beta) = X_{j,ab}(w, v) - \sum_{l \in S} \beta_{jl} E_{ab}(b_l) = 0 \qquad (j \in S,\ ab), \tag{5.2}$$

in the unknowns $t = (t_{T,k})$, $v \in (\mathbb{C}^3)^S$ and $\beta \in \mathbb{C}^{S \times S}$.

Corollary 5.4. *If* $\operatorname{rank} \mathcal{E} = s$*, the projection* $(t, v, \beta) \mapsto (t, v)$ *maps* $\ker \Gamma$ *isomorphically onto* $\ker J_P$ *(in star coordinates). Consequently* $\operatorname{rank} J_P = \operatorname{rank} \Gamma - s^2$.

Proof. For $(t, v) \in \ker J_P$ the coefficients β_{jl} exist and are unique because the rows of $\mathcal{E}$ are independent. The rank identity follows by comparing $\dim \ker \Gamma$ with the number $3L_d + 3s + s^2$ of unknowns. □

The four motions of Lemma 2.4 have the following description: a translation by b has $t_{T,k} = \nu_k \cdot b$, $v_j = b$, $\beta = 0$; the dilation has $t_{T,k} = -1$, $v_j = b_j$ and $\beta = (n-2)I$. This follows from $D_b E_{ab} = \sum_k (\nu_k \cdot b) V_{abk}$ and $D_x E_{ab}(x) = (n-2)E_{ab}(x) - \sum_k V_{abk}(x)$, which are identities of polynomials.

5.3. **Formal normal coordinates.** The entries of (5.2) are polynomials in $\nu = (\nu_1, \dots, \nu_n)$ and in B, once the $t_{T,k}$ are regarded as independent unknowns. For an arbitrary point $y = (\nu, B)$ of the affine space $Y_s = (\mathbb{C}^3)^n \times (\mathbb{C}^3)^S$, this defines the *formal* extended system $\Gamma^{\mathrm{f}}(y)$, a matrix with sM_d rows and $N_s = 3L_d + 3s + s^2$ columns. We stress that at a point y whose arrangement is not genuine, the triple points a_T need not exist and $\Gamma^{\mathrm{f}}(y)$ is not the differential of anything; it is merely a matrix of polynomials.

Let $Y_s^\circ \subset Y_s$ be the set of y such that the arrangement is genuine, $A \cup B$ consists of distinct points and $\operatorname{rank} \mathcal{E} = s$. It is open, and for $s < M_d$ it is nonempty: the evaluations at points of $\mathbb{A}^3$ span the dual of $I_A(\le d)$, and any prescribed point at which some E_{ab} does not vanish can be completed to s points with independent evaluations.

Lemma 5.5. *Let* $Z \subset Y_s$ *be an irreducible locally closed subset with* $Z \cap Y_s^\circ \neq \emptyset$.

(a) *If* $\operatorname{rank} \Gamma^{\mathrm{f}}(z_0) \ge N_s - 4$ *for some* $z_0 \in Z$*, then for a general point* $z \in Z$ *the configuration* $P = A \cup B$ *satisfies* $\operatorname{rank} J_P = 3r - 4$*, and* $U_{d,s}^{(-1)}$ *is nonempty.*

(b) *More generally, let* G *be a matrix of rational functions on* Z*, regular on a nonempty open subset* Z'*, and let* ρ, ρ' *be integers such that, for every* $z \in Z' \cap Y_s^\circ$*,* $\operatorname{rank} G(z) \ge \rho$ *implies* $\operatorname{rank} J_P \ge \rho'$ *for the configuration* $P = A \cup B$ *of* z*. If* $\operatorname{rank} G(z_0) \ge \rho$ *for one* $z_0 \in Z'$*, then* $\operatorname{rank} J_P \ge \rho'$ *for a general configuration* P *of* r *points.*

Proof. In (b) the condition $\operatorname{rank} G \geq \rho$ is open on Z' and nonempty, hence dense in the irreducible Z; so it meets the nonempty open set $Z' \cap Y_s^\circ$, and at a common point $\operatorname{rank} J_P \geq \rho'$. On the charts of Lemma 2.5 the condition $\operatorname{rank} J_P \geq \rho'$ is open on $(\mathbb{A}^3)^r$, which is irreducible. For (a) take $G = \Gamma^{\mathrm{f}}$ and $\rho = N_s - 4$: at a point of Y_s° the formal system is the extended system, so by Corollary 5.4 and Lemma 2.4, $4 \leq \dim\ker J_P = \dim\ker \Gamma^{\mathrm{f}} \leq 4$. Lemma 2.5 gives the conclusion for general configurations. □

Below, Z will be a linear family through a non-genuine (cyclic) point and Y_s itself, or the incidence variety of assigned points described next.

5.4. **Assigned points and the positions $s \leq d+2$.** For $S \subset [n]$ let $X_S \subset Y_s$ be the variety of (ν, B) with $\ell_i(b_i) = 0$ for $i \in S$; we then call b_i the point *assigned* to the plane i. On the affine chart where every assigned point is nonzero, the fibre over B is a product of affine planes and free factors, so X_S is irreducible. Put $C_{ij} = \ell_j(b_i)$; on the open set where $C_{ij} \neq 0$ for $i \neq j$, the point b_i lies on no other plane, and $\alpha_i = \prod_{j \neq i} C_{ij} \neq 0$. Then $E_{ab}(b_j) = 0$ unless $j \in \{a, b\}$, and $E_{ab}(b_a) = \alpha_a / C_{ab}$.

Theorem 5.6. *Let $d \geq 4$ and $0 \leq s \leq d+2$. For a general configuration P of $r = L_d + s$ points the map J_P is surjective,* $\operatorname{rank} J_P = s(M_d - s)$*, and*

$$\dim T^1_{-1}(C(Z)) = 3r - 4 - s(M_d - s) = -F_d(s).$$

In particular $\dim T^1_{-1}(C(Z)) = 4(d+1)$ *for general Z at $s = d+2$.*

Proof. We work on X_S with $|S| = s$ and use only the columns of the triple coordinates $t_{T,k}$; the auxiliary velocity columns are retained in J_P but are not needed for surjectivity. By the identification $Q(P) \cong \mathbb{C}^S$ used in the proof of Proposition 5.3, the target of J_P is $\operatorname{Hom}(\ker \mathcal{E}, \mathbb{C}^S)$, whose dual is $\ker \mathcal{E} \otimes (\mathbb{C}^S)^*$. A dual vector is thus a family $(\lambda_i)_{i \in S}$ with $\lambda_i \in \ker \mathcal{E}$, and it annihilates the column $t_{T,k}$ if and only if $\sum_{i \in S} \lambda_{i,ab} V_T(b_i) = 0$ for $ab = T \smallsetminus k$. Since $V_T(b_i) = 0$ for $i \notin T$ and $V_T(b_i) = \alpha_i / (C_{ij} C_{ik})$ for $T = \{i, j, k\}$, the substitution $\mu_{i,ab} = \alpha_i \lambda_{i,ab}$ turns these conditions and the conditions $\lambda_i \in \ker \mathcal{E}$ into

$$\sum_{i \in S \cap T} \frac{\mu_{i,uv}}{C_{ij} C_{ik}} = 0 \qquad (T \text{ a triple},\ uv \subset T,\ \{j,k\} = T \smallsetminus i), \tag{5.3}$$

$$\sum_{k \neq j} \frac{\mu_{i,jk}}{C_{jk}} = 0 \qquad (i, j \in S). \tag{5.4}$$

We call a solution a *stress*. Let G be the coefficient matrix of the linear system (5.3)–(5.4) in the sM_d unknowns μ; its entries are rational functions on X_S, regular where $C_{ij} \neq 0$ for $i \neq j$, and the absence of nonzero stresses means $\operatorname{rank} G = sM_d$. At points of $X_S \cap Y_s^\circ$ with $\operatorname{rank} \mathcal{E} = s$, the absence of stresses is equivalent to the surjectivity of the restriction of J_P to the triple-coordinate columns, and therefore implies that J_P itself is surjective, that is, $\operatorname{rank} J_P \geq sm$. By Lemma 5.5(b) with $Z = X_S$, $\rho = sM_d$ and $\rho' = sm$, it suffices to find one point of X_S with $C_{ij} \neq 0$ off the diagonal at which every stress vanishes. The set $X_S \cap Y_s^\circ$ is nonempty by a general choice of a genuine arrangement and of points on the planes. The formula for $\dim T^1_{-1}$ follows from Proposition 2.3, and the last assertion from (2.6). The case $s = 0$ is empty. We now produce stress-free points.

Case $1 \leq s \leq n-2$. Let $a \neq b$ be indices outside S. Applying (5.3) to triples with exactly one index i in S shows that μ_i vanishes on every edge $\{i, a\}$ with

$a \notin S$ and on every edge with both ends outside S. Applying it next to triples $\{i,j,a\}$ with $i,j \in S$, $a \notin S$, shows that every μ_i vanishes on each edge meeting the complement of S. For an edge $\{i,j\} \subset S$ the two triples $\{i,j,a\}$ and $\{i,j,b\}$ give a 2×2 system for $\mu_{i,ij}, \mu_{j,ij}$ that is nonsingular when

$$C_{ja}/C_{ia} \neq C_{jb}/C_{ib}; \tag{5.5}$$

then both vanish, and the triple $\{i,j,l\}$ gives $\mu_{l,ij} = 0$ for every other $l \in S$. Condition (5.5) holds simultaneously for all pairs for a general choice of the two unused planes. Here (5.4) was not used.

Case $s = n$. Put $x_{ab} = \mu_{a,ab}/C_{ab}$ for $a \neq b$. Equation (5.3) for the triple $\{a,b,k\}$ and the edge ab determines $\mu_{k,ab} = -C_{ka}C_{kb}(x_{ab}/C_{ak} + x_{ba}/C_{bk})$. Equations (5.4) then become $\sum_{k\neq i} x_{ik} = 0$ and

$$x_{ij} - \sum_{k\notin\{i,j\}} \Big(\frac{C_{ik}}{C_{jk}}x_{jk} + \frac{C_{ji}C_{ik}}{C_{jk}C_{ki}}x_{kj}\Big) = 0 \qquad (i \neq j).$$

Suppose C is symmetric, and put $Z_{ij} = x_{ij}/C_{ij}$, $Z_{ii} = 0$ and $c = Z^{\mathsf{t}}\mathbf{1}$; no symmetry of Z is assumed. Restoring the omitted terms $k = i$ in the column sums, the equations become

$$2(C\circ Z) = CZ^{\mathsf{t}} + C\,\mathrm{diag}(c), \qquad (C\circ Z)\mathbf{1} = 0, \qquad Cc = 0, \tag{5.6}$$

where $\circ$ is the entrywise product; the diagonal of the first identity is the row-sum condition, and the last follows by applying the first to $\mathbf{1}$.

Specialize to the circular point: $b_i = (\cos\theta_i, \sin\theta_i, 0)$ and $\ell_j(x) = 1 - \cos\theta_j\, x_1 - \sin\theta_j\, x_2$ with $\theta_i = 2\pi i/n$. Then $\ell_i(b_i) = 0$ and $C_{ij} = \frac{1}{2}(2 - \omega^{i-j} - \omega^{j-i})$ with $\omega = e^{2\pi\sqrt{-1}/n}$; this matrix is symmetric with nonzero off-diagonal entries. The equations (5.3), (5.4) are invariant under a common rescaling of C, so we may replace C by $2 - \omega^{i-j} - \omega^{j-i}$. All normals lie in a plane, so this point of X_S is not a genuine arrangement; it is used only as a point of X_S. With $e_p(i) = \omega^{pi}$, the matrix C has eigenvalues $\lambda_0 = 2n$, $\lambda_{\pm 1} = -n$ and $\lambda_p = 0$ otherwise. Write $Z = \sum z_{pq} e_p e_q^{\mathsf{t}}$, fix a total index k modulo n, and put $h_p = z_{p,k-p}$, $H = h_0$. Then (5.6) becomes

$$-2(h_{p-1} - 2h_p + h_{p+1}) = \lambda_p(h_{k-p} + H), \qquad \sum_p h_p = 0, \qquad \lambda_k H = 0. \tag{5.7}$$

For $p \notin \{0, \pm 1\}$ the right side vanishes, so h is affine on $1 \le p \le n-1$: with $A = h_1$, $B = h_{n-1}$ and $D = B - A$, $h_p = A + (p-1)D/(n-2)$, and $\sum h_p = 0$ gives $A + B = -2H/(n-1)$.

If $2 \le k \le n-2$, subtracting the equations at $p = 1$ and $p = -1$ gives $4nD/(n-2) = 0$, so $D = 0$; the equation at $p = 0$ then gives $2n(n-4)H = 0$, so $H = 0$ for $n \ge 5$, and $h = 0$. If $k = \pm 1$, then $H = 0$ and $A + B = 0$, and the equation at $p = 0$ gives $h_k = 0$, so $A = B = 0$. If $k = 0$, then $H = 0$, $A + B = 0$, and the equation at $p = 1$ reduces to $n(n-4)D/(2(n-2)) = 0$, so $D = 0$. Thus $Z = 0$ for $n \ge 5$, hence $x = 0$ and $\mu = 0$. (For $n = 4$ the stress space is two-dimensional. The case $s = n-1$ below needs $n - 1 \ge 5$, which is why we assume $d \ge 4$.)

Case $s = n - 1$. Let 0 be the unused index and $q = n - 1 \ge 5$. On the assigned indices take the circular point for q, and specialize ℓ_0 to the constant 1, so $C_{i0} = 1$. For assigned $i \neq j$ put $x_{ij} = \mu_{i,ij}/C_{ij}$; the triple $\{0,i,j\}$ gives $x_{ij} = -x_{ji}$, so

$Z_{ij} = x_{ij}/C_{ij}$ is skew-symmetric. With $a_j = \mu_{j,0j}$ the same triple gives $\mu_{i,0j} = -a_j$ for $i \neq j$, and (5.4) becomes

$$2(C \circ Z) + CZ + C\,\mathrm{diag}(Z\mathbf{1}) - \mathbf{1}a^{\mathsf{t}} = 0, \qquad a_i = (CZ)_{ii}; \tag{5.8}$$

for instance the (i,j) entry is $2C_{ij}Z_{ij} + (CZ)_{ij} + C_{ij}(Z\mathbf{1})_j - a_j$, which includes the term of the unused edge. In Fourier coordinates as before, skew symmetry gives $h_p = -h_{k-p}$ and $\sum h_p = 0$, and (5.8) gives, away from $p = 0$, $-2(h_{p-1} - 2h_p + h_{p+1}) + \lambda_p(h_p - H) = 0$. Hence h is affine on $1 \leq p \leq q-1$, and the equations at $p = \pm 1$ read

$$(q-2)(A - H) + \tfrac{2D}{q-2} = 0, \qquad (q-2)(B - H) - \tfrac{2D}{q-2} = 0.$$

Adding gives $A + B = 2H$, and with $H + (q-1)(A+B)/2 = 0$ this gives $H = 0$. Subtracting gives $q(q-4)D/(q-2) = 0$, so $D = 0$ and all modes vanish. Then $Z = 0$, $x = 0$, $a = 0$, and (5.3) gives $\mu = 0$.

Finally, at both circular points $\operatorname{rank}\mathcal{E} = s$. Indeed, let $\sum_i y_i E_{ab}(b_i) = 0$ for all edges ab. Since $E_{ab}(b_i) = 0$ unless $i \in \{a, b\}$ and $E_{ij}(b_i) = \alpha_i/C_{ij}$, the column of an assigned edge $\{i,j\}$ gives $\alpha_i y_i/C_{ij} + \alpha_j y_j/C_{ji} = 0$. Put $y'_i = \alpha_i y_i$; since C is symmetric, $y'_i + y'_j = 0$ for all assigned $i \neq j$, and three assigned indices force $y' = 0$, hence $y = 0$ because $\alpha_i \neq 0$. The required open conditions are therefore nonempty, and the theorem follows. □

Remark 5.7. Theorem 5.6 does not include the first position $s = d + 3$ at which $F_d(s) \geq 0$, and it implies nothing about that position; this is the content of Sections 6 and 7.

6. Cyclic arrangements and the special kernel

From now on until the end of Section 7, $n = d + 2$, $s = n + 1 = d + 3$ and $r = \binom{n}{3} + n + 1$. The auxiliary points are n points, one assigned to each plane, together with the origin o, and they will be moved off the planes by a homothety. The relevant configurations are not genuine, and all statements are about the formal extended system of Section 5.3.

6.1. **The cyclic family.** Let $G = \mu_n \subset \mathbb{C}^\times$ and fix a generator ω. Let $F = \{0, 1, 3, 7\}$ and consider coefficient vectors (a_1, a_3, a_7) in the affine plane $\Sigma = \{a_1 + a_3 + a_7 = -1\}$; put $a_0 = 1$ and

$$f(u) = \sum_{e \in F} a_e u^e, \qquad f(1) = 0.$$

For $i \in \mathbb{Z}/n$ let $t_i = \omega^i$ and

$$p_i = (t_i, t_i^3, t_i^7), \qquad \nu_i = (a_1 t_i^{-1}, a_3 t_i^{-3}, a_7 t_i^{-7}), \qquad \ell_i(x) = 1 + \nu_i \cdot x.$$

Then $\ell_j(p_i) = f(t_i/t_j)$; in particular $\ell_i(p_i) = 0$. For $\varepsilon \in \mathbb{C}$ let $y(\varepsilon) \in Y_{n+1}$ consist of the normals ν_i and the auxiliary points $b_i(\varepsilon) = (1 - \varepsilon)p_i$ ($i \in \mathbb{Z}/n$) and $b_o = o$, the origin; we write $b_j = b_j(0)$, so $b_i = p_i$. At $\varepsilon = 0$ the point p_i is assigned to the plane i. The linear map $\sigma(x) = (\omega x_1, \omega^3 x_2, \omega^7 x_3)$ satisfies $\sigma(p_i) = p_{i+1}$ and $\ell_{i+1} \circ \sigma = \ell_i$, and it commutes with homotheties. Hence $\mathbb{Z}/n$ acts on the unknowns and the equations of $\Gamma^{\mathrm{f}}(y(\varepsilon))$ compatibly, for every ε, and the system decomposes into its character components $\chi \in \mathbb{Z}/n$.

Put

$$r(x)=\frac{f(x)}{f(x^{-1})}\ (x\in G\smallsetminus\{1\}),\qquad r(1)=-1,\qquad \lambda_j=\sum_{x\in G} r(x)x^{-j}\quad (j\in\mathbb{Z}/n).$$

If $f'(1)\neq 0$ the value $r(1)=-1$ is the limit of $f(x)/f(x^{-1})$ at $x=1$. Let $\Omega_n\subset\Sigma$ be the set of coefficient vectors such that

$$\begin{aligned} &a_1a_3a_7\neq 0,\qquad f'(1)\neq 0,\qquad f(x)\neq 0\ \ (x\in G\smallsetminus\{1\}),\\ &\lambda_j\neq 0\ \ (j\in\mathbb{Z}/n),\qquad \gcd\bigl(f(u),u^7f(u^{-1})\bigr)=u-1. \end{aligned}\tag{6.1}$$

Lemma 6.1. *For every $n\geq 4$ the set Ω_n is a nonempty Zariski open subset of Σ.*

Proof. Each condition is the nonvanishing of a rational function or of a resultant, so it suffices to show that each is not identically violated; since Σ is irreducible, finitely many nonempty open sets meet. The last condition holds for $f=1+2u+3u^3-6u^7$: here $f'(1)=-31$, for $|u|>1$ we have $6|u|^7>1+2|u|+3|u|^3$, and on $|u|=1$ equality in $|1+2u+3u^3|\leq 6$ forces $u=1$; thus all roots other than 1 lie in the open unit disc, and those of $u^7f(u^{-1})$ other than 1 lie outside the closed disc.

Let $\zeta\in G$ be a primitive n-th root and let $\Lambda\subset\Sigma$ be the line $f(\zeta)=0$; it is a line because $\zeta^3-\zeta\neq 0$. If $\eta\in G\smallsetminus\{1,\zeta\}$ and $f(\eta)$ vanished identically on Λ, the vector $(1,\eta,\eta^3,\eta^7)$ would lie in the span of $(1,1,1,1)$ and $(1,\zeta,\zeta^3,\zeta^7)$; comparing the first three coordinates gives $(\eta-\zeta)(\eta+\zeta+1)=0$, so $\eta+\zeta=-1$, and two complex numbers of modulus one with sum -1 are the primitive cube roots of unity, which is impossible for $n\geq 4$. In the same way $f'(1)\equiv 0$ on Λ would force $\zeta^2+\zeta+1=3$. Hence a general point of Λ has $f'(1)\neq 0$ and no zero in G other than 1 and ζ. Near such a point, with transversal parameter $\tau=f(\zeta)$, exactly one summand of λ_j has a pole, namely $x=\zeta^{-1}$, with residue $f(\zeta^{-1})\zeta^j\neq 0$. So each λ_j is a nonzero rational function on Σ. Finally a_1,a_3,a_7 and $f(x)$ $(x\neq 1)$ are nonzero functions. □

6.2. **Star pivots and the residual system.** From now on we assume $n\geq 29$; Lemma 6.1 itself holds for all $n\geq 4$. At $\varepsilon=0$ write $C_{kl}=\ell_l(p_k)=f(t_k/t_l)$ and $\Pi=\prod_{x\in G\smallsetminus\{1\}}f(x)=\prod_{l\neq k}C_{kl}$, which is independent of k. The auxiliary points satisfy $E_{ab}(p_l)=0$ for $l\notin\{a,b\}$, $E_{ab}(p_a)=\Pi/C_{ab}$ and $E_{ab}(o)=1$, while $V_{abk}(p_k)=\Pi/(C_{ka}C_{kb})$ and $V_{abk'}(p_k)=0$ for $k'\neq k$. The rows of (5.2) are of three kinds: the *pivot rows* (k,ab) with $k\notin\{a,b\}$, the *assigned rows* (a,ab) and the *origin rows* (o,ab). The pivot row (k,ab) contains exactly one triple unknown, $t_{abk,k}$, with the nonzero coefficient $-V_{abk}(p_k)$, and each triple unknown occurs in exactly one pivot row. Solving,

$$t_{abk,k}=\nu_k\cdot v_k-C_{ka}C_{kb}\Bigl[\frac{\beta_{ka}}{C_{ab}}+\frac{\beta_{kb}}{C_{ba}}+\frac{\beta_{ko}}{\Pi}\Bigr].\tag{6.2}$$

Definition 6.2. The linear system in the unknowns (v,β) obtained by substituting (6.2) in the assigned and origin rows of $\Gamma^{\mathrm{f}}(y(0))$ is the *residual system* $\mathcal{R}$. Its kernel $\ker\mathcal{R}$ is the *special kernel*.

By construction

$$\ker\Gamma^{\mathrm{f}}(y(0))\cong\ker\mathcal{R},\qquad \operatorname{rank}\Gamma^{\mathrm{f}}(y(0))=3L_d+\operatorname{rank}\mathcal{R}.\tag{6.3}$$

The left kernel of $\Gamma^{\mathrm{f}}(y(0))$ is identified with that of $\mathcal{R}$ in the same way.

In the character χ the unknowns are

$$v_i=(V_1t_i^{\chi+1},V_3t_i^{\chi+3},V_7t_i^{\chi+7}),\quad \beta_{il}=t_i^{\chi}\beta(t_l/t_i),\quad \beta_{io}=\Pi c\,t_i^{\chi},\quad \beta_{oi}=\eta\,t_i^{\chi}/\Pi,$$

together with $\rho = \beta_{oo}$ if $\chi = 0$ and the component O_e of the origin velocity if $\chi = -e$, $e \in \{1,3,7\}$. Here $\beta\colon G \to \mathbb{C}$ is an arbitrary function; its value $\beta(1)$ is the diagonal coefficient. Thus there are $n+5$ unknowns in each character, plus one in the four *global* characters $\chi \in \{0,-1,-3,-7\}$; in total $n^2+5n+4 = 3(n+1)+(n+1)^2$, as it should be. We put $W_e = a_e V_e$, $\alpha = W_1 + W_3 + W_7$ and $\kappa_e = a_e O_e$.

Using rotation invariance, the assigned rows of the character χ are indexed by the separation $x = t_b \in G \smallsetminus \{1\}$ of the edge $\{0,b\}$ with assigned end 0, and the origin rows by the edges $\{0,b\}$ up to the reversal $b \leftrightarrow -b$ (for even n the diameter $b = n/2$ contributes only in even characters). Dividing the assigned row by $E_{0b}(p_0)$, the residual equations read, with $y = t_k$ running over $G \smallsetminus \{1,x\}$,

$$\sum_y \frac{\sum_{e>0} W_e y^{-e} - \alpha y^\chi}{f(y^{-1})} + \sum_y \frac{f(y)f(y/x)}{f(y^{-1})}\, y^\chi \Big[\frac{\beta(y^{-1})}{f(x^{-1})} + \frac{\beta(x/y)}{f(x)} + c\Big]$$

$$- \beta(1) - \beta(x)\frac{f(x^{-1})}{f(x)} - c\, f(x^{-1}) = 0, \tag{6.4}$$

$$\sum_y \big(\nu_k \cdot v_o - \alpha y^\chi\big) + \sum_y f(y)f(y/x)\, y^\chi \Big[\frac{\beta(y^{-1})}{f(x^{-1})} + \frac{\beta(x/y)}{f(x)} + c\Big]$$

$$- \frac{\eta}{f(x^{-1})} - \frac{\eta x^\chi}{f(x)} - \rho = 0. \tag{6.5}$$

Here $\nu_k \cdot v_o = \kappa_e y^{-e}$ in the character $-e$ and is zero otherwise, and ρ occurs only for $\chi = 0$.

6.3. **The assigned equations.** For $x \neq 1$ put $h(x) = f(x^{-1})\beta(x)$ and set $h(1) = 0$. Let

$$k_e = \sum_{x\neq 1} x^{-\chi-e}\frac{h(x)}{f(x)}\ (e \in F), \qquad \Theta = \sum_{y\neq 1} \frac{\sum_{e>0} W_e y^{-e} - \alpha y^\chi}{f(y^{-1})}, \qquad \vartheta = \Theta - \beta(1),$$

and let $\mathcal{C}_\chi h(x) = x^\chi \sum_{y\in G} r(x/y) y^{-\chi} h(y)$. For a function g on G write $\hat g_k = \sum_x g(x)x^{-k}$; then $\widehat{\mathcal{C}_\chi h}_k = \lambda_{k-\chi}\hat h_k$.

Lemma 6.3. *Multiplied by $f(x)$, the assigned equation* (6.4) *is equivalent to*

$$\mathcal{C}_\chi h(x) + r(x)\sum_{e\in F} a_e k_e x^{-e} + c\, f(x)\sum_{e\in F} a_e \lambda_{-\chi-e} x^{-e} + \vartheta f(x) - r(x)\Big[\sum_{e>0} W_e x^{-e} - \alpha x^\chi\Big] = 0. \tag{6.6}$$

The left side of (6.6) *at $x = 1$ equals $\sum_{e>0} W_e - \alpha = 0$. Hence the assigned equations are equivalent to* (6.6) *on all of G.*

Proof. We treat the terms of (6.4) after multiplication by $f(x)$. The first sum is $f(x)\Theta$ minus the omitted term $y = x$, which gives $-r(x)[\sum W_e x^{-e} - \alpha x^\chi]$; together with $-\beta(1)f(x)$ it gives the terms with ϑ and W. In the second sum write $f(y/x) = \sum_e a_e y^e x^{-e}$ and note $f(y)\beta(y^{-1}) = h(y^{-1})$. The $\beta(y^{-1})$ part becomes $r(x)\sum_e a_e x^{-e}\sum_{u\neq 1,x^{-1}} u^{-\chi-e}h(u)/f(u)$; the omitted $u = x^{-1}$ contributes a multiple of $\sum_e a_e = f(1) = 0$, so this is $r(x)\sum a_e k_e x^{-e}$. The $\beta(x/y)$ part, with $u = x/y$, is $x^\chi \sum_{u\neq 1,x} r(x/u)u^{-\chi}h(u) = \mathcal{C}_\chi h(x) + h(x)$, because $h(1) = 0$ and $r(1) = -1$; the extra $h(x)$ cancels the term $-\beta(x)f(x^{-1})$. The c part is $cf(x)\sum_e a_e x^{-e}\sum_{y\neq 1,x} r(y)y^{\chi+e}$, and $\sum_{y\neq 1,x} r(y)y^{\chi+e} = \lambda_{-\chi-e} + 1 - r(x)x^{\chi+e}$.

The term with $r(x)$ contributes $x^\chi f(1) = 0$, while the term $+1$ gives $cf(x)f(x^{-1})$, which cancels the last term $-cf(x)f(x^{-1})$ of (6.4). This proves (6.6) for $x \neq 1$. At $x = 1$ we have $f(1) = 0$, $r(1) = -1$ and, since $r(y^{-1}) = \sum_e a_e y^{-e}/f(y)$, $\mathcal{C}_\chi h(1) = \sum_{y\neq 1} r(y^{-1}) y^{-\chi} h(y) = \sum_e a_e k_e$; so the left side is $\sum_{e>0} W_e - \alpha$. □

Taking Fourier coefficients in (6.6) and putting $z_0 = -k_0$, $z_e = W_e - a_e k_e$ $(e > 0)$ and $H_k = \hat{h}_k$, we obtain, with indices modulo n and a_k the coefficient of u^k in f reduced modulo n,

$$H_k = \frac{\sum_{e\in F} z_e \lambda_{k+e} - n\vartheta a_k - nc\,\mathcal{B}_k}{\lambda_{k-\chi}} - \alpha, \qquad \mathcal{B}_k = \sum_{u,v\in F,\ u-v=k} a_u a_v \lambda_{-\chi-v}. \tag{6.7}$$

Proposition 6.4. *Let the coefficients lie in Ω_n. Given $(z_0, z_1, z_3, z_7, \alpha, \vartheta, c, \eta)$, define H by (6.7), h by inverse Fourier transformation, k_e as above and $W_e = z_e + a_e k_e$ for $e \in F$. Then $h(1) = \sum_{e\in F} W_e - \alpha$, and the solutions of all assigned equations of the character χ correspond bijectively to the data satisfying*

$$h(1) = 0, \qquad W_0 = z_0 + k_0 = 0, \tag{6.8}$$

the original unknowns being recovered by $V_e = W_e/a_e$, $\beta(x) = h(x)/f(x^{-1})$ for $x \neq 1$, $\beta(1) = \Theta - \vartheta$, with c, η and the global unknowns unchanged.

Proof. By construction the Fourier transform of (6.6), with $\sum_{e>0} W_e x^{-e}$ replaced by $\sum_{e\in F} W_e x^{-e}$, holds; evaluating its inverse transform at $x = 1$ as in Lemma 6.3 gives $h(1) = \sum_F W_e - \alpha$. Under (6.8) we get $W_0 = 0$ and $\alpha = W_1 + W_3 + W_7$, so (6.6) holds on G and the reconstructed unknowns solve the assigned equations by Lemma 6.3. Conversely a solution determines all eight quantities, and (6.7) is forced because every $\lambda_{k-\chi}$ is nonzero. □

The Fourier transform of the identity $r(x)f(x^{-1}) = f(x)$, valid on all of G, is

$$\sum_{e\in F} a_e \lambda_{k+e} = n\, a_k \qquad (k \in \mathbb{Z}/n). \tag{6.9}$$

Writing $z_e = z'_e + \vartheta a_e$, (6.9) removes ϑ from (6.7):

$$H_k = \tilde{H}_k - \alpha, \qquad \tilde{H}_k = \frac{\sum_{e\in F} z'_e \lambda_{k+e} - nc\,\mathcal{B}_k}{\lambda_{k-\chi}}. \tag{6.10}$$

The inverse transform of the constant α is α times the indicator of $x = 1$. Hence $h(1) = 0$ has the unique solution $\alpha = \tilde{h}(1)$; the k_e do not depend on α or ϑ; and $W_0 = 0$ has the unique solution $\vartheta = -z'_0 - k_0$.

Corollary 6.5. *For coefficients in Ω_n the assigned subsystem of every character has full row rank, and its solution space is parametrized isomorphically by the six unknowns $(z'_0, z'_1, z'_3, z'_7, c, \eta)$, together with $\rho' = \rho + n\alpha$ if $\chi = 0$ and κ_e if $\chi = -e$.*

6.4. **The origin equations.** For $j \in F$ put

$$\mathcal{A}_j = \sum_{e\in F} z'_e \lambda_{\chi+j+e} - nc\,\mathcal{B}_{\chi+j} - \eta\,[j = 0]\,\lambda_0, \tag{6.11}$$

and for the character χ let $S_\chi(x) = \sum_{u,v\in F,\, u+v+\chi\equiv 0} a_u a_v x^{-v}$. The ten sums $u + v$ with $u, v \in F$ are $0, 1, 2, 3, 4, 6, 7, 8, 10, 14$. We call χ *quadratic* if $\chi \in \{-2, -4, -6, -8, -10, -14\}$, global if $\chi \in \{0, -1, -3, -7\}$, and *ordinary* otherwise.

Proposition 6.6. *Let $n \geq 29$ and let the coefficients lie in Ω_n. On the solution space of the assigned equations, the origin equations of the character χ are equivalent to the following systems:*

(i) *ordinary χ: $\mathcal{A}_j = 0$ for $j \in F$;*
(ii) *quadratic χ: $\mathcal{A}_j = 0$ for $j \in F$, and $c = 0$;*
(iii) *$\chi = -e$, $e \in \{1, 3, 7\}$: $\mathcal{A}_j - (\kappa_e - nca_e)\lambda_j = 0$ for $j \in F$;*
(iv) *$\chi = 0$: $\mathcal{A}_j - \frac{1}{2}(\rho' - nc)\lambda_j = 0$ for $j \in F$.*

Proof. We evaluate the sums in (6.5). In the character $-e$, $\sum_{y \neq 1, x} \kappa_e y^{-e} = -\kappa_e(1 + x^{-e})$, and $\sum_{y \neq 1, x} \alpha y^\chi = \alpha(n[\chi = 0] - 1 - x^\chi)$. Using $f(y)\beta(y^{-1}) = h(y^{-1})$ and $f(y/x)\beta(x/y) = h(x/y)$, and the vanishing of $f(1)$ and $h(1)$, the two β sums are $f(x^{-1})^{-1}\sum_e a_e H_{\chi+e} x^{-e}$ and $x^\chi f(x)^{-1}\sum_e a_e H_{\chi+e} x^e$; the omitted terms contain $f(1)$ or $h(1)$. The c sum is $c\sum_{y \in G} f(y)f(y/x)y^\chi = nc\, S_\chi(x)$. Put

$$q_0 = H_\chi + \alpha - \eta, \qquad q_e = H_{\chi+e} + \alpha \ (e > 0), \qquad q(x) = \sum_{e \in F} a_e q_e x^e.$$

Multiplying (6.5) by $f(x)f(x^{-1})$, the terms with α combine, and we obtain on all of G (at $x = 1$ both sides vanish)

$$f(x)q(x^{-1}) + x^\chi f(x^{-1})q(x) = f(x)f(x^{-1})\big[n\alpha[\chi = 0] + \rho + \kappa_e(1 + x^{-e}) - nc\, S_\chi(x)\big], \tag{6.12}$$

with ρ and κ_e present only in the global characters 0 and $-e$.

If χ has a representative with $|\chi| \leq 14$, both sides of (6.12) are Laurent polynomials whose supports lie in an interval of length at most $28 < n$, so (6.12) is an identity of Laurent polynomials. If $15 \leq \chi \leq n - 15$ the right side is zero (no global or quadratic character lies in this range) and the supports $[-7, 7]$ and $\chi + [-7, 7]$ of the two terms on the left are disjoint modulo n; hence $f(x)q(x^{-1}) = 0$ and $q = 0$.

In the first case suppose the left side equals $f(x)f(x^{-1})\mathcal{T}(x)$. Then $f(x)$ divides $x^\chi f(x^{-1})q(x)$ up to a monomial, and since $\gcd(f, u^7 f(u^{-1})) = u - 1$, the polynomial $g = f/(1 - u)$ of degree six divides q. Write $q = (\mu' - \mu u)g$. From $f = (1 - u)g$ and the vanishing of the coefficients of f in degrees $4, 5, 6$, the coefficients of g in degrees $3, 4, 5, 6$ are all equal to $-a_7$; as q has no terms of degrees $4, 5, 6$, we get $(\mu' - \mu)a_7 = 0$, so $q = \mu(1 - u)g = \mu f$. Then $\mathcal{T} = \mu(1 + x^\chi)$. For ordinary χ, $S_\chi = 0$ and $\mathcal{T} = 0$, so $q = 0$. For quadratic χ, S_χ is $a_e^2 x^{-e}$ or $a_e a_{e'}(x^{-e} + x^{-e'})$ with $e, e' > 0$, which is not a multiple of $1 + x^\chi$; hence $c = 0$ and $q = 0$. For $\chi = -e$ we have $S_\chi = a_e(1 + x^\chi)$, so $q_j = \kappa_e - nca_e$ for all j. For $\chi = 0$, $S_0 = 1$ and $q_j = \frac{1}{2}(n\alpha + \rho - nc)$. Finally $q_j = \tilde{H}_{\chi+j} - \eta[j = 0]$ by (6.10), and $\lambda_j q_j = \mathcal{A}_j$ by (6.10) and (6.11); dividing by $\lambda_j \neq 0$ gives the four systems. □

6.5. **The rank of the origin systems.** Let $\mathcal{M}_\chi = (\lambda_{\chi+j+e})_{j \in \{1,3,7\},\, e \in F}$, a 3×4 matrix. The column of η in the systems of Proposition 6.6 is $-\lambda_0 \neq 0$ in the row $j = 0$ and zero in the other rows. Hence, if $\operatorname{rank} \mathcal{M}_\chi = 3$, the origin system has rank four in the ordinary and global characters and rank five in the quadratic ones (the extra equation $c = 0$ involves a column not used by the rank-four minor).

Theorem 6.7. *Let $n \geq 29$. For coefficients in a nonempty open subset of Ω_n we have $\operatorname{rank} \mathcal{M}_\chi = 3$ for every $\chi \in \mathbb{Z}/n$. Consequently the special kernel $\ker \mathcal{R}$ has dimension 2 in each of the $n - 10$ ordinary characters, 1 in each of the six quadratic*

characters and 3 *in each of the four global characters; in total*

$$\dim\ker\mathcal{R} = 2(n-10) + 6 + 3\cdot 4 = 2n-2.$$

The rest of this subsection proves Theorem 6.7. For each χ it suffices to show that some 3×3 minor of $\mathcal{M}_\chi$ is a nonzero rational function on Σ, since there are finitely many characters. We restrict to the curve

$$a = -\tfrac{1}{6}, \qquad g_b(u) = 1 + au + au^2 + b(u + u^2 + \cdots + u^6),$$
$$f_b(u) = (1-u)g_b(u) = 1 + (a+b-1)u - au^3 - bu^7,$$

and study minors near $b = 0$. The point $b = 0$ is not in Ω_n, since $a_7 = 0$ there; we only use that a minor with a nonzero coefficient in its expansion at $b = 0$ is not identically zero. Put $q_b(u) = u^6 g_b(u^{-1}) = u^4 h(u) + b\,q_1(u)$ with $h(u) = u^2 + au + a$ and $q_1 = 1 + u + \cdots + u^5$, so that $r_b(u) = -u^7 g_b(u)/q_b(u)$. At $b = 0$ the poles of r_0 are $v_1 = 1/2$ and $v_2 = -1/3$, with residues $R_1 = -21/160$ and $R_2 = -56/1215$. For small $b \neq 0$ there are two poles near v_1, v_2 and four small poles near 0, all simple for b in a punctured neighbourhood and all inside the unit disc; sums over the four small poles are analytic in b.

Lemma 6.8. *Let P_b be the polynomial part of r_b at infinity (of degree at most seven) and $L_j = \lambda_j/n$. For $0 \le j < n$,*

$$L_j = P_{b,j} + \sum_{poles\ v} \frac{R_v\, v^{n-j-1}}{1-v^n},$$

where R_v is the residue at v. For $j \ge 8$ the polynomial term vanishes, and for $b \neq 0$ and $0 \le j \le 6$ the right side equals $\sum_v R_v v^{-j-1}/(1-v^n)$, which is the same pole formula with j replaced by $n+j$. These identities extend to $b = 0$ by continuity.

Proof. On $|u| = 1$ expand $R/(u-v) = \sum_{k\ge 1} R v^{k-1} u^{-k}$ and average over G; this includes $u = 1$, where $r_b(1) = -1$ is the value of the rational function. For $b \neq 0$, r_b vanishes to order seven at 0, so its Taylor coefficients of degree $j \le 6$ vanish, which gives $P_{b,j} = \sum_v R_v v^{-j-1}$. □

Lemma 6.9. *For an integer $k \ge -2$ let $\mathcal{H}_k(b) = \sum_{v\ small} R_v v^{k-1}$. Then*

$$\mathcal{H}_k(b) = \gamma_k b^\ell + O(b^{\ell+1}), \qquad \ell = \left\lceil \frac{k+3}{4} \right\rceil, \qquad \gamma_k = (-1)^{\ell+1} a^{-\ell-1} c_\varrho(a,\ell),$$

where $\varrho = 4\ell - k - 3 \in \{0,1,2,3\}$ and

$$c_0 = 1, \quad c_1 = a-1, \quad c_2 = \frac{(\ell+1)(a-1)}{a}, \quad c_3 = \frac{\ell a^2 - 2\ell a + 2\ell - 2a + 2}{a}.$$

All four are nonzero at $a = -1/6$ for every $\ell \ge 1$. Moreover the cyclic small-pole sum $\sum_{v\ small} R_v v^{k-1}/(1-v^n) = \sum_{m\ge 0} \mathcal{H}_{k+mn}$ has the same leading term for $n \ge 29$.

Proof. The sum is the integral of $-u^{k+6} g_b(u)/q_b(u)$ over a small circle enclosing the four small poles and no other pole; $u = 0$ is not a pole since $k + 6 \ge 4$. On this circle $1/q_b = \sum_{t\ge 0} (-b)^t q_1^t/(u^{4t+4} h^{t+1})$, and the residue of the t-th term is $-(-b)^t [u^{4t-k-3}](g_b q_1^t/h^{t+1})$, which vanishes for $t < \ell$. At $t = \ell$ only g_0 contributes, since the b-part of g_b raises the order in b, and the coefficient of u^ϱ in $g_0 q_1^\ell h^{-\ell-1}$ is $a^{-\ell-1} c_\varrho$ by direct expansion through u^3. At $a = -1/6$, $c_1 = -7/6$, $c_2 = 7(\ell+1)$ and the numerator of c_3 is $85\ell/36 + 7/3 > 0$. The terms with $m \ge 1$ have larger order in b. □

Proof of Theorem 6.7. Write $k = n - \chi$ and take rows $j \in \{1, 3, 7\}$.

The range $7 \le \chi \le n - 12$. Take the columns $e \in \{0, 1, 7\}$. If $\chi \le n - 15$, every index $\chi + j + e$ lies in $[8, n - 1]$; if $n - 14 \le \chi \le n - 12$ only the entry $(7, 7)$ wraps, to $0, 1, 2$, and Lemma 6.8 gives the same pole formula. The contribution of the two old poles is an analytic matrix of rank at most two, equal at $b = 0$ to $D_1 v_1^{-j-e} + D_2 v_2^{-j-e}$ with $D_i = R_i v_i^{k-1}/(1 - v_i^n) \neq 0$. The cofactor of the entry $(7, 7)$ in this matrix is

$$\det \begin{pmatrix} v_1^{-1} & v_2^{-1} \\ v_1^{-3} & v_2^{-3} \end{pmatrix} D_1 D_2 \det \begin{pmatrix} 1 & v_1^{-1} \\ 1 & v_2^{-1} \end{pmatrix} = (-30)(-5) D_1 D_2 = 150 D_1 D_2.$$

By Lemma 6.9 the small-pole correction at $(7, 7)$, where the shift $j + e = 14$ is maximal, has order $\ell = \lceil (k - 11)/4 \rceil$, while all other corrections (shifts at most 10) have order at least $\ell + 1$. Since the old-pole matrix has determinant identically zero, the determinant of the minor is

$$150 D_1 D_2 \gamma_{k-14} b^\ell + O(b^{\ell+1}) \neq 0.$$

For $n - 14 \le \chi \le n - 12$ we have $k - 14 \in \{0, -1, -2\}$ and $\ell = 1$.

The range $0 \le \chi \le 6$. At $b = 0$ the polynomial part is $P_0 = u^3/6 + 7u^2/36 - 203u/216 - 161/1296$, and

$$\partial_b P_b|_{b=0} = -u^7 - \tfrac{7}{6}u^6 - \tfrac{49}{36}u^5 - \tfrac{307}{216}u^4 - \tfrac{1897}{1296}u^3 - \tfrac{12811}{7776}u^2 - \tfrac{41041}{46656}u + \tfrac{44093}{279936}.$$

For the columns listed in Table 1 the polynomial part has a single nonzero entry in the minor when $\chi \le 2$, and the determinant at $b = 0$ is $\Delta_\chi D_1 D_2$. For $3 \le \chi \le 6$ the polynomial part of P_0 does not enter, the old-pole part has rank two, and all small-pole corrections have order at least three since $n - (\chi + 14) \ge 9$; the coefficient of b in the determinant comes from $\partial_b P_b$ alone and equals $\Delta_\chi D_1 D_2$. The values Δ_χ, obtained from 3×3 determinants of these explicit entries and the old-pole cofactors, are nonzero and independent of n.

TABLE 1. Minors for $0 \le \chi \le 6$ (rows $j = 1, 3, 7$).

χ	columns e	order in b	Δ_χ
0	1,3,7	0	38329200
1	0,3,7	0	38329200
2	0,3,7	0	32853600
3	0,1,7	1	−26044200
4	0,1,3	1	119700
5	0,1,7	1	−32853600
6	0,1,3	1	−421200

The range $\chi = n - h$, $1 \le h \le 11$. Put $x = 2^{-n}$ and $y = (-3)^{-n}$, and let $t_j = [u^j] r_0(u)$, with $t_j = 0$ for $j < 0$. At $b = 0$ every entry of the minors below is

$$L_{n-h+j+e} = \sum_{i=1,2} \frac{R_i v_i^{h-j-e-1}}{1 - v_i^n} + t_{j+e-h},$$

by Lemma 6.8 and the identity $P_{0,j} - \sum_i R_i v_i^{-j-1} = t_j$ at wrapped indices. For the columns of Table 2, the determinant is the rational function $N_h/(K_h(x - 1)(y - 1))$ listed there. For $n \ge 29$ we have $0 < x < 10^{-3}$, $|y| < 10^{-3}$ and $|y/x| =$

$(2/3)^n < 10^{-3}$, and each N_h is nonzero: for $h = 1$ the perturbation of $-112x$ is less than $(2268+15120)x/1000 < 112x$; for $h = 2, 3$ the perturbations are less than $250047x/1000 < 21952x$ and $35721x/1000 < 3136x$; for $h = 5$ the perturbation of 1015 is less than $(28672 + 413343)/1000 < 1015$; the remaining numerators are constants. This covers both parities of n.

TABLE 2. Minors for $\chi = n - h$, $1 \le h \le 11$ (rows $j = 1, 3, 7$), with $x = 2^{-n}$, $y = (-3)^{-n}$.

h	columns e	N_h	K_h
1	0,1,3	$15120xy - 112x - 2268y$	5
2	0,1,3	$21952x - 250047y$	1620
3	0,1,3	$3136x - 35721y$	1620
4	0,1,3	-49	11664
5	0,1,7	$1015 - 28672x - 413343y$	233280
6	0,1,3	343	419904
7	0,1,3	49	419904
8	0,1,7	343	1259712
9	0,1,7	343	15116544
10	0,1,7	343	544195584
11	0,1,7	49	544195584

The three ranges cover $\mathbb{Z}/n$. For fixed n the finitely many nonzero minors define nonempty open subsets of Σ, which meet each other and Ω_n. The kernel dimensions follow from the ranks four and five of the origin systems and from Corollary 6.5: $6 - 4 = 2$, $6 - 5 = 1$ and $7 - 4 = 3$. □

7. The first opening and the lower boundary

7.1. **A perturbation lemma.**

Lemma 7.1. *Let $M(\varepsilon) = M_0 + \varepsilon M_1 + O(\varepsilon^2)$ be a matrix of analytic functions near $\varepsilon = 0$. Let the columns of N form a basis of $\ker M_0$ and let Q' be any matrix whose rows lie in the left kernel of M_0. Then $\operatorname{rank} M(\varepsilon) \ge \operatorname{rank} M_0 + \operatorname{rank}(Q' M_1 N)$ for all sufficiently small $\varepsilon \ne 0$.*

Proof. Let the rows of Q form a basis of the left kernel; then $Q' = TQ$ and $\operatorname{rank} Q'M_1N \le \operatorname{rank} QM_1N$. Choose invertible U, V such that the last rows of U are Q, the last columns of V are N, and

$$UM_0V = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}, \qquad UM(\varepsilon)V = \begin{pmatrix} I + \varepsilon X & \varepsilon Y \\ \varepsilon Z & \varepsilon QM_1N + O(\varepsilon^2) \end{pmatrix}.$$

The Schur complement of the upper left block is $\varepsilon(QM_1N + O(\varepsilon))$. □

We apply Lemma 7.1 to $M(\varepsilon) = \Gamma^{\mathrm{f}}(y(\varepsilon))$, character by character. By (6.3), N is the special kernel, lifted by (6.2). A left kernel vector ϖ of the residual system $\mathcal{R}$ lifts to one of M_0 by adding to it, on each pivot row (k, ab), the multiple $-\sum_j \varpi_{j,ab} V_{abk}(b_j)/V_{abk}(p_k)$, the sum running over the auxiliary indices $j \in \mathbb{Z}/n \cup$

$\{o\}$. Hence, if $S^1 = M_1 N$ denotes the first-order rows on special kernel vectors, the relevant first-order map is $\varpi \circ \mathcal{R}^1$ with the *corrected* rows

$$\mathcal{R}^1_{j,ab} = S^1_{j,ab} - \sum_{k \notin \{a,b\}} \frac{V_{abk}(b_j)}{V_{abk}(p_k)} S^1_{k,ab}. \tag{7.1}$$

For an assigned row this is divided by $E_{ab}(p_a)$; the derivative of this normalization does not matter on $\ker \mathcal{R}$. For the origin row $V_{abk}(o) = 1$, so $\mathcal{R}^1_o = S^1_o - \sum_k S^1_k / V_{abk}(p_k)$.

7.2. **The support of the origin rows.** Put $\mathbb{D} = F - F$ (modulo n) and

$$U_\chi = \mathbb{D} \cup (\chi + \mathbb{D}) \cup \bigcup_{u,v \in F,\ u+v+\chi \equiv 0} (\mathbb{D} - v). \tag{7.2}$$

We regard an origin row as a function of the separation $x \in G \smallsetminus \{1\}$, the values at x and x^{-1} being related by the reversal of the edge, and we extend it by zero at $x = 1$.

Lemma 7.2. *For every unknown vector of the character χ (not necessarily in the special kernel), the function $f(x)f(x^{-1})\mathcal{R}_o(x)$ has Fourier support in U_χ. Consequently, for every $k \notin U_\chi$, the functional $\mathcal{R}_o \mapsto (\widehat{f_+ f_- \mathcal{R}_o})_k$, where $f_\pm = f(x^{\pm 1})$, lies in the left kernel of $\mathcal{R}$.*

Proof. The computation in the proof of Proposition 6.6 did not use the assigned equations: the terms omitted from the β sums contain $f(1) = 0$, and $h(1) = f(1)\beta(1) = 0$ holds for every unknown vector. It shows that $f_+ f_- \mathcal{R}_o$ is a sum of $f(x)q(x^{-1}) + x^\chi f(x^{-1})q(x)$ with q supported on F, of $\eta\,(f(x) + x^\chi f(x^{-1}))$, and of $f_+ f_-$ times $\alpha(1 + x^\chi)$, constants, $\kappa_e(1 + x^{-e})$ (with $x^{-e} = x^\chi$) and $S_\chi(x)$. These have supports in $\mathbb{D}$, $\chi + \mathbb{D}$ and $\bigcup(\mathbb{D} - v)$. □

7.3. **The opening on the special kernel.** On the special kernel let $\hat\beta_j = \sum_t \beta(t) t^{-j}$ and $\mathsf{Q}_{uv} = \hat\beta_{\chi+u+v}$ for $u, v \in F$. Then $(\mathsf{Q}a)_u = \sum_t h(t) t^{-\chi-u} = H_{\chi+u}$, so the classification in Proposition 6.6 reads

$$\mathsf{Q}a = \mathsf{C}\mathbf{1} + \eta e_0, \qquad \mathsf{C} = \mathsf{L} - \alpha, \tag{7.3}$$

where $\mathsf{L} = 0$ in the ordinary and quadratic characters, $\mathsf{L} = \kappa_e - nca_e$ in the character $-e$ and $\mathsf{L} = \frac{1}{2}(n\alpha + \rho - nc)$ in the character 0. Put $\tilde{\mathsf{Q}} = \mathsf{Q} - \mathsf{C}(\mathbf{1}e_0^{\mathsf{t}} + e_0\mathbf{1}^{\mathsf{t}}) - \eta e_0 e_0^{\mathsf{t}}$. Then $\tilde{\mathsf{Q}}a = 0$, its block $\mathsf{S} = (\mathsf{Q}_{ef})_{e,f \in \{1,3,7\}}$ coincides with that of Q, and

$$\sum_{u,v \in F} a_u a_v \tilde{\mathsf{Q}}_{uv} t^u z^v = \sum_{e,f>0} a_e a_f \mathsf{S}_{ef} (t^e - 1)(z^f - 1). \tag{7.4}$$

Define regular functions on G by $\varphi_e(t) = a_e(t^e - 1)/f(t)$ for $t \neq 1$ and $\varphi_e(1) = a_e e / f'(1)$, $e \in \{1, 3, 7\}$; their sum is identically one. Put

$$T_{\mathsf{S}}(x) = \sum_{t \in G} t^\chi \varphi(t)^{\mathsf{t}} \mathsf{S} \varphi(t/x). \tag{7.5}$$

Proposition 7.3. *Let $n \geq 29$ and let the coefficients lie in Ω_n. On every special kernel vector, the Fourier coefficients outside U_χ of $f_+ f_- \mathcal{R}^1_o$ coincide with those of $f_+ f_- T_{\mathsf{S}}$.*

Proof. We take the edge $\{0,b\}$ with separation $x=t_b$ and write k for the internal labels $t_k\neq 1,x$. Call a function on G U_χ-*supported* if after multiplication by f_+f_- its Fourier support lies in U_χ; we must show that $\mathcal{R}^1_o - T_{\mathsf{S}}$ is U_χ-supported. The radial opening has velocity $-p_k$ at the k-th point and $\nu_k\cdot(-p_k)=1$. Let $u_k=t_{0bk,k}$ be the pivot unknown of this edge and, by (6.2),

$$D_k=\nu_k\cdot v_k-u_k=k^\chi f(k)f(k/x)\Big[\frac{\beta(1/k)}{f_-}+\frac{\beta(x/k)}{f_+}+c\Big]. \tag{7.6}$$

Put $\mathsf{H}=\sum_{t\neq1}1/f(t)$ and $V_k=V_{0bk}(p_k)$. Since $\frac{d}{d\varepsilon}\ell_j((1-\varepsilon)p_k)=1-C_{kj}$, differentiation of the products gives

$$\frac{S^1_k}{V_k}=D_k\Big[\mathsf{H}-\frac{1}{f(k)}-\frac{1}{f(k/x)}-(n-3)\Big]+\sum_{l\neq k}\frac{\nu_l\cdot v_k-u_l}{f(k/l)}-\sum_j\beta_{kj}\frac{E^1_j}{V_k}, \tag{7.7}$$

where $E^1_j=\frac{d}{d\varepsilon}E_{0b}((1-\varepsilon)p_j)$, and $S^1_o=-\sum_j\beta_{oj}E^1_j$ since the origin does not move. At the endpoints $E^1_0/E_0=\mathsf{H}-1/f_--(n-2)$ and $E^1_x/E_x=\mathsf{H}-1/f_+-(n-2)$.

Interior terms. The contribution of the interior j to the β sums in $-\sum_k S^1_k/V_k$ and in S^1_o is

$$T^{\mathrm{ex}}_{\mathsf{Q}}(x)=\sum_{t\neq1,x}t^\chi\frac{\sum_{u,v\in F}a_ua_v\mathsf{Q}_{uv}t^{u+v}x^{-v}-\eta}{f(t)f(t/x)}. \tag{7.8}$$

Assigned cancellations. Write $I=G\smallsetminus\{1,x\}$ for the set of internal labels and decompose $\nu_l\cdot v_k-u_l=D_l+\nu_l\cdot(v_k-v_l)$. In $\sum_{k\in I}S^1_k/V_k$ the second sum of (7.7) then splits into $\sum_{l\in I}D_l\sum_{k\in I,k\neq l}1/f(k/l)$ and the double velocity sum $\sum_{k\in I}\sum_{l\in I,l\neq k}\nu_l\cdot(v_k-v_l)/f(k/l)$. Hence, before the endpoint terms, the coefficient of D_k in $\sum_k S^1_k/V_k$ is $2\mathsf{H}-(n-3)-1/f(k)-1/f(k/x)-1/f(1/k)-1/f(x/k)$. The two special assigned equations at the endpoints 0 and x give exactly

$$\sum_{k\in I}\frac{D_k}{f(1/k)}=\beta(1)+\beta(x)\frac{f_-}{f_+}+cf_--\mathcal{V}_0,\qquad \mathcal{V}_0=\sum_{k\in I}\frac{\nu_k\cdot(v_0-v_k)}{f(1/k)},$$

$$\sum_{k\in I}\frac{D_k}{f(x/k)}=x^\chi\beta(1)+x^\chi\beta(1/x)\frac{f_+}{f_-}+cx^\chi f_+-\mathcal{V}_x,\quad \mathcal{V}_x=\sum_{k\in I}\frac{\nu_k\cdot(v_x-v_k)}{f(x/k)}.$$

Substituting these for the terms $-\sum_k D_k/f(1/k)$ and $-\sum_k D_k/f(x/k)$ adds $\mathcal{V}_0+\mathcal{V}_x$ to $\sum_k S^1_k/V_k$, and these are exactly the terms $k=1$ and $k=x$ missing from the double velocity sum. The velocity part of $\sum_k S^1_k/V_k$ is therefore

$$\sum_{l\in I}\sum_{k\in G,\,k\neq l}\frac{\nu_l\cdot(v_k-v_l)}{f(k/l)}.$$

Substituting $k=ly$ and using $\nu_l\cdot(v_{ly}-v_l)=l^\chi\big(y^\chi\sum_{e>0}W_ey^e-\alpha\big)$, the inner sum is $l^\chi J_\chi$ with $J_\chi=\sum_{y\neq1}(y^\chi\sum_{e>0}W_ey^e-\alpha)/f(y)$, and $\sum_{l\in I}l^\chi=n[\chi=0]-1-x^\chi$. Thus the velocity part of $\sum_k S^1_k/V_k$ equals $J_\chi(n[\chi=0]-1-x^\chi)$, in every character, including the global ones; it enters $\mathcal{R}^1_o=S^1_o-\sum_k S^1_k/V_k$ with a minus sign. Since f_+f_- is supported in $\mathbb{D}$, this term is U_χ-supported, whatever its sign.

Origin cancellation. The special origin equation gives

$$\sum_k D_k=\alpha(n[\chi=0]-1-x^\chi)+\nu_0\cdot v_o+\nu_x\cdot v_o+\eta\Big(\frac{1}{f_-}+\frac{x^\chi}{f_+}\Big)+\rho,$$

which is U_χ-supported; also $\sum_k k^\chi f(k)f(k/x) = nS_\chi(x)$. This accounts for all constant multiples of $\sum_k D_k$ and for the c part of the endpoint β sums. After these steps the terms that are not yet known to be U_χ-supported, besides (7.8), are

$$\sum_k D_k\Big[\frac{1}{f(k)}+\frac{1}{f(k/x)}\Big] - \sum_k k^\chi f(k)f(k/x)\Big[\frac{\beta(1/k)}{f_-^2}+\frac{\beta(x/k)}{f_+^2}\Big] + \beta(1)(1+x^\chi) + \beta(x)\frac{f_-}{f_+} + x^\chi\beta(1/x)\frac{f_+}{f_-} + c(f_- + x^\chi f_+) + \eta\Big(\frac{1}{f_-^2}+\frac{x^\chi}{f_+^2}\Big). \tag{7.9}$$

In the first sum, cancelling $f(k)$ or $f(k/x)$ against (7.6) allows the sum to be extended over all k; the omitted endpoint terms are $\beta(1)+\beta(x)f_-/f_+ + cf_-$ and $x^\chi\beta(1)+x^\chi\beta(1/x)f_+/f_- + cx^\chi f_+$, which enter with a minus sign and cancel the three β terms and the term $c(f_- + x^\chi f_+)$ of (7.9); this c term is cancelled, not discarded. The completed c sums equal $c\sum_k k^\chi(f(k/x)+f(k)) = nc\sum_{e\in F,\, e+\chi\equiv 0} a_e(x^{-e}+1)$, which vanishes outside the global characters and is U_χ-supported in them. The completed β sums are U_χ-supported: the substitution $y = x/k$ turns them into x^χ times a fixed moment or a polynomial supported on F, and the $\beta(1/k)$ sums are supported on $-F$; the exterior factors $f_\pm$ then give $\mathbb{D}$ or $\chi+\mathbb{D}$. For the second sum, (7.3) gives

$$\sum_k k^\chi\beta(1/k)f(k)f(k/x) = \mathsf{C}f_- + \eta, \qquad \sum_k k^\chi\beta(x/k)f(k)f(k/x) = x^\chi(\mathsf{C}f_+ + \eta),$$

where the omitted endpoints have a vanishing factor $f(1)$. The η terms cancel the last term of (7.9), and the C terms are U_χ-supported. Hence $\mathcal{R}^1_o - T^{\mathrm{ex}}_{\mathsf{Q}}$ is U_χ-supported.

The affine part of Q. By definition of $\tilde{\mathsf{Q}}$, the numerator in (7.8) equals its $\tilde{\mathsf{Q}}$ version plus $\mathsf{C}(f(t)+f(t/x))$, which contributes $\mathsf{C}[(1+x^\chi)K_\chi - x^\chi/f_+ - 1/f_-]$ with $K_\chi = \sum_{t\neq 1} t^\chi/f(t)$, a U_χ-supported function. By (7.4) the remaining summand is $t^\chi\varphi(t)^{\mathsf{t}}\mathsf{S}\varphi(t/x)$, which is regular also at $t=1$ and $t=x$; adding these two terms changes the sum by $\varphi(1)^{\mathsf{t}}\mathsf{S}\varphi(x^{-1}) + x^\chi\varphi(x)^{\mathsf{t}}\mathsf{S}\varphi(1)$, which after multiplication by f_+f_- is supported in $\mathbb{D}\cup(\chi+\mathbb{D})$. Thus $\mathcal{R}^1_o - T_{\mathsf{S}}$ is U_χ-supported.

Finally, every U_χ-supported remainder obtained above contains a factor f_+ or f_- that is not divided out, so it vanishes at $x=1$, as does $f_+f_-T_{\mathsf{S}}$; hence the Fourier transformation of functions extended by zero at $x=1$ introduces no extra term. □

7.4. **Three moments and the detector.** Since $\sum_e \varphi_e = 1$, a matrix S of the form $\mathbf{1}w^{\mathsf{t}} + w\mathbf{1}^{\mathsf{t}}$ contributes to $\widehat{T_{\mathsf{S}}}$ only at the indices 0 and χ, hence only to U_χ after multiplication by f_+f_-. Eliminating $\varphi_7 = 1-\varphi_1-\varphi_3$, only the symmetric 2×2 matrix m with entries

$$\begin{aligned} \mathsf{m}_{11} &= \hat\beta_{\chi+2} - 2\hat\beta_{\chi+8} + \hat\beta_{\chi+14},\\ \mathsf{m}_{13} &= \hat\beta_{\chi+4} - \hat\beta_{\chi+8} - \hat\beta_{\chi+10} + \hat\beta_{\chi+14},\\ \mathsf{m}_{33} &= \hat\beta_{\chi+6} - 2\hat\beta_{\chi+10} + \hat\beta_{\chi+14} \end{aligned} \tag{7.10}$$

is visible outside U_χ. Let $\hat\varphi(k) = (\hat\varphi_1(k), \hat\varphi_3(k))$ with $\hat\varphi_e(k) = \sum_t \varphi_e(t)t^{-k}$. Then $\widehat{T_{\mathsf{S}k}} = \hat\varphi(k-\chi)^{\mathsf{t}}\mathsf{m}\hat\varphi(-k)$ modulo the two invisible indices, and the outside-support

coefficients of $f_+ f_- T_{\mathsf{S}}$ are given by the *detector*

$$(\mathcal{O}_\chi \mathsf{m})_k = \sum_{u,v\in F} a_u a_v\, \hat\varphi(k-u+v-\chi)^{\mathsf{t}}\, \mathsf{m}\, \hat\varphi(-k+u-v), \qquad k \notin U_\chi. \tag{7.11}$$

Let $\mathcal{W}_\chi$ denote the linear map from the six- or seven-dimensional unknown space of Corollary 6.5 to $(\mathsf{m}_{11}, \mathsf{m}_{13}, \mathsf{m}_{33})$, and $\mathcal{P}_\chi$ the origin system of Proposition 6.6. Combining Lemma 7.1, Lemma 7.2 and Proposition 7.3, the rank of the first opening on the special kernel of χ is at least, writing $[\mathcal{P};\mathcal{X}]$ for the matrix obtained by placing the rows of $\mathcal{X}$ below those of $\mathcal{P}$,

$$\operatorname{rank}\big(\mathcal{O}_\chi \mathcal{W}_\chi|_{\ker \mathcal{P}_\chi}\big) = \operatorname{rank}\begin{pmatrix}\mathcal{P}_\chi \\ \mathcal{O}_\chi\mathcal{W}_\chi\end{pmatrix} - \operatorname{rank}\mathcal{P}_\chi. \tag{7.12}$$

Lemma 7.4. *Let $P_{11} = (u-u^7)^2$, $P_{13} = (u-u^7)(u^3-u^7)$, $P_{33} = (u^3-u^7)^2$ and $\Psi_A(j) = \sum_t t^{-j} P_A(t)/f(t)$, with value 0 at $t=1$. In the coordinates of Corollary 6.5,*

$$\mathsf{m}_A = \frac{1}{n}\sum_{j\in\mathbb{Z}/n} \frac{\Psi_A(j)}{\lambda_j}\Big[\sum_{e\in F} z'_e \lambda_{\chi+j+e} - nc\,\mathcal{B}_{\chi+j}\Big]. \tag{7.13}$$

Proof. For instance $\mathsf{m}_{11} = \sum_t \beta(t) t^{-\chi}(t^{-1}-t^{-7})^2$; the term $t=1$ vanishes, and $\beta(t) = h(t)/f(t^{-1})$ otherwise. Inserting $h(t) = \frac{1}{n}\sum_k H_k t^k$ gives $\mathsf{m}_A = \frac{1}{n}\sum_j H_{\chi+j}\Psi_A(j)$. By (6.10) the constant $-\alpha$ contributes $\sum_j \Psi_A(j) = 0$, and the rest is (7.13). □

The required opening ranks are 2 in the ordinary and global characters and 1 in the quadratic characters; equivalently $\operatorname{rank}[\mathcal{P}_\chi;\mathcal{O}_\chi\mathcal{W}_\chi] = 6$ in every character. In a global character the remaining one-dimensional kernel is spanned by a translation or by the dilation: in the character $-e$ the vector with $z' = u^e$ (that is, $z'_e = 1$) and $\kappa_e = 1$, and in the character 0 the vector with $z' = 1$ and $\rho' = 2$, have zero moments.

7.5. The detector is injective for $n \geq 89$.

Proposition 7.5. *Let $n \geq 89$. On a nonempty open subset of Ω_n, $\mathcal{O}_\chi$ is injective on symmetric 2×2 matrices for every χ.*

Proof. Write $g = f/(1-u)$. We add to Ω_n the open conditions that the six roots $\alpha_1,\dots,\alpha_6$ of g are simple and nonzero, that the thirty ratios α_i/α_j $(i\neq j)$ are distinct, and that none of them is a root of $f(u)f(u^{-1})$. These conditions are nonempty: for $g = 1+3u+3u^2+6u^3+6u^4+6u^5+6u^6$ the polynomial $\operatorname{Res}_z(g(z),g(uz))/(u-1)^6$ has degree 30, is squarefree and is coprime to f and to its reciprocal, as one verifies modulo 101; a nonzero reduction of these resultants certifies the corresponding characteristic zero conditions.

The functions $\varphi_1 = -a_1/g$ and $\varphi_3 = -a_3(1+u+u^2)/g$ are proper rational functions, and for $0 \leq k < n$

$$\hat\varphi(k) = \sum_i U_i \alpha_i^{-k}, \qquad U_i = -\frac{n\alpha_i^{n-1}}{(1-\alpha_i^n)g'(\alpha_i)}\big(a_1,\ a_3(1+\alpha_i+\alpha_i^2)\big)^{\mathsf{t}},$$

by the same averaging as in Lemma 6.8. Choose one of the intervals $8 \leq k \leq \chi-8$ or $\chi+8 \leq k \leq n-8$. On it every shifted index $k-u+v$ in (7.11) lies strictly between 0 and χ, or strictly between χ and n, so (7.11) is a sum of the thirty exponentials $(\alpha_j/\alpha_i)^k$: the diagonal terms are killed by $f(1)^2 = 0$, and the coefficient of $(\alpha_j/\alpha_i)^k$

is a nonzero multiple of $U_i^{\mathsf t}\mathsf m U_j\, f(\alpha_i/\alpha_j)f(\alpha_j/\alpha_i)$, the wrap factor being $\alpha_i^{\chi}\alpha_j^{-n}$, or $\alpha_i^{\chi-n}\alpha_j^{-n}$ on the lower interval. These intervals avoid $\mathbb D\cup(\chi+\mathbb D)$; in a quadratic character $\chi=n-(u+v)$ the extra sets $\mathbb D-v$ avoid $8\le k\le\chi-8$, and in a global character there is no extra set. The longer interval has $\max(\chi,n-\chi)-15\ge 30$ elements for $n\ge 89$, so if $\mathcal O_\chi\mathsf m=0$ the Vandermonde matrix of thirty consecutive indices forces $U_i^{\mathsf t}\mathsf m U_j=0$ for all $i\ne j$. At least three of the projective points $[a_1 : a_3(1+\alpha_i+\alpha_i^2)]$ are distinct, since a quadratic polynomial takes each value at most twice. A symmetric bilinear form on $\mathbb C^2$ vanishing on all pairs of three pairwise independent vectors is zero; hence $\mathsf m=0$. □

7.6. **The fixed matrices.** For each character χ we regard $\mathcal P_\chi$ (the origin system of Proposition 6.6, in the columns $z_0', z_1', z_3', z_7', c, \eta$ and, in a global character, κ_e or ρ') and $\mathcal W_\chi$ (the three rows (7.13)) as matrices whose entries are functions of the coefficient vector $a\in\Sigma$. Their entries are polynomials in n, a_1, a_3, a_7, the λ_j, the λ_j^{-1} and the values $\Psi_A(j)=\sum_{t\ne1}t^{-j}P_A(t)/f(t)$; no entry involves a division by a_1, a_3 or a_7. Hence both matrices are regular on the open set $\Sigma'\subset\Sigma$ where $f(x)\ne0$ for $x\in G\smallsetminus\{1\}$ and $\lambda_j\ne0$ for all j, and every minor is a rational function on Σ, regular on Σ'. The divisions by a_e occur only in the passage from the six or seven fixed coordinates back to the original unknowns ($V_e=W_e/a_e$, $O_e=\kappa_e/a_e$), that is, in Corollary 6.5, which is applied on $\Omega_n=\Sigma'\cap\{a_1a_3a_7\ne0,\ f'(1)\ne0,\ \gcd(f,u^7f(u^{-1}))=u-1\}$. Consequently, if a minor of $[\mathcal P_\chi;\mathcal W_\chi]$ (or of $[\mathcal P_\chi;\mathcal O_\chi\mathcal W_\chi]$) is nonzero at one point of Σ', or has a nonzero coefficient in its expansion along an analytic curve whose general point lies in Σ', then it is a nonzero rational function on Σ, hence nonzero on a dense open subset of Σ, which meets Ω_n; at the points of this intersection the rank statement is a statement about the special kernel. In the next two subsections we use two such curves: the *cubic line* $a=(-\frac76+b,\frac16,-b)$, which lies in Ω_n for small $b\ne0$ and whose point $b=0$ lies in Σ' but not in Ω_n, and the *six-small-pole line* $a=(-1-\epsilon,0,\epsilon)$, which lies entirely in the plane $\{a_3=0\}$ and hence outside Ω_n, but whose general point lies in Σ'.

7.7. **The bulk characters.**

Proposition 7.6. *Let $n\ge29$ and $7\le\chi\le n-16$. On a nonempty open subset of Σ, $\operatorname{rank}[\mathcal P_\chi;\mathcal W_\chi]=6$.*

All characters in this range are ordinary. We use the same curve as before,

$$g_b(t)=1-\tfrac{t}{6}-\tfrac{t^2}{6}+b(t+\cdots+t^6),\quad f_b=(1-t)g_b,\quad a_1=-\tfrac76+b,\ a_3=\tfrac16,\ a_7=-b,$$

and study the fixed matrices near $b=0$; the point $b=0$ lies in Σ', as shown below, so by Section 7.6 a minor with a nonzero coefficient in its expansion at $b=0$ is a nonzero rational function on Σ. Put $K=n-\chi\ge16$, $\ell=\lceil(K-15)/4\rceil\ge1$ and $\varrho=4\ell-K+15\in\{0,1,2,3\}$.

Fourier denominators. At $b=0$, $g_0=(1-t/2)(1+t/3)$; the old poles of r_0 are $v_1=1/2$, $v_2=-1/3$ with residues $R_1=-21/160$, $R_2=-56/1215$, and the polynomial part is $P_0=t^3/6+7t^2/36-203t/216-161/1296$. For $4\le j<n$, $\lambda_j=n\sum_i R_i v_i^{n-j-1}/(1-v_i^n)$, which is nonzero because the ratio of the second summand to the first is at most $(1792/5103)/(1-3^{-29})<1/2$; for $j\le3$ the nonzero coefficient $P_{0,j}$ dominates the two tails, each of modulus less than 10^{-4}.

The values $f_0(x)$, $x \neq 1$, are nonzero since 2 and -3 are not roots of unity. Put

$$D_1 = \frac{nR_1 v_1^{\chi-1}}{1-v_1^n}, \quad T = \frac{R_2}{R_1}\Big(-\frac{2}{3}\Big)^{\chi-1}\frac{1-2^{-n}}{1-(-3)^{-n}}, \quad d_k = 2^k + T(-3)^k.$$

Then $\lambda_{K+k} = D_1 d_k$ whenever $4 \le K+k \le n-1$, and in the bulk range

$$0 < |T| \le \frac{1792}{5103}\Big(\frac{2}{3}\Big)^6 \frac{1}{1-3^{-29}} < \frac{1}{30}, \tag{7.14}$$

uniformly in n and χ; in particular $d_k > 0$ for $k < 0$.

The limiting kernel. Factor $g_b = G_b H_b$ analytically, where G_b is the quadratic factor with roots near $2, -3$ and H_b the quartic factor whose roots tend to infinity, both with constant term one. Let $Z_b = t^7 - \mathrm{rem}_{G_b}(t^7)$. A polynomial supported on F and divisible by G_b is uniquely of the form $z = \mathsf{A} f_b + \mathsf{B} Z_b$. At $b = 0$,

$$Z_0 = t^7 - 463t + 798,$$

$$l_Z(t) = -t g_0(t)\,\frac{Z_0}{g_0}(t^{-1}),$$

$$l_Z(t) = 6t^{-4} - 7t^{-3} + 42t^{-2} - 84t^{-1} + 336 - 840t + 78t^2 + 133t^3.$$

The three origin rows $j = 1, 3, 7$ have rank two at $b = 0$ and require exactly that z be divisible by g_0; their c coefficients vanish since all $\chi + j$ lie strictly between 7 and $n-7$; and the row $j = 0$ determines η. On the resulting three-dimensional space with coordinates $(\mathsf{A}, \mathsf{B}, c)$, the numerators in (7.13) are the Fourier transforms of $r_0(t) z(t^{-1}) = \mathsf{A} f_0 + \mathsf{B} l_Z$, supported in $[-4, 3]$, and of the c term, supported in $[-3, 3]$; so only the indices $j = K + k$, $-4 \le k \le 3$, occur, all between 12 and $n-4$, and the polynomial parts of P_A/f_0 (of degree at most 11) do not contribute. The two old poles give two moment conditions

$$\mathcal{A}_i \mathsf{A} + \mathcal{B}_i \mathsf{B} - \mathcal{C}_i (cD_1) = 0 \qquad (i = 1, 2), \tag{7.15}$$

with, for $\alpha_1 = 2$, $\alpha_2 = -3$,

$$\mathcal{A}_i = \sum_k \frac{(f_0)_k \alpha_i^{-k}}{d_k}, \quad \mathcal{B}_i = \sum_k \frac{(l_Z)_k \alpha_i^{-k}}{d_k}, \quad \mathcal{C}_i = \sum_{u,v \in \{0,1,3\}} a_u a_v \frac{d_{-v} \alpha_i^{-u+v}}{d_{u-v}}.$$

The two residue vectors are independent, since $t^2 + t^4$ takes the values 20 and 90 at 2 and -3. A direct rational simplification gives

$$\det(\mathcal{A}, \mathcal{C}) = \frac{-9475375\, T(8736T^2 + 1277T - 1134)}{10077696(T+1)(3T-2)(4T+9)(9T+4)(27T-8)}, \tag{7.16}$$

nonzero under (7.14). So (7.15) has rank two, every nonzero limiting kernel vector has $\mathsf{B} \neq 0$, and

$$\frac{\mathsf{A}}{\mathsf{B}} = -9870, \qquad \frac{cD_1}{\mathsf{B}} = \mathsf{c}(T), \tag{7.17}$$

where

$$\mathsf{c}(T) = \frac{-7776\, \mathsf{p}(T)}{175T(2T-3)(8T-27)(16T+81)(8736T^2 + 1277T - 1134)}$$

and

$$\mathsf{p}(T) = 609140736T^6 - 2807074304T^5 - 4400288360T^4 + 13769852070T^3 - 1887377085T^2 - 2495601009T + 589545216.$$

These identities are obtained by forming the finite sums in (7.15). Thus the extended matrix has rank exactly five at $b = 0$, with this one-dimensional kernel.

A moving third moment. Let $\alpha_1(b), \alpha_2(b)$ be the roots of G_b, $\mathsf{b}_i = \alpha_i^2 + \alpha_i^4$, and

$$P_\perp(t,b) = t^2(1-t^2)^2\big(t^2 + t^4 - \mathsf{b}_1\big)\big(t^2 + t^4 - \mathsf{b}_2\big),$$

which lies in the span of P_{11}, P_{13}, P_{33} (at $b = 0$ with coefficients $(1800, -3710, 1911)$) and vanishes at both old roots. Then $P_\perp / f_b = N_b / H_b$ with N_b an analytic polynomial of degree at most 11, and

$$N_0 = 6t^{11} + 42t^9 - 36t^8 - 378t^7 - 504t^6 - 1770t^5 - 1260t^4 + 2100t^3 + 1800t^2.$$

The coefficient of $b^{\ell'}$ in $P_\perp / f_b$ has degree at most $11 + 4\ell'$; for $z = \mathsf{A} f_b + \mathsf{B} Z_b$, $r_b(t) z(t^{-1}) = -t g_b(t)[z/G_b](t^{-1})/H_b(t^{-1})$ has b^m coefficient with exponents between $-4 - 4m$ and 7; and the c numerator has lowest exponent -3 at order zero and -7 from order one on. In (7.13) a term of orders ℓ' and m can pair only if $K - 4 - 4m \le j \le 11 + 4\ell'$, which forces $\ell' + m \ge \ell$. Hence the moving moment vanishes through order $\ell - 1$ on the whole family, and the positive origin equations vanish on it through order ℓ, their first possible alias having order $\lceil (K-11)/4 \rceil = \ell + 1$.

Leading coefficients. The four highest nonconstant coefficients of $\partial_b H_b|_0$ are $-6t^4 + 0t^3 - 42t^2 + 36t$, and one obtains for $j = 0, 1, 2, 3$

$$[b^{\ell'} t^{11+4\ell'-j}]\, \frac{P_\perp}{f_b} = 6^{\ell'+1}\big(1, 0, 7(\ell'+1), -6(\ell'+1)\big)_j,$$
$$[b^m t^{-4-4m+j}]\, r_b z(t^{-1}) = \mathsf{B}\, 6^m\big(6, -7, 42(m+1), -84 - 85m\big)_j;$$

the A part begins four exponents later. The c terms that can occur at this order are $a_3\lambda_{K-3}$, $a_1 a_3 \lambda_{K-3}$, $a_1\lambda_{K-1}$ at the exponents $-3, -2, -1$ (order zero) and $-\lambda_{K-7}$, $-a_1\lambda_{K-7}$, 0 at $-7, -6, -5$ (order one). For $\varrho = 1$ the first pair cancels because $6a_3 = 1$; for $\varrho = 3$ the surviving difference cancels because $6a_1 = -7$; for $\varrho = 0$ there is no c term; for $\varrho = 2$ the retained term is $-nca_1 6^\ell[\lambda_{K-3}/\lambda_{K-2} - \lambda_{K-7}/\lambda_{K-6}]$. Normalizing $\mathsf{B} = 1$ and substituting (7.17), the coefficient of b^ℓ of the moving moment is $n6^\ell \mathsf{S}_\varrho(T,\ell)/D_1$, where, with sums over $m = 0, \dots, \ell$,

$$\mathsf{S}_0 = 36 \sum \frac{1}{d_{-4-4m}}, \qquad \mathsf{S}_1 = -42 \sum \frac{1}{d_{-3-4m}},$$
$$\mathsf{S}_2 = 252 \sum \Big[\frac{\ell - m + 1}{d_{-4-4m}} + \frac{m+1}{d_{-2-4m}}\Big] + \frac{7}{6}\mathsf{c}(T)\Delta(T),$$
$$\mathsf{S}_3 = -6 \sum \Big[\frac{36(\ell - m + 1)}{d_{-4-4m}} + \frac{49(\ell - m + 1)}{d_{-3-4m}} + \frac{84 + 85m}{d_{-1-4m}}\Big],$$

where

$$\Delta = \frac{d_{-3}}{d_{-2}} - \frac{d_{-7}}{d_{-6}} = \frac{-1950T}{(4T+9)(64T+729)}.$$

All d_k occurring here are positive, so $\mathsf{S}_0 > 0$ and $\mathsf{S}_1, \mathsf{S}_3 < 0$. For S_2, $1/d_k \ge \frac{30}{31} 2^{-k}$, and the terms $m = 0, 1$ alone contribute more than $252 \cdot 420 \cdot 30/31 > 102000$, while the numerator of $\frac{7}{6}\mathsf{c}\Delta$ is $101088\,\mathsf{p}(T)$ with $|\mathsf{p}(T)| < 7 \cdot 10^8$ for $|T| < 1/30$, and its denominator $(2T-3)(4T+9)(8T-27)(16T+81)(64T+729)(8736T^2+1277T-1134)$ has factors of absolute values greater than $2, 8, 26, 80, 726, 1080$; hence $|\frac{7}{6}\mathsf{c}\Delta| < 2712$ and $\mathsf{S}_2 > 0$.

Proof of Proposition 7.6. At $b = 0$ choose five independent rows and columns: two positive origin rows, the η row and the two old moment rows; their pivot minor stays invertible for small b. In the family $z = \mathsf{A}(b)f_b + Z_b$ solve the two chosen moment equations for $\mathsf{A}(b), c(b)$, using (7.16), and solve the η row. The resulting vector satisfies the chosen origin equations to order ℓ, so its correction to the exact five-pivot kernel changes it only at order $\ell+1$. The moving moment vanishes on the family through order $\ell-1$, and its coefficient of b^ℓ depends only on $\mathsf{A}(0) = -9870$, $\mathsf{B} = 1$ and $c(0)D_1 = \mathsf{c}(T)$; it is the nonzero number above. Hence the Schur complement of a sixth row with respect to the five pivots is nonzero at order ℓ, and some 6×6 minor has a nonzero coefficient of b^ℓ. □

7.8. **The boundary characters.**

Proposition 7.7. *Let $n \geq 89$ and $\chi \in \{0, 1, \dots, 6\} \cup \{n-15, \dots, n-1\}$. On a nonempty open subset of Σ, $\operatorname{rank}[\mathcal{P}_\chi; \mathcal{W}_\chi] = 6$.*

These are 22 characters. We write $\chi = -h$ for the negative ones and use the two lines of Section 7.6: the six-small-pole line for $h \in \{1, 2, \dots, 10, 14\}$ and $\chi = 0$, and the cubic line for $\chi \in \{1, \dots, 6\}$ and $h \in \{11, 12, 13, 15\}$. Throughout we use the normalized quantities $l_j = \lambda_j/n$, $\psi_A = \Psi_A/n$ and $\bar{c} = nc$; in the character $-h$ the moment rows are

$$\mathsf{W}_A = \sum_{j \in \mathbb{Z}/n} \psi_A(j)\, \frac{\sum_e z'_e l_{j-h+e} - \bar{c}\,\bar{\mathcal{B}}_j}{l_j}, \qquad \bar{\mathcal{B}}_j = \sum_{u-v=j-h} a_u a_v l_{h-v},$$

and the origin rows are the numerators for $j = 1, 3, 7$, with $(\bar{c}a_h - \kappa)l_j$ added for $h \in \{1, 3, 7\}$, $(\bar{c} - \rho')l_j/2$ added for $h = 0$, and the extra equation $\bar{c} = 0$ for the quadratic h. The row $j = 0$ has coefficient $-l_0$ on η and determines η.

The six-small-pole line. Put $a_1 = -1 - \epsilon$, $a_3 = 0$, $a_7 = \epsilon$, so that $f_\epsilon = (1-u)(1-\epsilon A(u))$ with $A = u + \cdots + u^6$; with $Q(u) = u^{-1} + \cdots + u^{-6}$ we have $r_\epsilon(u) = -u(1-\epsilon A(u))/(1-\epsilon Q(u))$. Let $\mathsf{L}_m(\epsilon)$ be the coefficient of u^m in

$$-u + \sum_{q \geq 1} \epsilon^q \big[-uQ(u)^q + uA(u)Q(u)^{q-1}\big],$$

a Laurent polynomial at each order, with exponents at most 7. Then $l_j = \sum_{k \in \mathbb{Z}} \mathsf{L}_{j+kn}$; in particular $l_1 = -1 + O(\epsilon)$, $l_0 = -\epsilon + O(\epsilon^2)$ and $l_j = \epsilon + O(\epsilon^2)$ for $2 \leq j \leq 7$. For $8 \leq j < n$ the valuation of l_j is $q_j = \lceil (n-j+1)/6 \rceil$ and its leading coefficient is $-\binom{q_j + \varrho_j - 1}{\varrho_j}$ with $\varrho_j = 6q_j - n + j - 1 \in \{0, \dots, 5\}$. So every l_j is a unit in $\mathbb{Q}((\epsilon))$; moreover f_ϵ does not vanish on $G \smallsetminus \{1\}$ near $\epsilon = 0$ and $f'_\epsilon(1) = -(1 - 6\epsilon)$. Write $n = 6\mathsf{N} + \mathsf{s}$, $0 \leq \mathsf{s} \leq 5$, $\mathsf{N} \geq 14$. For fixed $j \geq 8$ and $\delta_j = \lceil (\mathsf{s} - j + 1)/6 \rceil$, before a second alias enters, the coefficient of $\epsilon^{\mathsf{N}+\delta}$ in l_j is the polynomial in N

$$d_{j,\delta}(\mathsf{N}) = -C(\mathsf{N} + \delta, 6\delta - \mathsf{s} + j - 1) + \sum_{a=1}^{6} C(\mathsf{N} + \delta - 1, 6\delta - \mathsf{s} + j - a - 7),$$

where $C(q, \mathsf{d}) = 0$ for $\mathsf{d} < 0$ and otherwise $C(q, \mathsf{d}) = \sum_{i \leq \mathsf{d}/6} (-1)^i \binom{q}{i} \binom{q + \mathsf{d} - 6i - 1}{\mathsf{d} - 6i}$; this follows by reversing the powers in Q^q. Finally $P_A/f_\epsilon = p_A(u) \sum_q \epsilon^q A(u)^q$ with $p_A = P_A/(1-u)$ of degree at most 13.

Replacing every l_m in the positive origin rows by L_m gives a formal 3-row matrix over $\mathbb{Q}((\epsilon))$. For $h = 2, \dots, 10, 14$ choose the first three-column minor of least

valuation (Table 3), and form the cofactor kernel vectors: the free entry is this determinant, the pivot entries are the negatives of the determinants with the corresponding pivot column replaced, the other free entries are zero, and the common power of ϵ is divided out.

TABLE 3. Least-valuation pivot minors in the positive origin rows (columns numbered from 0 in the order $z'_0, z'_1, z'_3, z'_7, \bar{c}$).

h	2	3	4	5	6	7	8	9	10	14
columns	012	012	023	023	013	013	023	023	023	023
valuation	2	1	3	2	3	1	3	2	3	5

Every entry of the three origin rows has index at most 12, so a missing cyclic alias begins at order at least $\mathsf{N}-1$; for the $\bar{\mathcal{B}}$ terms, $u-v=j-h$ implies $h-v=j-u\le 7$. Replacing the formal entries by the actual cyclic entries changes the cofactor vectors only by $O(\epsilon^{\mathsf{N}-6})$, which is $O(\epsilon^8)$, and the chosen minors remain nonzero, so the corrected vectors lie in the actual origin kernel. After multiplication by ϵ^{N} the moment rows are regular, and through order six their values on these vectors are given by the finite sums

$$\sum_{j=8}^{h+7} \psi_A(j)\, \frac{\sum_e z'_e \mathsf{L}_{j-h+e} - \bar{c}\sum_{u-v=j-h} a_u a_v \mathsf{L}_{h-v}}{D_j(\epsilon;\mathsf{N},\mathsf{s})}, \tag{7.18}$$

where D_j is the series $\epsilon^{-\mathsf{N}} l_j$ computed from $d_{j,\delta}$: indices $j\le 7$ contribute only from order $\mathsf{N}-1$ on, indices $j>h+7$ have numerators without ordinary Laurent coefficient, cyclic corrections to the numerators start at order $\mathsf{N}-2$, a second alias in a denominator is at least $\mathsf{N}-1$ orders later than the first, and the degree bound on p_A excludes any ψ alias. All these bounds exceed six for $n\ge 89$.

For the quadratic $h\in\{2,4,6,8,10,14\}$ take the unique cofactor vector and its P_{11} moment, multiplied by ϵ^{N}; its first nonzero coefficient has order at most three. For $h\in\{3,5,7,9\}$ take the first two free cofactor vectors and the 2×2 determinant of the rows $\epsilon^{\mathsf{N}}\mathsf{W}_A$ on them, with $A\in\{11,13\}$ for $h=3,7,9$ and $A\in\{11,33\}$ for $h=5$; its first nonzero order is at most six. All $36+24$ orders and leading coefficients, for every residue s, are listed in Tables 5 and 6 of Appendix B; for instance, for $\mathsf{s}=0$ and $h=2$ the coefficient is $2/(\mathsf{N}(\mathsf{N}-1))$ at order three. In each case all lower coefficients vanish. Each listed coefficient is a rational function of N which does not vanish for real $\mathsf{N}\ge 14$, by the sign test of Appendix B. In the translation characters $h=3,7$ the third free vector is the global direction and has zero moments. Since the positive origin rows have rank three over $\mathbb{Q}((\epsilon))$ and η is determined, this gives rank six.

For $h=1$ the pivot columns are 012 with valuation two; the first two cofactor vectors K_0, K_1 correspond to the free z'_7 and $\bar{c}$, the third is global. Only $j=8$ produces a moment pole of size $\epsilon^{-\mathsf{N}}$, with numerator $C_8(z,\bar{c}) = z'_0\mathsf{L}_7 - \bar{c}\epsilon\mathsf{L}_1$. With $C_i = C_8(K_i)$ the vector $K_* = (C_1K_0 - C_0K_1)/\epsilon^2$ is regular, lies in the formal origin kernel, kills C_8, and begins with $(z'_0, z'_1, z'_3, z'_7, \bar{c}, \kappa) = (-1,2,0,0,1,0)$. The normalized moment $\epsilon^{\mathsf{N}}\mathsf{W}_{11}(K_0)$ has a nonzero leading coefficient of order at most three in every residue class (Table 7). For the second row use the exact cyclic projection $\mathsf{W}_* = \mathsf{W}_{13} - [\psi_{13}(8)/\psi_{11}(8)]\mathsf{W}_{11}$, where $\psi_{11}(8)$ is a unit with constant

term -1; it removes the whole summand $j = 8$ on the full source. On K_* the first nonzero coefficient of W_* has order zero, except for $\mathsf{s} = 3$, where it has order one. Both leading coefficients are listed in Table 7 for every residue s, and they are nonzero by the same sign test. The cyclic origin cofactors differ from the formal ones only at order at least $\mathsf{N} - 4$; solving the first moment pivot on K_0, of order at most three, changes the vector by $O(\epsilon^{\mathsf{N}-7})$, and since the projected row has valuation at least -2, its displayed coefficient is unchanged because $\mathsf{N} - 9 \geq 5$. Thus the two moments are independent modulo the global direction.

For $\chi = 0$ subtract the global direction to make $z_0' = 0$, keep the columns $(z_1', z_3', z_7', \bar{c}, \rho')$ and only the two positive origin rows $j = 1, 3$, whose least pivot is in columns $(0, 3)$ with valuation one. The cofactor construction gives K_0, K_1, K_2 (free columns z_3', z_7', ρ'); put $K_2' = K_2 - \epsilon^2 K_1$ and use the moment rows W_{11}, $\mathsf{W}_{13} - \mathsf{W}_{11}$, $\mathsf{W}_{33} - 2\mathsf{W}_{13} + \mathsf{W}_{11}$, whose numerators after division by $1 - u$ have degrees at most $13, 9, 5$. Multiply (K_0, K_1, K_2') by $(\epsilon, \epsilon^2, \epsilon^{-1})$ and the three rows by ϵ^{-r_i} with $(r_0, r_1, r_2) = (0, 1, 1)$ for $\mathsf{s} = 0, 1$ and $(0, 0, 1)$ for $\mathsf{s} \geq 2$. All entries become regular. The resulting 3×3 matrices M of constant terms and their determinants are listed in Appendix B; for example, for $\mathsf{s} = 2$, $\det \mathsf{M} = 15/(\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4))$. All six determinants are nonzero for $\mathsf{N} \geq 14$. Thus two origin rows and three moments have rank five after the global gauge, and the η row gives rank six; the third positive origin row is not needed.

The cubic line. For the remaining ten characters use f_b of the bulk proof at $b = 0$, with $K = n - \chi$, $D = R_1(1/2)^{\chi-1}/(1 - 2^{-n})$, T and d_k as there, so that $l_{K+k} = D d_k$ for indices in $[4, n-1]$, and $z = \mathsf{A} f_0 + \mathsf{B} Z_0$ for z divisible by g_0, with $r_0 z(u^{-1}) = \mathsf{A} f_0 + \mathsf{B} l_Z$. The two old-pole moment equations and their kernel are (7.16)–(7.17), with $\bar{c}D$ in place of cD_1. Two facts about wrapped indices are needed: the Taylor coefficients of r_0 at 0 vanish in degrees $0, 1, 2$, so the old-pole formula for l_{K+k} continues across $n, n+1, n+2$; and those of P_A/f_0 vanish in degrees $0, 1$ and equal $(1, 0, 0)$ in degree two, so its old-pole formula continues across $n, n+1$ and acquires this extra vector at $n+2$. The two old-pole residue vectors together with either $(1, 0, 0)$ or $(1, 1, 1)$ are independent. For $1 \leq \chi \leq 6$ the true value of T lies in the interval $T_\chi(1 \pm 10^{-8})$, $T_\chi = (1792/5103)(-2/3)^{\chi-1}$, because $(2^{-29} + 3^{-29})/(1 - 3^{-29}) < 10^{-8}$.

For $\chi = 1, 2$, the origin rows $j = 3, 7$ force divisibility by g_0, and all moment contributions lie in the two old-pole directions, except that for $\chi = 1$ the wrapped index $n+2$ contributes the numerator of the remaining positive origin row $(\mathsf{A} + 798\mathsf{B} - \bar{c} l_K)/6$ times the independent vector $(1, 0, 0)$. In both cases zero moments force that row and both old-pole equations, which is impossible unless $9072 + \mathsf{c}(T)(1 + T) = 0$; this expression and (7.16) are nonzero on the corresponding T intervals, by the interval bounds of Appendix B.

For $\chi = 3, 4, 5, 6$ the three positive origin rows at $b = 0$ involve only the old-pole directions and impose divisibility by g_0, and the extended matrix has rank five with kernel (7.17). The left annihilator of the two old positive-origin columns on the rows $j = 1, 3, 7$ is $(468, -133, 1)$. Moving z along the family divisible by the analytic factor G_b, the first b-derivatives of $r_b z(u^{-1})$ in the degrees $4, 5, 6, 7$ are $-504, -546, -468, 9072$, and the c contributions to these degrees are $\bar{c} a_3 l_{K-3}$, 0,

$\bar{c}a_1 l_{K-1}$, $\bar{c}d_K$. The projected first-order obstructions are

$$\chi = 3:\ -173628 + \mathsf{c}(T)\big[78d_{-3} + \tfrac{931}{6}d_{-1}\big], \quad \chi = 4:\ -1462104 - 133\,\mathsf{c}(T)d_0,$$
$$\chi = 5:\ 468\big[-468 - \tfrac{7}{6}\mathsf{c}(T)d_{-1}\big], \qquad \chi = 6:\ 468\big[9072 + \mathsf{c}(T)d_0\big],$$

each nonzero on its T interval (Appendix B). Since the projected row annihilates every column at $b = 0$, these are the first Schur obstructions after the five pivots, independently of the derivative of the lifted kernel vector.

For $h \in \{11, 12, 13, 15\}$ ($\chi = n - h \geq 74$, so $|T| < 10^{-4}$), the first two positive origin rows force divisibility by g_0. The polynomial parts of the three moment functions agree in the degrees $8, \dots, 11$ (coefficients $-36, 42, 0, 6$), so all polynomial contributions lie in the direction $(1, 1, 1)$, independent of the old-pole directions, and a zero-moment vector satisfies the two equations of (7.17) and a remaining scalar equation. For $h = 11$ the third positive origin row is $6\mathsf{B} = 0$; for $h = 15$ the scalar is $36\mathsf{B}/(Dd_{-4})$; for $h = 12$, after multiplication by D/B, it is $-216/d_{-4} - 294/d_{-3} - 504/d_{-1} < 0$ (the c terms $\mp 7\bar{c}$ cancel); for $h = 13$ it is $F_{13}(T) = 252/d_{-4} + 252/d_{-2} + \frac{7}{6}\mathsf{c}(T)d_{-3}/d_{-2}$, and $TF_{13}(T)$ has numerator and denominator bounded away from zero on $|T| \leq 10^{-4}$ (Appendix B). In each case $\mathsf{B} = 0$, and then the invertible old system forces $\mathsf{A} = \bar{c} = 0$. This completes the proof of Proposition 7.7.

7.9. **The lower boundary.**

Theorem 7.8. *Let $d \geq 27$ and $s = d + 3$. Then a general configuration P of $r = \binom{d+2}{3} + d + 3$ points satisfies* $\operatorname{rank} J_P = 3r - 4$, *and $T^1_j(C(Z)) = 0$ for all $j < 0$.*

Proof. Let $n = d+2 \geq 29$. Consider first $n \geq 89$. Fix coefficients in the intersection of the finitely many nonempty open subsets of Σ given by Lemma 6.1, Theorem 6.7 and Propositions 7.5, 7.6 and 7.7. Since $\mathcal{O}_\chi$ is injective, (7.12) gives opening ranks 2, 1 and 2 on the special kernels of the ordinary, quadratic and global characters, so the total opening rank is

$$2(n - 10) + 6 + 2 \cdot 4 = 2n - 6.$$

For $29 \leq n \leq 88$ the same ranks are certified directly by $\operatorname{rank}[\mathcal{P}_\chi; \mathcal{O}_\chi \mathcal{W}_\chi] = 6$ for every character (Appendix A); there the injectivity of $\mathcal{O}_\chi$ is not used. By (6.3), Theorem 6.7 and Lemma 7.1, for small $\varepsilon \neq 0$,

$$\operatorname{rank} \Gamma^{\mathrm{f}}(y(\varepsilon)) \geq 3L_d + (n^2 + 5n + 4) - (2n - 2) + (2n - 6) = N_{n+1} - 4.$$

Let $Z \subset Y_{n+1}$ be the affine subspace in which the last auxiliary point is the origin. It is irreducible, contains $y(\varepsilon)$, and meets Y°_{n+1}, since the evaluation at the origin is the nonzero vector $(1, \dots, 1)$ and can be completed by n general points. Lemma 5.5(a) gives $\operatorname{rank} J_P = 3r - 4$ for a general configuration; explicitly, after eliminating the injective $(n + 1)^2$ coefficients β,

$$\operatorname{rank} J_P = 3\tbinom{n}{3} + n^2 + 5n - (n + 1)^2 = 3\tbinom{n}{3} + 3n - 1 = 3r - 4.$$

Then $T^1_{-1} = 0$ by Lemma 2.4. Since $m = M_d - d - 3 \geq 4$, Propositions 3.1 and 3.2 give $T^1_j = 0$ for all $j < 0$ on the intersection of the two nonempty open sets. $\square$

8. Coupled Horace transitions

8.1. **Setting.** Let $e \geq 2$, $d = e+1$ and $D = \binom{e+2}{2} = M_e$. Choose integers $0 \leq s' < D$ and $0 \leq c \leq e+2$, and put

$$a = \tbinom{e+2}{3} + s', \quad b = D + c, \quad k = D - s', \quad t = e + 2 - c, \quad s = s' + c, \quad m = k + t.$$

Then $a + b = L_d + s$ and $m = M_d - s$. Let $H = \{z = 0\}$, let A be a distinct points of $\mathbb{A}^3 \setminus H$ and B be b distinct points of H, and assume:

- (H1) evaluation of F_{e-1} on A is injective and evaluation of F_e on A is surjective;
- (H2) evaluation of $F_e(H)$ (plane polynomials of degree at most e) on B is injective and evaluation of $F_{e+1}(H)$ on B is surjective.

These are nonempty open conditions. Put $E_A = K(A) \subset F_e$, of dimension k, and $V_A = V_{e-1}(A)$. Restriction of E_A to H is injective, since an element divisible by z would give an equation of degree $e-1$ on A. Let $L_A = E_A|_H$, $L_B = \mathrm{ev}_B(L_A)$, $W = \mathrm{ev}_B\, F_e(H)$, and

$$Q_A = \mathbb{C}^A/V_A \ (\dim s'), \qquad Q_B = \mathbb{C}^B/L_B \ (\dim s), \qquad C_B = \mathbb{C}^B/W \ (\dim c).$$

The union $\Gamma = A \cup B$ has no equation of degree e (restrict to B, divide by z, and use (H1)), evaluation of F_{e+1} on Γ is surjective (interpolate on B, then correct on A by z times a polynomial of degree e), and restriction to H gives an exact sequence

$$0 \to zE_A \to K(\Gamma) \to I_{B,H}(\leq e+1) \to 0, \tag{8.1}$$

where $I_{B,H}(\leq e+1)$ is the t-dimensional space of plane polynomials of degree at most $e+1$ vanishing on B. Thus $\Gamma \in U^{\mathrm{ev}}_{d,s}$.

8.2. **The coupled identity.** A velocity of Γ is a triple (v_A, u, w) with $v_A \in (\mathbb{C}^3)^A$, tangential components $u \in (\mathbb{C}^2)^B$ and normal components $w \in \mathbb{C}^B$. Define the *normal constraint*

$$R_1(w)(h) = [w\, h(B)] \in C_B \quad (h \in E_A), \qquad W_1 = \ker R_1, \qquad \rho_1 = \operatorname{rank} R_1,$$

where $w\, h(B)$ is the coordinatewise product. For $w \in W_1$ let $g_{w,h} \in F_e(H)$ be the unique plane polynomial with values $w\, h(B)$, regarded as a polynomial independent of z, and put

$$\mathcal{K}(w)(h) = \big[g_{w,h}(A)/z(A)\big] \in Q_A, \qquad J_A(v_A)(h) = [D_{v_A} h(A)] \in Q_A.$$

Thus J_A is the moving differential of A in degree e; let j_A be its rank.

Lemma 8.1. *The linearized conditions of all equations zh, $h \in E_A$, hold if and only if $w \in W_1$ and $J_A(v_A) = \mathcal{K}(w)$. Constant vectors lie in W_1, and $\mathcal{K}(\lambda \mathbf{1}) = 0$.*

Proof. The derivative data of zh are $z(A) D_{v_A} h(A)$ on A and $w\, h(B)$ on B. A correcting polynomial $G \in F_e$ must restrict on H to $g_{w,h}$, which requires $w \in W_1$, so $G = g_{w,h} + zq$ with $q \in F_{e-1}$; comparing values on A gives the condition. For $w = \lambda \mathbf{1}$ write $h = h_0 + z h_1$ with $h_0 = h|_H$; then $g = \lambda h_0$ and $h_0(A)/z(A) = -h_1(A) \in V_A$. □

Let π be the quotient by $\operatorname{im} J_A$, $\tau = \operatorname{rank}(\pi\mathcal{K})$ and $\mathcal{Z} = \{(v_A, w) : w \in W_1,\ J_A(v_A) = \mathcal{K}(w)\}$. Projection to w gives an exact sequence $0 \to \ker J_A \to \mathcal{Z} \to \ker(\pi\mathcal{K}) \to 0$, so $\dim \mathcal{Z} = (3a - j_A) + (b - \rho_1 - \tau)$.

For $f \in I_{B,H}(\leq e+1)$ choose a lift $F_f = f + zH_f \in K(\Gamma)$, and a right inverse T_A of evaluation of F_e on A. Define on $\mathcal{Z} \oplus (\mathbb{C}^2)^B$

$$\mathcal{T}(v_A, w, u)(f) = \big[D_u f(B) + w\, H_f(B) - \mathrm{ev}_B\, T_A(D_{v_A} F_f(A))\big] \in Q_B.$$

Any correcting polynomial of degree e for F_f must interpolate the derivative data on A, and two such differ by E_A, whose values on B form L_B. Hence the remaining linearized conditions are exactly $\mathcal{T} = 0$. Changing T_A or the lift changes $\mathcal{T}$ by terms that vanish on $\mathcal{Z}$, so $\rho_2 = \operatorname{rank}\mathcal{T}$ is well defined. By (8.1),

$$\operatorname{rank} J_\Gamma = j_A + \rho_1 + \tau + \rho_2. \tag{8.2}$$

This is an identity at one configuration; it involves no independence assumption on its summands.

Proposition 8.2. *Suppose that* $\ker J_A$ *consists of the four motions, that* $W_1 = \mathbb{C}\mathbf{1}$*, and that the operator*

$$(u,\eta) \longmapsto \big(f \mapsto [D_u f(B) + \eta\, N_A(f)(B)] \in Q_B\big), \tag{8.3}$$

where $N_A(f) = H_f|_H \bmod L_A$*, on* $(\mathbb{C}^2)^B \oplus \mathbb{C}$ *is injective. Then* $\operatorname{rank} J_\Gamma = 3(a+b)-4$*.*

Proof. By Lemma 8.1, $\mathcal{K} = 0$ on W_1, so $\tau = 0$, $\rho_1 = b-1$, and $\mathcal{Z} = \ker J_A \oplus \mathbb{C}\mathbf{1}$ has dimension five. Together with u this source has dimension $2b+5$, and contains the four motions of Γ. Subtracting the normal translation we may take $w = 0$; subtracting the two tangential translations and the dilation, $v_A = \lambda e_z$. On this $(2b+1)$-dimensional representative, $D_{v_A}F_f = \lambda\partial_z F_f$, which has degree at most e and interpolates its own values on A, and $\partial_z F_f|_H = H_f|_H$. Hence $\mathcal{T}$ becomes (8.3) with $\eta = -\lambda$. By (8.2), $\operatorname{rank} J_\Gamma = (3a-4) + (b-1) + (2b+1)$. □

Remark 8.3. Since $\tau \le b-\rho_1-1$, $\tau \le s'k - j_A$, $\rho_1 \le ck$ and $\rho_2 \le st$, $\operatorname{rank} 3(a+b)-4$ requires $st \ge 2b+1+(3a-4-j_A)$; in particular a positive residual defect $3a-4-j_A$ cannot be ignored. For $c = 1$ and $s' > 0$ we have $\dim W_1 \ge s'+1 > 1$, so the separated form of Proposition 8.2 is impossible; we use only $c \ge 2$.

The conditions of Proposition 8.2 other than the one on $\ker J_A$ will be called (R1) ($W_1 = \mathbb{C}\mathbf{1}$) and (R2) (injectivity of (8.3)). On the evaluation pivot charts both are rank conditions on matrices of rational functions of (A,B), hence open in the irreducible space of pairs (A,B).

Corollary 8.4. *Fix* (e,s',c)*. Suppose that a general configuration in degree* e *and position* s' *has* $T^1_{-1} = 0$*, and that (R1) and (R2) hold for one pair* (A,B) *satisfying (H1), (H2), with* A *arbitrary. Then a general configuration in degree* $e+1$ *and position* $s'+c$ *has* $T^1_{-1} = 0$*.*

Proof. The three open conditions on pairs (A,B) (residual rank, (R1), (R2)) are nonempty and meet. At a common point Proposition 8.2 gives $\operatorname{rank} J_\Gamma = 3(a+b)-4$, and this is an open condition on all configurations of $a+b$ points. □

8.3. **Cartesian residuals and the normal operator.** For a lower set $\mathsf{T} \subset \{(i,j) : i+j \le e\}$ with $|\mathsf{T}| = k$ let Λ_T consist of all triples of degree at most $e-1$ and the triples $(i,j,e-i-j)$ with $(i,j) \notin \mathsf{T}$. Choose pairwise distinct *plane nodes* $\xi_0,\dots,\xi_{e+1}$ and pairwise distinct nonzero *normal nodes* $\zeta_0,\dots,\zeta_e$, and let $A_{\xi,\zeta} = \{(\xi_i,\xi_j,\zeta_l) : (i,j,l) \in \Lambda_\mathsf{T}\}$. Put $N_i(x) = \prod_{v<i}(x-\xi_v)$ and $Z_l(z) = \prod_{v<l}(z-\zeta_v)$. Unless stated otherwise $\xi_v = v$; we write A_ζ in that case.

Lemma 8.5. *For all nodes as above,* $A = A_{\xi,\zeta}$ *satisfies (H1),* $|A| = a$*, and* $L_A = \operatorname{span}\{x^i y^j : (i,j) \in \mathsf{T}\}$*. For a plane polynomial* $f = \sum f_{ij} N_i(x) N_j(y)$ *of degree*

at most $e+1$, written in the Newton basis of the plane nodes, with lift $f+zH_f$ vanishing on A,

$$N_A(f) = -\sum_{i+j\le e,\ (i,j)\notin \mathsf{T}} w_{e-i-j+1} f_{ij} N_i(x) N_j(y) \bmod L_A, \tag{8.4}$$

where $w_v = \sum_{l<v} \zeta_l^{-1}$.

Proof. The polynomials $N_iN_jZ_l$, $(i,j,l)\in\Lambda_{\mathsf{T}}$, form a triangular interpolation basis on A, as in Lemma 2.1, since the nodes on each axis are distinct. For $(i,j)\in\mathsf{T}$ the polynomial $N_iN_jZ_{e-i-j}$ vanishes on A; these k polynomials span E_A, and their restrictions to H are nonzero multiples of N_iN_j, which span the monomial space of the lower set T because the change from Newton to monomial basis is triangular. Interpolate $-f(x,y)/z$ on A by the Newton basis. Its coefficients are products of divided differences; the divided difference of f over the nodes $\xi_0,\dots,\xi_i$ and $\xi_0,\dots,\xi_j$ is f_{ij}, and $[\zeta_0,\dots,\zeta_l](1/z) = (-1)^l/(\zeta_0\cdots\zeta_l)$ by induction from the recurrence. Since $Z_l(0) = (-1)^l\zeta_0\cdots\zeta_{l-1}$, the restriction to H receives $-f_{ij}/\zeta_l$ from each $(i,j,l)\in\Lambda_{\mathsf{T}}$. For $(i,j)\notin\mathsf{T}$ all $l\le e-i-j$ occur; for $(i,j)\in\mathsf{T}$ the contribution lies in L_A; and $i+j=e+1$ gives nothing. □

Formula (8.4) is linear in $(w_1,\dots,w_{e+1})$, and the map from $(\zeta_0^{-1},\dots,\zeta_e^{-1})$ to w is triangular and invertible. Hence, if a maximal minor of the (R2) matrix with the normal column (8.4) is nonzero at some *formal* weight vector w, it is a nonzero linear polynomial in w and is nonzero for some pairwise distinct nonzero nodes. This is how formal normal operators are used below.

Below, T is often a *first-row* lower set: the first k pairs (i,j) in the lexicographic order, that is, ordered by increasing i and then by increasing j.

8.4. **Plane points and canonical modules.** In this and the following subsections $S=\mathbb{C}[X_0,X,Y]$ denotes the homogeneous coordinate ring of the projective closure of the plane H, with affine coordinates $x=X/X_0$, $y=Y/X_0$ (the closure of H in $\mathbb{P}^3$ is the plane $X_3=0$, with homogeneous coordinates X_0, $X=X_1$, $Y=X_2$). Let B be $b=D+c$ general points of $H=\{X_0=1\}$, with homogeneous coordinate ring R_B; its h-vector is $(1,2,\dots,e+1,c)$. By Sauer [8, Proposition 1.7], for points in a dense open set the ideal I_B has $\max(t,c+1)$ minimal generators, where $t = e+2-c = \dim(I_B)_{e+1}$. Hence:

- if $c\le(e+1)/2$ (*low* c), with $l=e+1-2c\ge 0$, the minimal resolution is $0\to S(-e-2)^l\oplus S(-e-3)^c\to S(-e-1)^t\to I_B\to 0$;
- if $c\ge(e+1)/2$ (*high* c), with $q=2c-e-1\ge 0$, it is $0\to S(-e-3)^c\to S(-e-1)^t\oplus S(-e-2)^q\to I_B\to 0$.

In both cases the graded ranks are forced by the number of generators and the Hilbert series. Let ω be the canonical module of R_B, graded so that dt_i has degree one on the normalization. By the residue description [9, Section 1.3.2],

$$\omega_j = (\mathrm{ev}_B\, S_{-j})^\perp \subset \mathbb{C}^b,$$

multiplication by a form being coordinatewise multiplication by its values; thus $\omega_j=0$ for $j<-e$, $\omega_{-e}=W^\perp$ has dimension c, and $\omega_0=\mathbf{1}^\perp$. For a subspace $\mathcal{U}$ of polynomials put $\mathcal{U}_B=\mathrm{ev}_B\,\mathcal{U}$. Since $\langle w\cdot l,\theta\rangle=\langle w, l\cdot\theta\rangle$, we have

$$\{w : w\,L_B\subset W\} = (W^\perp\cdot L_B)^\perp, \tag{8.5}$$

and $W^\perp\cdot L_B\subset\mathbf{1}^\perp$ because $L_B\subset W$. So (R1) holds as soon as $L_A\cdot\omega_{-e}=\omega_0$.

Lemma 8.6. *Let $c \le (e+1)/2$, and let $\mathcal{P}$ be the affine space of $t \times (t-1)$ matrices $H = [\mathsf{A} \mid \mathsf{B}_q]$ over S whose first l columns consist of linear forms and whose last c columns consist of quadratic forms. Let $\mathcal{P}_{\mathrm{red}}$ be the set of H whose maximal minors generate the ideal of b distinct reduced points of the affine plane $X_0 \neq 0$.*

(a) *$\mathcal{P}_{\mathrm{red}}$ is a nonempty open subset of $\mathcal{P}$, and the map from $\mathcal{P}_{\mathrm{red}}$ to the space of b-point subsets of the affine plane is dominant.*

(b) *Let $\mathcal{U} \subset \mathcal{P}$ be a nonempty open subset defined by the nonvanishing of minors in the entries of H, and suppose that at every point of $\mathcal{U} \cap \mathcal{P}_{\mathrm{red}}$ a given rank condition on point evaluations of B holds. Then this rank condition holds for a general set B of b points.*

Proof. (a) The space $\mathcal{P}$ is affine, hence irreducible. All entries have positive degree, so at every point of $\mathbb{P}^2$ the values of the entries can be prescribed independently, and the universal evaluation map is a submersion onto the space of $t \times (t-1)$ matrices. In that space the matrices of rank $t-2$ form a smooth stratum of codimension two and those of rank at most $t-3$ have codimension six. Hence, for general H, the degeneracy locus $\{p : \operatorname{rank} H(p) \le t-2\}$ is a finite reduced subscheme of $\mathbb{P}^2$ which does not meet the line $X_0 = 0$. Its ideal of maximal minors then has height two, the Hilbert–Burch theorem gives the resolution of Section 8.4 with fixed graded ranks, and therefore the locus consists of $b = D + c$ reduced points. These are open conditions, so $\mathcal{P}_{\mathrm{red}}$ is a nonempty open subset. By Sauer's count, the ideal of b general points has exactly t minimal generators, all of degree $e+1$, and the low-c resolution; so it is generated by the maximal minors of some $H \in \mathcal{P}$, which lies in $\mathcal{P}_{\mathrm{red}}$. This proves dominance; for ordered configurations one passes to the finite étale cover given by the orderings.

(b) Since $\mathcal{P}$ is irreducible, $\mathcal{U} \cap \mathcal{P}_{\mathrm{red}}$ is a nonempty open subset of $\mathcal{P}_{\mathrm{red}}$. Its image under the dominant map of (a) is constructible and dense, hence contains a nonempty open set of configurations, and at every point of the image the rank condition holds. □

Lemma 8.7 (high c)**.** *Let $c \ge (e+1)/2$ and let L_A contain all monomials y^j $(j \le e)$ and xy^j $(j \le e-1)$. Then (R1) holds for general B.*

Proof. Every element of ω_{-e} is a minimal generator, and by Sauer's count the type of R_B is $c = \dim \omega_{-e}$, so ω is generated in degree $-e$. Dualizing the resolution presents ω by c generators of degree $-e$, t quadratic and q linear relations. Choose coordinates with $P_\infty = (0:1:0) \notin B$. The maximal minors of the Hilbert–Burch matrix generate I_B and do not all vanish at P_∞, so the presentation matrix has rank c there. Multiply the linear relations by X (which is 1 at P_∞) and choose c relations independent at P_∞; constant combinations give, for a basis $\gamma_1, \dots, \gamma_c$ of ω_{-e},

$$X^2\gamma_i + \sum_j \big(X a_{ij}(X_0, Y) + b_{ij}(X_0, Y)\big)\gamma_j = 0.$$

Repeated substitution reduces every $p\gamma_i$, with p of degree e, to a combination of the γ_j with coefficients in $\mathbb{C}[X_0, Y]_e \oplus X\mathbb{C}[X_0, Y]_{e-1}$, which is contained in the homogenization of L_A. Hence $L_A \cdot \omega_{-e} = \omega_0$. □

Lemma 8.8 (low c)**.** *Let $e \ge 3$, $1 \le c \le (e+1)/2$, $l = e+1-2c$ and $q_0 = \lceil l/c \rceil + 1$. For general B and every integer $\nu > l$, the multiplication map $\mathcal{U}_{\nu,q_0} \otimes \omega_{-e} \to \omega_{\nu-e}$ is*

surjective, where $\mathcal{U}_{\nu,q_0}$ is spanned by the monomials x^iy^j with $i \le q_0$ and $i+j \le \nu$. In particular

$$\mathcal{U}_{e,q_0} \cdot W^\perp = \mathbf{1}^\perp, \qquad \mathcal{U}_{e-2,q_0} \cdot W^\perp = (\mathrm{ev}_B\, F_2)^\perp \quad (c \ge 2),$$

and (R1) holds whenever $L_A \supset \mathcal{U}_{e,q_0}$; for a first-row lower set this means $k \ge (q_0+1)(e+1) - q_0(q_0+1)/2$.

Proof. Put $\mathcal{N}_\nu = \omega_{\nu-e}$. Dualizing the low-$c$ resolution, $\mathcal{N}$ is generated by c elements g_i of degree 0 and l elements h_j of degree 1, with t relations of degree 2; the presentation matrix has c quadratic and l linear rows. We first exhibit a presentation with the required property, not necessarily of reduced points. Partition l into c integers a_i differing by at most one. For $a_i > 0$ take lines $\ell_0, \dots, \ell_{a_i}$ in general position avoiding P_∞, and let L_i be the $a_i \times (a_i+1)$ matrix whose row j has ℓ_{j-1} in column $j-1$ and $-\ell_j$ in column j. Its maximal minors are, up to sign, the products of all but one of the lines, so L_i drops rank exactly at the pairwise intersections of these lines, and only by one; choose the arrangements for different i with disjoint intersection sets. Add a quadratic first row to obtain a square block P_i with $F_i = \det \mathsf{P}_i$ nonzero at P_∞, the F_i pairwise without common components (possible since the linear minors have no common curve). Form $\mathcal{B} = [\mathrm{diag}(\mathsf{P}_1, \dots, \mathsf{P}_c) \mid v]$ with a column v, quadratic in the g rows and linear in the h rows, chosen so that at each of the finitely many rank-drop points of the L_i the linear part of v is outside the image. Then the $l \times (l+c+1)$ linear submatrix $\mathcal{L}$ has full row rank at every point of $\mathbb{P}^2$, and for generic v the maximal minors of $\mathcal{B}$ have no common factor (one of them is $\prod F_i$, and at a general point of a component of F_i some other minor is nonzero), so they generate an ideal of height two and Hilbert–Burch applies; the cokernel of $\mathcal{B}$ is the corresponding shifted canonical module.

Quotient $\mathcal{N}$ by the g_i. The quotient is presented by the linear matrix $\mathcal{L}$. For $l > 0$ choose $l+2$ general constant combinations of its columns; since $\mathcal{L}$ is surjective at every point and the bad choices have codimension three at each point of the two-dimensional $\mathbb{P}^2$, the resulting $l \times (l+2)$ matrix is surjective everywhere, its maximal minors have grade three, and by the Buchsbaum–Rim criterion [6, Introduction and Theorem 8.4] its cokernel Q' has the exact resolution

$$0 \to S(-l-3)^l \to S(-l-2)^{l+2} \to S(-2)^{l+2} \to S(-1)^l \to Q' \to 0.$$

Thus $\operatorname{reg} Q' \le l$ and $Q'_\nu = 0$ for $\nu > l$; the same holds for the quotient of $\mathcal{N}$, which is a quotient of Q'. For $l = 0$ the quotient is zero. Hence $\mathcal{N}_\nu$ is generated by the g_i for $\nu > l$. The adjugate of P_i gives $F_i g_i = 0$, and since $F_i(P_\infty) \ne 0$ its leading X term is a nonzero multiple of X^{a_i+2}; repeated substitution bounds the X exponent of the coefficients by $a_i + 1 \le q_0$, preserving degrees. This proves the surjectivity for the witness.

The presentation matrix of $\mathcal{N}$ is the transpose of a Hilbert–Burch matrix, so the witness is a point of the space $\mathcal{P}$ of Lemma 8.6. On the height-two locus the graded ranks of the resolutions are constant, so surjectivity in a fixed degree is the nonvanishing of a minor in the entries of H, a nonempty open condition on $\mathcal{P}$. At points of $\mathcal{P}_{\mathrm{red}}$ the residue description turns it into a rank condition on point evaluations of B. By Lemma 8.6(b), surjectivity in the finitely many degrees $l+1, \dots, e+1$ holds for general B; for $\nu \ge e+1$, $\omega_{\nu-e}$ has dimension b and multiplication by X_0 is an isomorphism, so the higher degrees follow without raising the X exponent. □

8.5. **The second condition by syzygies.**

Lemma 8.9 (column count)**.** *Let T be a first-row lower set of size k with $2e+1 \le k < D$, let m_0 be its last row index and h_0 the last j in that row, and let Δ_p be the set of pairs of total degree at most $p \ge 1$. Then*

$$\left|(\mathsf{T}+\Delta_p)\cup\Delta_{m_0+1+p}\right| = k+p(e+1)+\tfrac{p(p+1)}{2}+1+[h_0=0] \le k+p(e+1)+\tfrac{p(p+1)}{2}+2.$$

Proof. Here $1 \le m_0$ and $m_0+1 \le e$. The set T has full columns $i < m_0$ of height $e-i$ and a last column m_0 of height h_0. In $\mathsf{T}+\Delta_p$ the columns $i \le m_0+p-1$ have height $e+p-i$ (use a full column and the maximal vertical shift) and the last column m_0+p has height h_0. Adjoining Δ_{m_0+1+p} raises the height of that column to $\max(h_0,1)$, adds the column m_0+p+1 of height zero, and changes nothing else since $m_0+1 \le e$. Counting gives the formula. □

Proposition 8.10. *Let T be a first-row lower set with $2e+1 \le k < D$, L_A its monomial span, and suppose that the plane trace generators $f_1,\dots,f_t$ of $I_B(\le e+1)$ admit polynomial syzygies of degree at most p spanning their syzygy space over $\mathbb{C}(x,y)$, for general B, and that $t \ge 2$. If*

$$k + p(e+1) + \tfrac{p(p+1)}{2} + 2 \le b, \tag{8.6}$$

then (R2) holds for the Cartesian residual A_ζ of T, for suitable nodes ζ and general B.

Proof. Let $\ell' = m_0+1 \le e$, take the plane nodes $\xi_v = v$, and use the formal weight vector with $w_{e-\ell'+1} = -1$ and all other $w_v = 0$; then (8.4) is the projection $P_{\ell'}(f) = \sum_{i+j=\ell'} f_{ij}N_iN_j \bmod L_A$ onto Newton total degree ℓ'. Suppose $(D_{u_p}f_i(p) + \eta P_{\ell'}(f_i)(p))_{p\in B} \in \mathrm{ev}_B\, L_A$ for all i. Choose $g_i \in L_A$ with these values and put $G_i = g_i - \eta P_{\ell'}(f_i) \in L_A + F_{\ell'}$, so that $G_i(p) = D_{u_p}f_i(p)$. Differentiating a syzygy $\sum a_if_i = 0$ at $p \in B$ gives $\sum a_i(p)G_i(p) = 0$, and $\sum a_iG_i$ lies in $(L_A+F_{\ell'})F_p$, whose dimension is at most b by Lemma 8.9 and (8.6). For general B evaluation is injective on this fixed space, so $\sum a_iG_i = 0$ for all syzygies, and $G = \varrho f$ for a rational function ϱ. The f_i have no common factor: a common factor of degree j, $1 \le j \le e+1$, would contain at most $\binom{j+2}{2}-1$ of the general points, and dividing it out would place the t-dimensional space $I_B(\le e+1)$ in a space of dimension at most $\max(0, t-j(e+1-j)) < t$, or of dimension at most one if $j = e+1$. So ϱ is a polynomial, and since $\deg G_i \le e < e+1 = \deg f_i$, $G = 0$.

Let λ be the coefficient functional of $N_{\ell'}(x)N_0(y)$. Since $(\ell',0) \notin \mathsf{T}$, $P_{\ell'}(f) \in L_A$ implies $\lambda(f) = 0$. On the other hand λ does not vanish on $I_B(\le e+1)$ for general B: choosing $\dim F_{e+1}$ points whose evaluations form a basis of the dual, some b of them have evaluation span not containing λ. If $\eta \ne 0$ then $P_{\ell'}(f_i) = g_i/\eta \in L_A$ for all i, contradicting $\lambda \ne 0$ on $I_B(\le e+1)$. So $\eta = 0$, $g_i = 0$ and $D_{u_p}f_i(p) = 0$. Finally, at a general point p the derivatives of $I_B(\le e+1)$ span the cotangent space: the polynomials of degree at most $e+1$ vanishing to order two at p have codimension three, and $b-1$ further general points impose independent conditions on them because $t \ge 2$. Hence $u = 0$. Passing from the formal weights to actual nodes as explained after Lemma 8.5, (R2) holds for some A_ζ, and then on a nonempty open set. □

Lemma 8.11. *Let $c \ge 2$.*

(a) *If $l = e+1-2c \ge 0$, the syzygies of the f_i are spanned by the columns of the Hilbert–Burch matrix, of degrees one and two; so $p = 2$.*

(b) *If $l < 0$ and $t = e + 2 - c \geq 3$, put $q = 2c - e - 1$, $h'_0 = \lfloor (t-1)/2 \rfloor$ and $\mathsf{h} = \lceil q/h'_0 \rceil$. Then the syzygies among $f_1, \dots, f_t$ of degree at most $p = \mathsf{h} + 2$ span their syzygy space over the fraction field.*

Proof. (a) is the low-c resolution. In (b) the Hilbert–Burch matrix is a $t \times c$ quadratic block U over a $q \times c$ linear block V, with $t + q = c + 1$. The block V has rank q: otherwise its kernel would have dimension at least t, U would be injective on it, and its image would be annihilated by the nonzero row $(f_1, \dots, f_t)$. Syzygies among the f_i alone are $U\kappa$ with $\kappa \in \ker V$. For general V the kernel of V is spanned over the fraction field by polynomial vectors of degree h: for a witness, partition q into h'_0 balanced parts a_i and take a block diagonal matrix with $a_i \times (a_i + 2)$ blocks, everywhere of full rank on $\mathbb{P}^2$, adding a zero column if $t - 1$ is odd; by the Buchsbaum–Rim complex each block has kernel generated in coefficient degree $a_i \leq \mathsf{h}$, and empty blocks contribute free vectors. Surjectivity of the coefficient-degree-h kernel map is open, a fraction-field spanning minor persists on an open set, and this open set meets the plane point presentations. □

Proposition 8.12 (low-c and high-c transitions)**.** *Let $e \geq 3$ and $c \geq 2$.*

(a) *If $l = e + 1 - 2c \geq 0$, put $q_0 = \lceil l/c \rceil + 1$, $p = 2$ and $k_{\mathrm{req}} = (q_0 + 1)(e + 1) - q_0(q_0 + 1)/2$.*

(b) *If $l < 0$ and $t \geq 3$, put p as in Lemma 8.11(b) and $k_{\mathrm{req}} = 2e + 1$.*

Let s' be a position such that a general configuration in degree e and position s' has $T^1_{-1} = 0$ (so that $s' \geq e + 3$), and suppose

$$p(e+1) + \tfrac{p(p+1)}{2} + 2 - c \leq s' \leq D - k_{\mathrm{req}}. \tag{8.7}$$

Then a general configuration in degree $e + 1$ and position $s = s' + c$ has $T^1_{-1} = 0$.

Proof. Let T be the first-row lower set of size $k = D - s' \geq k_{\mathrm{req}} \geq 2e + 1$. By Lemmas 8.8 and 8.7, (R1) holds for general B. The lower bound in (8.7) is (8.6), so Proposition 8.10 and Lemma 8.11 give (R2); here $t \geq 2$ in case (a) because $t = l + c + 1$, and $k < D$ because $s' > 0$. Corollary 8.4 applies. □

8.6. **The sparse transition.**

Proposition 8.13 (sparse transition)**.** *Let $c \geq 2$ and $e \geq 2c^2 + 2c - 3$, and put $l = e + 1 - 2c$, $a_0 = \lfloor l/(c+1) \rfloor$, $q_0 = \lceil l/c \rceil + 1$ and*

$$K_{\min} = (q_0 + 1)(e - 1) - \tfrac{q_0(q_0+1)}{2} + 2e - 2 - 2q_0.$$

If a general configuration in degree e and position s' has $T^1_{-1} = 0$, and $e + c + 1 \leq s' \leq D - K_{\min}$, then a general configuration in degree $e + 1$ and position $s' + c$ has $T^1_{-1} = 0$.

The residual. Let $\mathbb{D}_0 = \{a_0, 2a_0 + 1, \dots, (c+1)a_0 + c\}$, a subset of $\{2, \dots, e-1\}$. Let $\mathsf{T}_{\max}$ consist of all pairs of degree at most $e-1$ except the $c+1$ pairs $(e-1-\delta, \delta)$ with $\delta \in \mathbb{D}_0$, together with $(e, 0)$; it is a lower set of size $D - e - c - 1$. Let $\mathsf{T}_{\min}$ be the union of the pairs of degree at most $e - 2$ with $i \leq q_0$, of the axis $(i, 0)$, $i \leq e$, and of the pairs $(i, 1)$, $i \leq e - 2$; it is a lower set of size $K_{\min}$ contained in $\mathsf{T}_{\max}$. Every size between $K_{\min}$ and $D - e - c - 1$ is realized by a lower set T with $\mathsf{T}_{\min} \subset \mathsf{T} \subset \mathsf{T}_{\max}$; this is exactly the range $e + c + 1 \leq s' \leq D - K_{\min}$. Use the Cartesian residual A_ζ of T.

The binary data. Write the low-c Hilbert–Burch matrix as $[\mathsf{A} \mid \mathsf{B}_q]$, with l linear and c quadratic columns, and let a subscript 0 denote restriction to the line

$X_0 = 0$. Let $\Lambda_{e+1} \subset \mathbb{C}[X,Y]_{e+1}$ be the t-dimensional space of leading forms of $I_B(\le e+1)$, with c-dimensional quotient Q. We use Lemma 8.6 and the following open conditions on the binary restriction, each shown to be nonempty by a witness.

(B1) *Admissibility*: Λ_{e+1} is base-point-free, that is, the forms in Λ_{e+1} have no common zero on $\mathbb{P}^1$, and $\mathbb{C}[X,Y]_1\Lambda_{e+1} = \mathbb{C}[X,Y]_{e+2}$. A witness is the monomial space whose quotient positions (exponents of Y) are $1, 3, \dots, 2c-1$.

(B2) *Detection*: the maps $p_2 \mapsto ([X^{e-1}p_2], [X^{e-2}Yp_2]) \in \mathsf{Q}^2$ on binary quadrics and $p_1 \mapsto [X^e p_1] \in \mathsf{Q}$ on binary linear forms are injective. A witness is any Λ_{e+1} on which the two coefficient functionals $c_0 + c_2$ and c_1 vanish (c_j the coefficient of $X^{e+1-j}Y^j$): for $p_2 = \alpha X^2 + \beta XY + \gamma Y^2$ they read $(\alpha+\gamma, \beta)$ on $X^{e-1}p_2$ and (β, α) on $X^{e-2}Yp_2$, and for $p_1 = \delta X + \epsilon Y$ they read (δ, ϵ) on $X^e p_1$.

(B3) *The rank conditions in the proof of (R2) below*: $\mathsf{A}_0^{\mathsf{t}}$ has no constant kernel vector, $\pi_{\mathbb{D}_0}$ is injective on the kernel of $\mathsf{A}_0^{\mathsf{t}}$ in degree $e-1$, and the matrix $[\pi_{\mathbb{D}_0} T_0 \mid \pi_{\mathbb{D}_0} f_2]$ has rank $ct+1$. A witness is the binary chain matrix A_0 with $c+1$ blocks of balanced lengths $a_i \ge a_0 \ge 1$ summing to l, together with the bidiagonal matrix B_0 and the lift described there.

Conditions (B1) and (B2) are open in the Grassmannian $\mathrm{Gr}(t, \mathbb{C}[X,Y]_{e+1})$, which is irreducible, so they hold simultaneously on a nonempty open subset. On the open set (B1) the Hilbert numerator of the binary ideal generated by Λ_{e+1} yields a minimal binary Hilbert–Burch presentation with l linear and c quadratic columns; placing it in $\mathcal{P}$ with entries independent of X_0 gives a point at which the minors expressing (B1) and (B2) do not vanish. For (B3) the chain witness is realized inside $\mathcal{P}$: the quadratic entries that pair with the kernel vector v_i of a chain of length a_i span all binary forms of degree $a_i + 2$, so the bidiagonal matrix below is realized by actual quadratic columns, and the linear entries span the binary forms of degree $a_i + 1$, so the lift $X_0\mathsf{B}_1$ used below is realized as well. Each of (B1)–(B3) is therefore a nonempty open condition on $\mathcal{P}$, and by Lemma 8.6, together with the finitely many open conditions on point evaluations used below, they hold for general B.

Proof of Proposition 8.13: condition (R1). By Lemma 8.8 and $L_A \supset \mathcal{U}_{e-2,q_0}$, a vector w with $wL_B \subset W$ lies in $(\mathcal{U}_{e-2,q_0} \cdot W^{\perp})^{\perp} = \mathrm{ev}_B F_2$, so w is the evaluation of a polynomial p of degree at most two. If $g \in L_A$ and pg has degree $e+1$, then $w \cdot g(B) \in W$ means that pg agrees on B with a polynomial of degree at most e, so the homogeneous part of degree $e+1$ of pg lies in Λ_{e+1}. Testing with $g = x^{e-1}$ and $g = x^{e-2}y$, which lie in $\mathsf{T}_{\min}$, condition (B2) kills the quadratic part of p, and testing with $g = x^e$ kills its linear part. So w is constant. □

Proof of Proposition 8.13: condition (R2). Two degenerations. Take the plane nodes $\xi_v = \epsilon v$ and the formal weights $w_2 = -1$, $w_v = 0$ for $v \ne 2$. By Lemma 8.5, which holds for every $\epsilon \ne 0$, the normal column (8.4) is then the projection $P_{e-1}^{(\epsilon)}(f) = \sum_{i+j=e-1} f_{ij} N_i N_j \bmod L_A$ onto Newton total degree $e-1$ for the scaled Newton basis. As $\epsilon \to 0$ the scaled Newton basis and the Newton coefficients of f tend polynomially to the monomial basis and the monomial coefficients, while L_A is the same monomial space for all ϵ; hence $P_{e-1}^{(\epsilon)}$ tends to the projection P_{e-1}^{hom} onto the homogeneous part of degree $e-1$, and the (R2) matrix with this normal column is polynomial in ϵ. It therefore suffices to show that the (R2)

operator with normal column $P_{e-1}^{\hom}$ is injective; then a $(2b+1)$-minor is nonzero for some $\epsilon \neq 0$, it is a nonzero linear polynomial in w for these plane nodes, and it is nonzero for suitable pairwise distinct nonzero normal nodes ζ, by the remark after Lemma 8.5.

The homogeneous operator. Suppose $D_u f_i(B) + \eta P_{e-1}^{\hom}(f_i)(B) \in \mathrm{ev}_B L_A$, choose $g_i \in L_A$ and put $G_i = g_i - \eta P_{e-1}^{\hom}(f_i) \in F_{e-1} + \mathbb{C}x^e$, so $G_i(B) = D_u f_i(B)$. The space $(F_{e-1}+\mathbb{C}x^e)F_1$ has dimension $D+2 \le b$, so differentiating the linear syzygies gives $\mathsf{A}^{\mathsf{t}}G = 0$ identically. The degree e part of G is αx^e with $\alpha \in \mathbb{C}^t$; since $\mathsf{A}_0^{\mathsf{t}}$ has no constant kernel vector by (B3) (at the chain witness because every chain has positive length), $\alpha = 0$ and the homogenization of G (in degree e) is divisible by X_0.

The quadratic syzygies give $\mathsf{B}_q^{\mathsf{t}}G \in I_B$ in degree $e+2$, so there is a $t \times c$ matrix C of linear forms with $\mathsf{B}_q^{\mathsf{t}}G + \mathsf{C}^{\mathsf{t}}f = 0$. Let $\delta f(\mathsf{C})$ be the derivative of the maximal minors of $[\mathsf{A} \mid \mathsf{B}_q + \epsilon\mathsf{C}]$ at $\epsilon = 0$; it has degree e, and $\mathsf{A}^{\mathsf{t}}\delta f = 0$, $\mathsf{B}_q^{\mathsf{t}}\delta f + \mathsf{C}^{\mathsf{t}}f = 0$. The polynomial kernel of $[\mathsf{A} \mid \mathsf{B}_q]^{\mathsf{t}}$ is generated by the primitive vector f of degree $e+1$, so $G = \delta f(\mathsf{C})$. At $X_0 = 0$ these derivative minors vanish, so each column of C_0 is a linear syzygy of the binary f_0, hence equals A_0 times a constant vector by the binary Hilbert–Burch theorem; subtracting this column-operation gauge we get $\mathsf{C} = X_0\mathsf{C}^0$ with C^0 constant. Thus

$$G = X_0\,\delta f(\mathsf{C}^0), \qquad (G/X_0)|_{X_0=0} = T_0(\mathsf{C}^0) := \delta f_0(\mathsf{C}^0), \tag{8.8}$$

and T_0 is injective on the ct-dimensional space of constant C^0, since a kernel vector would give constant syzygies of the minimal binary ideal. Let f_2 be the coefficient of X_0^2 in f and $\pi_{\mathbb{D}_0}$ the vector of coefficients of the deleted monomials $x^{e-1-\delta}y^{\delta}$, $\delta \in \mathbb{D}_0$, in each component. Comparing the parts of degree $e-1$ of $G = g - \eta P_{e-1}^{\hom}(f)$, and using that $g \in L_A$ has no deleted coefficients, we get

$$\pi_{\mathbb{D}_0}\big(T_0(\mathsf{C}^0) + \eta f_2\big) = 0. \tag{8.9}$$

The detector is injective on the binary kernel. At the chain witness of (B3), the kernel of $\mathsf{A}_0^{\mathsf{t}}$ on $\mathbb{C}[X,Y]_{e-1}^t$ is $\bigoplus_i v_i\mathbb{C}[X,Y]_{e-1-a_i}$ with $v_i = (X^{a_i}, X^{a_i-1}Y, \dots, Y^{a_i})$ on the i-th block. Deleting the coefficient of Y-degree h in all components of one block detects the coefficients of the cofactor in the degrees $[\mathsf{h}-a_i, \mathsf{h}]$. For the minimal length a_0 the $c+1$ intervals attached to $\mathbb{D}_0$ are consecutive from 0 to $(c+1)a_0 + c$. Writing $e = (c+1)a_0 + 2c - 1 + \varrho$ with $0 \le \varrho \le c$, the required last degree is $e - 1 - a_0 = ca_0 + 2c - 2 + \varrho$, which is at most $(c+1)a_0 + c$ because $a_0 \ge 2c - 2 \ge c - 2 + \varrho$; this uses $e \ge 2c^2 + 2c - 3$. Longer chains only widen the intervals.

A normal witness. Let $w_i = a_i + 2$ and take, in the quotient by the chain columns, the $(c+1) \times c$ bidiagonal quadratic matrix R with $\mathsf{R}_{i,i} = X^{w_i}$ $(0 \le i < c)$ and $\mathsf{R}_{i,i-1} = Y^{w_i}$ $(1 \le i \le c)$. Its omitted-row minors are, up to sign, $F_i = \prod_{h<i} X^{w_h} \prod_{h>i} Y^{w_h}$, and the component of f_0 in the block i is v_iF_i; since F_0 and F_c are pure powers of Y and X, this is a genuine binary Hilbert–Burch matrix, realized by quadratic columns B_0. A constant variation changes an entry in row u and column v; deleting row 1, a nonzero cofactor has $u \le v < 1$ or the reverse, and its ratio to F_1 does not decrease the Y exponent (for $u < 1$) or the X exponent (for $u > 1$). Hence the monomial $F_1/(XY)$ does not occur in the constant tangent image in the component 1. Now take the lift $\mathsf{B}_q = \mathsf{B}_0 + X_0\mathsf{B}_1$ whose only nonzero entries of B_1 replace $\mathsf{R}_{0,0}$ by X^{w_0-1} and $\mathsf{R}_{2,1}$ by Y^{w_2-1}, realized by linear entries of the actual blocks; the coefficient of X_0^2 in the component 1 is exactly

$F_1/(XY)$. Therefore $f_2 \notin \operatorname{im} T_0$, and since $\pi_{\mathbb{D}_0}$ is injective on the binary kernel, the matrix $[\pi_{\mathbb{D}_0} T_0 \mid \pi_{\mathbb{D}_0} f_2]$ has rank $ct+1$. This lift need not define reduced points; it only certifies a nonzero polynomial minor, which persists on an open subset of the irreducible presentation space, where it meets the open sets of reduced point sets, of the low-c resolution and of the injectivity of evaluation on $(F_{e-1} + \mathbb{C}x^e)F_1$.

At such a presentation (8.9) forces $\eta = 0$ and $\mathsf{C}^0 = 0$, then $G = 0$ by (8.8), $g = 0$ and $D_u f_i(B) = 0$, and $u = 0$ as in Proposition 8.10.

By the first step this gives (R2) for an actual Cartesian residual $A_{\xi,\zeta}$, and Corollary 8.4 gives the proposition. □

9. A rational seed and the propagation to large degrees

9.1. **The seed $(d, s) = (11, 14)$.**

Proposition 9.1. *For $(d, s) = (11, 14)$, $r = 300$, a general configuration satisfies* $\operatorname{rank} J_P = 896 = 3r - 4 = sm$, *so* $T^1_{-1} = 0$.

Proof. Let $n = 13$. Appendix C describes thirteen planes $\ell_i = 1 + \nu_i \cdot x$ with rational normals, points p_i with $\ell_i(p_i) = 0$ and directions w_i with $\nu_i \cdot w_i = 1$; the third coordinates are fixed by $\nu_{i,3} = (-1 - \nu_{i,1}p_{i,1} - \nu_{i,2}p_{i,2})/p_{i,3}$ and $w_{i,3} = (1 - \nu_{i,1}w_{i,1} - \nu_{i,2}w_{i,2})/\nu_{i,3}$, so these identities hold over $\mathbb{Q}$. Every three normals are independent and no four planes meet. Let A be the 286 triple points and, for $\lambda \in \{1, 2\}$, let B_λ consist of the points $p_i + \lambda w_i$ and the origin. One checks over $\mathbb{Q}$ that each triple point lies on exactly its three planes, that no point of B_λ lies on a plane, and that the 300 points are distinct and span $\mathbb{A}^3$.

By Proposition 5.3, after choosing a basis $\mathcal{K}$ of $\ker \mathcal{E}$ ($\mathcal{E}$ the 14×78 edge-product evaluation matrix), J_P is the 896×900 matrix with rows (j, α) and entries

$$J_{(j,\alpha),(T,k)} = -V_T(b_j)\mathcal{K}_{T\smallsetminus k,\alpha},$$

$$J_{(j,\alpha),(j,\beta)} = \sum_{ab} \mathcal{K}_{ab,\alpha}\partial_\beta E_{ab}(b_j), \qquad J_{(j,\alpha),(j',\beta)} = 0 \ \ (j' \neq j).$$

All coordinates and the pivots used to compute $\mathcal{K}$ have denominators prime to $P = 32003$. Modulo P one finds $\operatorname{rank} \mathcal{E} = 14$ and $\operatorname{rank} J = 896$, with a selected 896×896 minor whose determinant is 19954 for $\lambda = 1$ and 31377 for $\lambda = 2$ modulo P; either configuration alone suffices for the argument, and both are recorded in the supplementary material (Appendix E). The same pivots define a rational basis $\mathcal{K}$ reducing to the modular one, so the corresponding rational minor has a denominator prime to P and a nonzero reduction; it is therefore nonzero. Hence $\operatorname{rank} J_P = 896$ over $\mathbb{Q}$ and over $\mathbb{C}$, and the kernel consists of the four motions. Lemma 2.5 gives the statement for general configurations. □

9.2. **Finite propagation up to degree 84.** For each degree d we write $[\sigma_d^-, \sigma_d^+]$ for an interval of positions s on which a general configuration has already been shown to satisfy $\operatorname{rank} J_P = 3r - 4$. Starting from Proposition 9.1, we apply the transitions of Propositions 8.13 and 8.12 degree by degree. At degree e the input is an interval $[\sigma_e^-, \sigma_e^+]$ of positions already proved; for each admissible c and each transition, its residual interval is intersected with $[\sigma_e^-, \sigma_e^+]$ and shifted by c. This is a finite integer computation, and each of its steps is an instance of the cited propositions, whose hypotheses are inequalities between explicit integers. The resulting

188 transitions are listed in Appendix D. Their lower endpoints are always $2d-8$, and the last row is

$$d = 84: \qquad 160 \le s \le 2640, \qquad M_{84} - 2640 = 1015 = 7\cdot 84 + 427. \tag{9.1}$$

9.3. **The analytic induction.**

Theorem 9.2. *For every $d \ge 84$ and $2d-8 \le s \le M_d - 7d - 427$, a general configuration satisfies* $\operatorname{rank} J_P = 3r-4$.

Proof. The case $d = 84$ is (9.1). Let $e \ge 84$ and assume the statement for $[\sigma_e^-, \sigma_e^+] = [2e-8, D-7e-427]$, $D = M_e$. We cover $[2(e+1)-8, M_{e+1}-7(e+1)-427] = [2e-6, \sigma_e^+ + e - 5]$ by three transitions.

Sparse, $c = 2$. Here $e \ge 9$ and $q_0 = \lfloor e/2 \rfloor$. Writing K_2 for $K_{\min}$, for $e = 2h$ we have $K_2 = (3h^2+5h-6)/2$ and $D - K_2 - (2e-2) = (h-3)(h-4)/2$, and for $e = 2h+1$, $K_2 = (3h^2+7h)/2$ and $D - K_2 - (2e-2) = (h-2)(h-3)/2$. In either case $K_2 \ge 3e^2/8 \ge 7e + 427$, so $D - K_2 \le \sigma_e^+$; also $\sigma_e^- \ge e+3$ and $D - K_2 \ge \sigma_e^-$. Proposition 8.13 gives $I_2 = [2e-6, D-K_2+2]$.

Low c, $c_H = \lfloor (e+1)/2 \rfloor$. Then $l \in \{0,1\}$, $p = 2$ and $k_{\mathrm{req}} \le 3e$, which is at most $k = D - s'$ for all $s' \le \sigma_e^+$. The lower bound of (8.7) is $2e + 7 - c_H \le 2e - 8$ since $c_H \ge 15$. Proposition 8.12 gives $I_H = [2e - 8 + c_H, \sigma_e^+ + c_H]$. There is no gap after I_2: with $\Delta = D - K_2 - (2e-2)$ we need $\Delta \ge c_H - 9$, and $\Delta - c_H + 9$ equals $(h^2 - 9h + 30)/2$ for $e = 2h$ and $(h^2 - 7h + 22)/2$ for $e = 2h+1$, both positive.

High c, $c_7 = e-5$. Then $t = 7$, $q = e-11$, $h_0' = 3$, $p = \lceil (e-11)/3 \rceil + 2 \le e/3 - 1$, $k_{\mathrm{req}} = 2e+1$, and the image is $I_7 = [T_e, \sigma_e^+ + e - 5]$ with $T_e = p(e+1) + p(p+1)/2 + 2 \ge 3e - 13$. Since $T_e \le 7e^2/18 - 5e/6 + 1$ and $\sigma_e^+ + c_H \ge e^2/2 - 5e - 426$,

$$(\sigma_e^+ + c_H) - (T_e - 1) \ge e^2/9 - 25e/6 - 426,$$

which equals 8 at $e = 84$ and increases. So I_7 meets I_H, and $I_2 \cup I_H \cup I_7 \supset [2e-6, \sigma_e^+ + e - 5]$. □

10. Proof of the main theorems

Proposition 10.1. *A general configuration satisfies* $\operatorname{rank} J_P = 3r-4$ *in each of the following cases:*

(a) *$d \ge 27$ and $d+3 \le s \le 2d-24$;*
(b) *$d \ge 28$ and $d+3 \le s \le 3d-52$.*

Proof. (a) Put $\mathtt{t} = s - d - 3 \ge 0$ and $d_0 = d - \mathtt{t} = 2d + 3 - s \ge 27$. Start from (d_0, d_0+3), given by Theorem 7.8, and apply the sparse transition with $c = 2$ exactly $\mathtt{t}$ times; for $\mathtt{t} = 0$ there is nothing to do. Before the j-th step the residual has degree $e = d_0 + j - 1 \ge 27$ and position $s' = d_0 + 2j + 1$, so $s' - (e+3) = j - 1 \ge 0$ and $(2e-2) - s' = d_0 - 5 \ge 22$. By the formulas for $D - K_2 - (2e-2)$ in the proof of Theorem 9.2, $D - K_2 \ge 2e - 2 \ge s'$, and $e \ge 9$; so the hypotheses of Proposition 8.13 hold, and the residual is the position reached at the previous step. The final position is s.

(b) At $d = 28$ the interval $[31, 32]$ is covered by (a). Suppose (b) holds in degree $e \ge 28$, and apply the sparse transition with $c = 3$ to $s' \in [e+4, 3e-52]$; here $e \ge 21$ and the lower bound is $e + c + 1$. With $q_0 = \lceil (e-5)/3 \rceil + 1 \le (e+1)/3$ and $K_{\min}$ increasing in q_0 in this range,

$$D - K_{\min} \ge \frac{2e^2 - 5e + 47}{9} \ge 3e - 52,$$

the last difference being $(2X^2 + 80X + 1187)/9$ with $e = X + 28$. This gives $[d+6, 3d-52]$ in degree $d = e+1$, and (a) gives $[d+3, 2d-24] \supset [d+3, d+5]$ for $d \geq 29$. □

Proof of Theorems 1.1, 1.2 and 1.3. The rank assertions are Theorem 7.8, Proposition 10.1 and, for $d \geq 84$, the union of Proposition 10.1(b) with Theorem 9.2: the two intervals overlap because $(3d-52)-(2d-9) = d-43 > 0$, so together they cover $d+3 \leq s \leq M_d - 7d - 427$. In each case the open set of Lemma 2.5 is nonempty, and Proposition 2.6 translates the rank statement into the dimension of the image of Φ. By Lemma 2.4, $T^1_{-1} = 0$. In all cases $m \geq 4$: at $s = d+3$, $m = (d^2+d-4)/2$; in the band (a), $m \geq (d^2 - d + 50)/2$; in the band (b), $m \geq (d^2 - 3d + 106)/2$; in the broad range, $m \geq 7d + 427$. Propositions 3.1 and 3.2 give $T^1_j = 0$ for all $j < 0$ on the intersection with the open set where Ψ_P is injective, which is nonempty by irreducibility. The cones are reduced, have an isolated singularity at the vertex and no equations of degree less than $d \geq 27$, so Proposition 4.1 excludes algebraic and analytic smoothings. The statement at $s = d+2$ is Theorem 5.6.

For the density, every integer $r \geq 1$ can be written as $L_d + s$ with $0 \leq s < M_d$, since $L_{d+1} - L_d = M_d$. In the layer $d \geq 84$ the omitted positions number $M_d - (M_d - 8d - 429) = 8d + 429$. Up to the layer D the omissions are $O(D^2)$ among $L_{D+1} = \Theta(D^3)$ integers, and a partial last layer below a bound R contributes $O(D^2) = O(R^{2/3})$ further omissions. Hence the set of branch numbers covered has natural density one. □

Remark 10.2. At $d = 84$ the covered interval is $87 \leq s \leq 2640$, that is $102427 \leq r \leq 104980$; the broad range of Theorem 9.2 alone begins at $s = 160$. The first case of Theorem 1.2(a) is $d = 27$, $r = 3684$, with $\operatorname{rank} J_P = 11048$. The uniform part of the lower-boundary proof starts at $d = 87$, $r = 113654$; the degrees $27 \leq d \leq 86$ are covered by the finite certificates of Appendix A.

Remark 10.3. The same arguments give further finite ranges: Proposition 9.1 at $(d, s) = (11, 14)$ and Table 11 for $12 \leq d \leq 83$ provide positions $\sigma^-_d \leq s \leq \sigma^+_d$ (with $\sigma^-_d = 2d-8$) at which a general configuration has $\operatorname{rank} J_P = 3r - 4$. In all of them $m \geq 4$ (in fact $m \geq 64$), so Propositions 3.1, 3.2 and 4.1 give $T^1_j(C(Z)) = 0$ for all $j < 0$ and the absence of smoothings for general Z at these positions. We do not claim the lower endpoint $2d - 8$ in degrees outside this table other than through Theorem 9.2.

Remark 10.4. The results leave open the positions $M_d - 7d - 426 \leq s \leq M_d - d - 6$ near the upper end of the window (2.7) ($6d + 421$ positions in each degree $d \geq 84$), the degrees below those covered, the question whether J_P has the expected rank $\min(3r - 4, sm)$ at a general configuration for every position (d, s), and the smoothability of the cones when $T^1_{-1} \neq 0$. In particular Theorem 5.6 makes no assertion about smoothability at $s = d + 2$.

Appendix A. Finite characteristic-zero certificates

This appendix describes the computation used in the proof of Theorem 7.8 for $29 \leq n \leq 88$, that is, $27 \leq d \leq 86$; it covers all 3510 characters of these 60 degrees.

Data. The coefficient vector is $(a_1, a_3, a_7) = (2, 3, -6)$ throughout, so $f = 1 + 2u + 3u^3 - 6u^7$, which satisfies the last condition of (6.1) and has no root on the unit circle other than 1 (Lemma 6.1). For each n a prime $p \equiv 1 \pmod{n}$ and an element

$\omega \in \mathbb{F}_p$ of exact order n are chosen; the order is checked at every prime divisor of n. For $n = 87$ the first suitable prime has a vanishing Fourier denominator, and $p = 32713$ is used instead.

Checks of the denominator conditions. The computation verifies, modulo p, that $f(\omega^j) \neq 0$ for $j \neq 0$ (hence $\Pi \neq 0$), that every λ_j is nonzero, that $f'(1) = -31 \neq 0$, that 2, n and all incidence pivots are nonzero, and that the Euclidean algorithm gives $\gcd(g, u^6 g(u^{-1})) = 1$ for $g = f/(1-u)$. It also records a nonzero minor of the auxiliary evaluation matrix on the $n+1$ edges $\{0, j\}$ $(1 \leq j < n)$, $\{1, 2\}$, $\{1, 3\}$; after division of the assigned rows by Π its endpoint entries are $1/f(\omega^{a-b})$ and its origin row consists of ones.

Rank statements. For each character χ the computation forms the origin system $\mathcal{P}_\chi$ of Proposition 6.6 on the six unknowns of Corollary 6.5 (seven in a global character, with c, η and ρ' or κ_e retained), the three moment rows $\mathcal{W}_\chi$ from (7.13), and the detector $\mathcal{O}_\chi$ from (7.11) on all indices outside U_χ, including the collision sets. It certifies

$$\operatorname{rank} \mathcal{P}_\chi = \begin{cases} 4 & \text{ordinary and global } \chi, \\ 5 & \text{quadratic } \chi, \end{cases} \qquad \operatorname{rank} \begin{pmatrix} \mathcal{P}_\chi \\ \mathcal{O}_\chi \mathcal{W}_\chi \end{pmatrix} = 6,$$

each rank being witnessed by a selected square minor, with its row and column indices and its nonzero determinant modulo p. Since $\operatorname{rank}[\mathcal{P}; \mathcal{X}] - \operatorname{rank} \mathcal{P}$ is the rank of $\mathcal{X}$ on $\ker \mathcal{P}$, this gives opening ranks 2, 1 and 2 in the three types of characters without using the injectivity of $\mathcal{O}_\chi$, which may fail for small n (for instance for $n = 29$, $\chi = -14$ its rank outside U_χ is one, which suffices for that one-dimensional special kernel).

Characteristic zero. Let $\mathbb{Z}[\zeta_n]$ be localized at n and at all coefficients and denominators occurring in the computation. Since $p \nmid n$ and ω has order n, $\zeta_n \mapsto \omega$ defines a ring homomorphism to $\mathbb{F}_p$, under which these denominators map to the nonzero recorded residues. Every selected minor is the image of the corresponding minor over $\mathbb{Q}(\zeta_n)$, which is therefore nonzero. The modular residues are not used as rational coordinates: the characteristic-zero witness is the cyclotomic point, and the transfer to genuine configurations is that of Lemma 5.5.

Result. Table 4 lists the data. In every degree the special kernel has dimension $2n - 2$, the opening has rank $2n - 6$, the remaining kernel has dimension four, and $\operatorname{rank} J_P = 3\binom{n}{3} + 3n - 1 = 3r - 4$.

Appendix B. Leading coefficients for the boundary characters

This appendix lists the data used in the proof of Proposition 7.7. Throughout, $n = 6\mathsf{N} + \mathsf{s}$ with $0 \leq \mathsf{s} \leq 5$ and $\mathsf{N} \geq 14$, and the *order* of a Laurent series in ϵ is the exponent of its first nonzero coefficient; in each case all coefficients of lower order vanish.

Normalizations on the six-small-pole line. For $h \in \{2, \dots, 10, 14\}$ the cofactor kernel vectors are those of Table 3: for each free column the free entry is the pivot minor, the pivot entries are the negatives of the minors with the corresponding pivot column replaced, and the common power of ϵ is divided out. Table 5 gives, for the quadratic characters, the leading term of $\epsilon^{\mathsf{N}} \mathsf{W}_{11}$ on the unique cofactor vector. Table 6 gives, for $h \in \{3, 5, 7, 9\}$, the leading term of the determinant of the 2×2 matrix $(\epsilon^{\mathsf{N}} \mathsf{W}_A(K_i))$, where K_0, K_1 are the first two free cofactor vectors and

TABLE 4. Finite certificates for $29 \le n \le 88$: $n = d + 2$, $r = \binom{n}{3} + n + 1$, the prime p, the element $\omega \in \mathbb{F}_p$ of order n, and $\operatorname{rank} J_P = 3r - 4$. In every case the special kernel has dimension $2n - 2$ and the opening rank is $2n - 6$.

n	d	r	p	ω	$3r-4$	n	d	r	p	ω	$3r-4$
29	27	3684	32191	18740	11048	59	57	32569	32569	6207	97703
30	28	4091	32191	205	12269	60	58	34281	32341	16632	102839
31	29	4527	32117	5220	13577	61	59	36052	32941	30384	108152
32	30	4993	32257	2292	14975	62	60	37883	32117	19658	113645
33	31	5490	32077	8638	16466	63	61	39775	32257	227	119321
34	32	6019	32029	25858	18053	64	62	41729	32257	624	125183
35	33	6581	32341	11264	19739	65	63	43746	33151	24488	131234
36	34	7177	32077	4077	21527	66	64	45827	32077	11439	137477
37	35	7808	32117	17815	23420	67	65	47973	32027	7157	143915
38	36	8475	32377	7278	25421	68	66	50185	32029	6271	150551
39	37	9179	32059	13433	27533	69	67	52464	32569	18953	157388
40	38	9921	32321	27732	29759	70	68	54811	32341	1612	164429
41	39	10702	32063	3482	32102	71	69	57227	32377	15197	171677
42	40	11523	32089	14261	34565	72	70	59713	32257	23701	179135
43	41	12385	32251	19979	37151	73	71	62270	32413	25819	186806
44	42	13289	32077	12182	39863	74	72	64899	32117	27518	194693
45	43	14236	32401	12258	42704	75	73	67601	32251	14038	202799
46	44	15227	32063	2282	45677	76	74	70377	32377	1183	211127
47	45	16263	32713	4574	48785	77	75	73228	32341	8220	219680
48	46	17345	32257	12683	52031	78	76	76155	32059	2712	228461
49	47	18474	32341	29183	55418	79	77	79159	32233	30818	237473
50	48	19651	32051	28637	58949	80	78	82241	32321	27827	246719
51	49	20877	32029	23450	62627	81	79	85402	32077	31795	256202
52	50	22153	32189	24068	66455	82	80	88643	32063	29394	265925
53	51	23480	32119	4239	70436	83	81	91965	32371	18905	275891
54	52	24859	32077	3568	74573	84	82	95369	32089	2318	286103
55	53	26291	32341	24631	78869	85	83	98856	33151	9733	296564
56	54	27777	32089	12505	83327	86	84	102427	32251	25757	307277
57	55	29318	32377	24283	87950	87	85	106083	32713	9871	318245
58	56	30915	32191	22087	92741	88	86	109825	32297	10780	329471

A runs over the indicated pair of rows. Table 7 gives the two leading terms used for $\chi = -1$, with

(B.1) $\mathsf{p}_3(\mathsf{N}) = 24\mathsf{N}^7 + 125\mathsf{N}^6 + 69\mathsf{N}^5 - 475\mathsf{N}^4 - 549\mathsf{N}^3 + 350\mathsf{N}^2 + 17736\mathsf{N} - 20160.$

For $\chi = 0$ the matrix M is the constant term of the 3×3 matrix whose (i,k) entry is ϵ^{-r_i} times the i-th row of $(\mathsf{W}_{11},\ \mathsf{W}_{13} - \mathsf{W}_{11},\ \mathsf{W}_{33} - 2\mathsf{W}_{13} + \mathsf{W}_{11})$ evaluated on the k-th vector of $(\epsilon K_0, \epsilon^2 K_1, \epsilon^{-1} K_2')$; the row weights (r_0, r_1, r_2) are given with each matrix.

The sign test. Each of the $36 + 24 + 12 + 6 = 78$ rational functions of N listed below (the coefficients in Tables 5, 6 and 7, and the six determinants $\det \mathsf{M}$) is written in lowest terms. After the substitution $\mathsf{N} = X + 14$ its numerator has all coefficients of one sign and a nonzero constant term, and its denominator has positive coefficients. Hence none of these functions vanishes for real $\mathsf{N} \ge 14$. This

is a statement about finitely many explicit polynomials in X, not a test at finitely many values of N.

Table 5: Quadratic characters $\chi = -h$: order and first nonzero coefficient of $\epsilon^{\mathsf{N}}\mathsf{W}_{11}$ on the cofactor kernel vector, for $n = 6\mathsf{N} + \mathsf{s}$.

h	s	order	coefficient
2	0	3	$\dfrac{2}{\mathsf{N}(\mathsf{N}-1)}$
2	1	3	$\dfrac{1}{\mathsf{N}-1}$
2	2	3	1
2	3	2	$\dfrac{120}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
2	4	2	$\dfrac{24}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
2	5	2	$\dfrac{6}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)}$
4	0	2	$\dfrac{48}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)}$
4	1	2	$\dfrac{12}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)}$
4	2	2	$\dfrac{4}{\mathsf{N}(\mathsf{N}-1)}$
4	3	2	$\dfrac{2}{\mathsf{N}-1}$
4	4	2	2
4	5	1	$\dfrac{240}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
6	0	2	$-\dfrac{48(\mathsf{N}+8)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
6	1	2	$-\dfrac{12(\mathsf{N}+6)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)}$
6	2	2	$-\dfrac{4(\mathsf{N}+4)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)}$
6	3	2	$-\dfrac{2(\mathsf{N}+2)}{\mathsf{N}(\mathsf{N}-1)}$
6	4	2	$-\dfrac{2\mathsf{N}}{\mathsf{N}-1}$
6	5	1	$-\dfrac{240}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
8	0	1	$-\dfrac{2(\mathsf{N}-2)}{\mathsf{N}(\mathsf{N}-1)}$
8	1	1	$-\dfrac{2(\mathsf{N}-2)}{\mathsf{N}-1}$
8	2	0	$-\dfrac{240}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$

h	s	order	coefficient
8	3	0	$-\dfrac{48(\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
8	4	0	$-\dfrac{12(\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
8	5	0	$-\dfrac{4(\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)}$
10	0	1	$-\dfrac{6(\mathsf{N}-2)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)}$
10	1	1	$-\dfrac{2(\mathsf{N}-2)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)}$
10	2	1	$-\dfrac{\mathsf{N}-2}{\mathsf{N}(\mathsf{N}-1)}$
10	3	1	$-\dfrac{\mathsf{N}-2}{\mathsf{N}-1}$
10	4	0	$-\dfrac{120}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
10	5	0	$-\dfrac{24(\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
14	0	3	$-\dfrac{2(3\mathsf{N}-8)}{(\mathsf{N}-2)(\mathsf{N}-1)}$
14	1	3	$-\dfrac{6\mathsf{N}-17}{\mathsf{N}-2}$
14	2	2	$-\dfrac{720}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
14	3	2	$-\dfrac{24(6\mathsf{N}-7)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
14	4	2	$-\dfrac{12(3\mathsf{N}-4)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)}$
14	5	2	$-\dfrac{6(2\mathsf{N}-3)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)}$

Table 6: Characters $\chi = -h$, $h \in \{3,5,7,9\}$: order and first nonzero coefficient of the 2×2 determinant of the rows $\epsilon^{\mathsf{N}}\mathsf{W}_A$ on the first two cofactor kernel vectors.

h	rows	s	order	coefficient
3	(P_{11}, P_{13})	0	6	$-\dfrac{6(\mathsf{N}-2)}{\mathsf{N}^2(\mathsf{N}-1)^2(\mathsf{N}+1)}$
3	(P_{11}, P_{13})	1	6	$-\dfrac{2(\mathsf{N}-2)}{\mathsf{N}(\mathsf{N}-1)^2}$
3	(P_{11}, P_{13})	2	5	$-\dfrac{120}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
3	(P_{11}, P_{13})	3	5	$-\dfrac{24(\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$

h	rows	s	order	coefficient
3	(P_{11}, P_{13})	4	4	$-\dfrac{720(\mathsf{N}-1)}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)^2(\mathsf{N}+4)}$
3	(P_{11}, P_{13})	5	4	$-\dfrac{48(\mathsf{N}-1)}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)}$
5	(P_{11}, P_{33})	0	6	$-\dfrac{2880}{\mathsf{N}^2(\mathsf{N}-1)^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)}$
5	(P_{11}, P_{33})	1	6	$-\dfrac{144}{\mathsf{N}^2(\mathsf{N}-1)^2(\mathsf{N}+1)^2(\mathsf{N}+2)}$
5	(P_{11}, P_{33})	2	6	$-\dfrac{12}{\mathsf{N}^2(\mathsf{N}-1)^2(\mathsf{N}+1)}$
5	(P_{11}, P_{33})	3	6	$-\dfrac{2}{\mathsf{N}(\mathsf{N}-1)^2}$
5	(P_{11}, P_{33})	4	6	$-\dfrac{1}{\mathsf{N}-1}$
5	(P_{11}, P_{33})	5	5	$-\dfrac{120}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
7	(P_{11}, P_{13})	0	4	$-\dfrac{120}{\mathsf{N}(\mathsf{N}-1)^2(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
7	(P_{11}, P_{13})	1	4	$-\dfrac{24}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)}$
7	(P_{11}, P_{13})	2	3	$-\dfrac{720}{\mathsf{N}^2(\mathsf{N}-1)(\mathsf{N}+1)^2(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
7	(P_{11}, P_{13})	3	3	$-\dfrac{48}{\mathsf{N}^2(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
7	(P_{11}, P_{13})	4	3	$-\dfrac{6(\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3+50\mathsf{N}^2-96\mathsf{N}+120)}{\mathsf{N}^2(\mathsf{N}-1)(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)(\mathsf{N}+4)}$
7	(P_{11}, P_{13})	5	3	$-\dfrac{2(\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3+50\mathsf{N}^2-216)}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
9	(P_{11}, P_{13})	0	3	$-\dfrac{36}{\mathsf{N}^2(\mathsf{N}-1)^2(\mathsf{N}+1)}$
9	(P_{11}, P_{13})	1	3	$-\dfrac{6}{\mathsf{N}(\mathsf{N}-1)^2}$
9	(P_{11}, P_{13})	2	3	$-\dfrac{3(\mathsf{N}+5)(\mathsf{N}^4+5\mathsf{N}^3+10\mathsf{N}^2+24)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
9	(P_{11}, P_{13})	3	2	$-\dfrac{360}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
9	(P_{11}, P_{13})	4	1	$-\dfrac{8640}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)^2(\mathsf{N}+4)}$
9	(P_{11}, P_{13})	5	1	$-\dfrac{432}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)}$

TABLE 7. The character $\chi = -1$: the first nonzero coefficient of $\epsilon^{\mathsf{N}}\mathsf{W}_{11}$ on K_0, and of the projected row W_* on K_*, with their orders; p_3 is the polynomial in (B.1).

s	order	$\epsilon^{\mathsf{N}}\mathsf{W}_{11}(K_0)$	order	$\mathsf{W}_*(K_*)$
0	3	$\dfrac{1}{\mathsf{N}-1}$	0	$\dfrac{24(6\mathsf{N}-7)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
1	3	1	0	$\dfrac{24(6\mathsf{N}-7)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
2	2	$\dfrac{120}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$	0	$\dfrac{24}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)}$
3	2	$\dfrac{24}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$	1	$\dfrac{\mathsf{p}_3(\mathsf{N})}{30\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)}$
4	2	$\dfrac{6}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)}$	0	$\dfrac{120(\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$
5	2	$\dfrac{2}{\mathsf{N}(\mathsf{N}+1)}$	0	$\dfrac{24(6\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}$

The character zero.

$\mathsf{s} = 0$. Row weights $(0, 1, 1)$; the constant matrix has entries

$$\mathsf{M}_{00} = -\frac{9(\mathsf{N}-2)(5\mathsf{N}+3)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{01} = -\frac{60}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{02} = -\frac{\mathsf{N}^5+5\mathsf{N}^4+5\mathsf{N}^3-5\mathsf{N}^2-6\mathsf{N}-840}{2\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{10} = \frac{(\mathsf{N}-2)(7\mathsf{N}^7+76\mathsf{N}^6+254\mathsf{N}^5+240\mathsf{N}^4-177\mathsf{N}^3-316\mathsf{N}^2+21516\mathsf{N}+12960)}{240\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{11} = \frac{5(\mathsf{N}^6+3\mathsf{N}^5-5\mathsf{N}^4-15\mathsf{N}^3+4\mathsf{N}^2+12\mathsf{N}+288)}{12\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{12} = -\frac{35\mathsf{N}^6+93\mathsf{N}^5-235\mathsf{N}^4-585\mathsf{N}^3+200\mathsf{N}^2+492\mathsf{N}+10080}{12\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{20} = \frac{\mathsf{N}^4+2\mathsf{N}^3-\mathsf{N}^2-38\mathsf{N}+24}{2\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)},$$

$$\mathsf{M}_{21} = 0,$$

$$\mathsf{M}_{22} = 0,$$

and

$$\det \mathsf{M} = \frac{5(\mathsf{N}-2)(\mathsf{N}^4+2\mathsf{N}^3-\mathsf{N}^2-38\mathsf{N}+24)}{48\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)}.$$

$\mathsf{s} = 1$. Row weights $(0, 1, 1)$; the constant matrix has entries

$$\mathsf{M}_{00} = -\frac{9(\mathsf{N}-2)(5\mathsf{N}+3)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{01} = -\frac{60}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$

$$\mathsf{M}_{02} = -\frac{\mathsf{N}^5+5\mathsf{N}^4+5\mathsf{N}^3-5\mathsf{N}^2-6\mathsf{N}-840}{2\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$
$$\mathsf{M}_{10} = \frac{(\mathsf{N}-2)(5\mathsf{N}^7+38\mathsf{N}^6+78\mathsf{N}^5-20\mathsf{N}^4-155\mathsf{N}^3-18\mathsf{N}^2+4392\mathsf{N}+2592)}{48\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$
$$\mathsf{M}_{11} = \frac{3(\mathsf{N}^6+3\mathsf{N}^5-5\mathsf{N}^4-15\mathsf{N}^3+4\mathsf{N}^2+12\mathsf{N}+160)}{4\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$
$$\mathsf{M}_{12} = -\frac{21\mathsf{N}^6+59\mathsf{N}^5-125\mathsf{N}^4-335\mathsf{N}^3+104\mathsf{N}^2+276\mathsf{N}+3360}{4\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$
$$\mathsf{M}_{20} = \frac{(\mathsf{N}-2)(\mathsf{N}^2+2\mathsf{N}+3)}{2\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)},$$
$$\mathsf{M}_{21} = 0,$$
$$\mathsf{M}_{22} = 0,$$

and

$$\det \mathsf{M} = \frac{3(\mathsf{N}-2)^2(\mathsf{N}^2+2\mathsf{N}+3)}{16\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)}.$$

$\mathsf{s}=2$. Row weights $(0,0,1)$; the constant matrix has entries

$$\mathsf{M}_{00} = -\frac{3(15\mathsf{N}^3+9\mathsf{N}^2+8\mathsf{N}-52)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{01} = -\frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{02} = -\frac{\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3+50\mathsf{N}^2+24\mathsf{N}-840}{2\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{10} = \frac{30(\mathsf{N}-2)}{\mathsf{N}(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{11} = \frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{12} = -\frac{420}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{20} = \frac{1}{2},$$
$$\mathsf{M}_{21} = 0,$$
$$\mathsf{M}_{22} = 0,$$

and

$$\det \mathsf{M} = \frac{15}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)}.$$

$\mathsf{s}=3$. Row weights $(0,0,1)$; the constant matrix has entries

$$\mathsf{M}_{00} = -\frac{9(5\mathsf{N}^3-\mathsf{N}^2-4\mathsf{N}+20)}{\mathsf{N}(\mathsf{N}-1)(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{01} = -\frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{02} = -\frac{\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3+50\mathsf{N}^2+24\mathsf{N}-840}{2\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{10} = \frac{6(7\mathsf{N}^2+\mathsf{N}-18)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{11} = \frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$

$$\mathsf{M}_{12} = -\frac{420}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{20} = \frac{\mathsf{N}^4+9\mathsf{N}^3+26\mathsf{N}^2-36\mathsf{N}+120}{2\mathsf{N}(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{21} = -\frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{22} = \frac{420}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$

and

$$\det \mathsf{M} = \frac{15(\mathsf{N}^4+6\mathsf{N}^3+11\mathsf{N}^2+30\mathsf{N}-24)}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)^2(\mathsf{N}+4)}.$$

$\mathsf{s}=4$. Row weights $(0,0,1)$; the constant matrix has entries

$$\mathsf{M}_{00} = -\frac{9(\mathsf{N}-1)(5\mathsf{N}+8)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{01} = -\frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{02} = -\frac{\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3+50\mathsf{N}^2+24\mathsf{N}-840}{2\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{10} = \frac{3(5\mathsf{N}-1)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)},$$
$$\mathsf{M}_{11} = 0,$$
$$\mathsf{M}_{12} = 0,$$
$$\mathsf{M}_{20} = \frac{\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3-94\mathsf{N}^2+72\mathsf{N}+336}{2\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{21} = -\frac{120}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{22} = \frac{840}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$

and

$$\det \mathsf{M} = \frac{180(5\mathsf{N}-1)}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)^2(\mathsf{N}+4)}.$$

$\mathsf{s}=5$. Row weights $(0,0,1)$; the constant matrix has entries

$$\mathsf{M}_{00} = -\frac{9(\mathsf{N}-1)(5\mathsf{N}+8)}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{01} = -\frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{02} = -\frac{\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3+50\mathsf{N}^2+24\mathsf{N}-840}{2\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{10} = \frac{3}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)},$$
$$\mathsf{M}_{11} = 0,$$
$$\mathsf{M}_{12} = 0,$$
$$\mathsf{M}_{20} = \frac{\mathsf{N}^5+10\mathsf{N}^4+35\mathsf{N}^3-64\mathsf{N}^2-102\mathsf{N}+240}{2\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$
$$\mathsf{M}_{21} = -\frac{60}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$

$$\mathsf{M}_{22} = \frac{420}{\mathsf{N}(\mathsf{N}+1)(\mathsf{N}+2)(\mathsf{N}+3)(\mathsf{N}+4)},$$

and

$$\det \mathsf{M} = \frac{90}{\mathsf{N}^2(\mathsf{N}+1)^2(\mathsf{N}+2)^2(\mathsf{N}+3)(\mathsf{N}+4)}.$$

The cubic line. Let $\mathsf{c}(T)$ be as in (7.17) and $d_k = 2^k + T(-3)^k$. The obstruction functions of the proof of Proposition 7.7 are, as identities of rational functions,

$$\begin{aligned}
\mathcal{F}_1 = \mathcal{F}_2 &= 9072 + \mathsf{c}(T)(1+T) = -\frac{1296(27T-8)\,\mathsf{q}_1(T)}{175\,\Delta_0(T)},\\
\mathcal{F}_3 &= -173628 + \mathsf{c}(T)\big[78d_{-3} + \tfrac{931}{6}d_{-1}\big] = \frac{36\,\mathsf{q}_3(T)}{175\,\Delta_0(T)},\\
\mathcal{F}_4 &= -1462104 - 133\,\mathsf{c}(T)d_0 = \frac{216\,\mathsf{q}_4(T)}{25\,\Delta_0(T)},\\
\mathcal{F}_5 &= 468\big[-468 - \tfrac{7}{6}\mathsf{c}(T)d_{-1}\big]\\
&= -\frac{16848(9T+4)\,\mathsf{q}_5(T)}{25T(8T-27)(16T+81)(8736T^2+1277T-1134)},\\
\mathcal{F}_6 &= 468\big[9072 + \mathsf{c}(T)d_0\big] = 468\,\mathcal{F}_1,\\
\mathcal{F}_{n-13} &= TF_{13}(T) = \frac{216\,\mathsf{q}_{13}(T)}{25(2T-3)(16T+81)(8736T^2+1277T-1134)},
\end{aligned}$$

where $\Delta_0(T) = T(2T-3)(8T-27)(16T+81)(8736T^2+1277T-1134)$ and

$$\begin{aligned}
\mathsf{q}_1 &= 135364608T^6 - 549788672T^5 - 1798393680T^4 + 3550155660T^3\\
&\quad + 1385393220T^2 - 1202021127T - 442158912,\\
\mathsf{q}_3 &= 71854241218 56T^7 - 464906703482 88T^6 + 417002960096T^5\\
&\quad + 282658729970760T^4 - 324936129057390T^3 - 5712883184349T^2\\
&\quad + 60310985662062T - 11121180954624,\\
\mathsf{q}_4 &= 416652263424T^7 - 1881844058112T^6 - 505611852617 6T^5\\
&\quad + 13871807147640T^4 - 477371748510T^3 - 5379518823921T^2\\
&\quad - 44678717262T + 403248927744,\\
\mathsf{q}_5 &= 406093824T^5 - 2011489280T^4 - 1965486960T^3 + 9368244540T^2\\
&\quad - 5531611905T + 884317824,\\
\mathsf{q}_{13} &= 152285184T^5 - 310586240T^4 + 462770950T^3 - 369645495T^2\\
&\quad - 145657845T + 65505024.
\end{aligned}$$

For $1 \le \chi \le 6$ the value of T lies in the interval $[T^-, T^+]$ of Table 9, which contains $T_\chi(1 \pm 10^{-8})$; for $h = 13$ the interval is $[-10^{-4}, 10^{-4}]$.

Horner interval evaluation. For a polynomial $a_m T^m + \cdots + a_0$ with rational coefficients and a rational interval $[T^-, T^+]$, start with $V = [0,0]$ and replace V successively by $V \cdot [T^-, T^+] + [a_i, a_i]$ for $i = m, \dots, 0$, where the product of two intervals is the interval spanned by the four products of endpoints. The result is a rational interval containing all values of the polynomial on $[T^-, T^+]$. Applied to the numerator and the denominator of each $\mathcal{F}_\chi$ and of $\det(\mathcal{A}, \mathcal{C})$ in (7.16), it gives exact rational bounds, and Table 8 lists these bounds rounded outward to integers. The

exact rational bounds for the numerators and denominators of the seven functions $\mathcal{F}_\chi$ are stored in the boundary cubic certificate (Appendix E). For $\det(\mathcal{A},\mathcal{C})$ the certificate program checks the nonvanishing but does not store the exact bounds; the integer bounds in Table 8 are reproduced by applying the procedure above to the expression (7.16). None of the intervals contains 0, so all the functions are nonzero on the relevant intervals. For $h = 13$ the factor T is removed from the numerator of $\det(\mathcal{A},\mathcal{C})$; since $T \neq 0$, this suffices.

TABLE 8. The cubic line: outward integer bounds, obtained by Horner interval evaluation on the intervals of Table 9, for the numerator $\mathcal{N}$ and the denominator $\mathcal{D}$ of the obstruction function $\mathcal{F}_\chi$ (first block) and of $\det(\mathcal{A},\mathcal{C})$ (second block). For $h = 13$ the function is $TF_{13}(T)$, and the factor T is removed from the numerator of $\det(\mathcal{A},\mathcal{C})$.

χ	$\mathcal{N}(\mathcal{F}_\chi) \in$	$\mathcal{D}(\mathcal{F}_\chi) \in$
1	$[1093854649755, 1093854946711]$	$[115903399, 115903422]$
2	$[-2512434530478, -2512434417064]$	$[302411944, 302411954]$
3	$[-104962328052535, -104962320384825]$	$[-113951301, -113951295]$
4	$[76006215754635, 76006216235690]$	$[21601255, 21601257]$
5	$[-42459944942637, -42459944579737]$	$[3778287, 3778288]$
6	$[-2153885545348037, -2153885544034488]$	$[64562671, 64562674]$
$n-13$	$[14145938176, 14152232192]$	$[6887950, 6890150]$

χ	$\mathcal{N}(\det) \in$	$\mathcal{D}(\det) \in$
1	$[-1303490736, -1303490536]$	$[-1422521450, -1422521141]$
2	$[-2116596437, -2116596365]$	$[4559499630, 4559499719]$
3	$[1067577420, 1067577461]$	$[3514487454, 3514487528]$
4	$[-1155769381, -1155769354]$	$[5934244452, 5934244468]$
5	$[659494726, 659494742]$	$[5076129515, 5076129536]$
6	$[-514586161, -514586149]$	$[6006176585, 6006176592]$
$n-13$	$[10743864416, 10746286084]$	$[5804067075, 5805438717]$

TABLE 9. The intervals for T on the cubic line.

χ	T^-	T^+
1	$\frac{11111111}{31640625}$	$\frac{100000001}{284765625}$
2	$-\frac{200000002}{854296875}$	$-\frac{22222222}{94921875}$
3	$\frac{44444444}{284765625}$	$\frac{400000004}{2562890625}$
4	$-\frac{800000008}{7688671875}$	$-\frac{88888888}{854296875}$
5	$\frac{177777776}{2562890625}$	$\frac{1600000016}{23066015625}$
6	$-\frac{3200000032}{69198046875}$	$-\frac{355555552}{7688671875}$
$n-13$	$-\frac{1}{10000}$	$\frac{1}{10000}$

Appendix C. The rational seed

Table 10 lists the rational data used in Proposition 9.1. The planes are $\ell_i = 1 + \nu_i \cdot x$, the points p_i satisfy $\ell_i(p_i) = 0$, and the auxiliary sets are $B_\lambda = \{p_i + \lambda w_i\} \cup \{0\}$ for $\lambda = 1, 2$. The 286 triple points are computed exactly over $\mathbb{Q}$. All denominators are prime to $P = 32003$; modulo P, the 896×896 minor selected in the matrix of Proposition 9.1 has determinant 19954 for $\lambda = 1$ and 31377 for $\lambda = 2$. The four motions lie in the kernel: a translation by b has star coordinates $t_{T,k} = \nu_k \cdot b$ and auxiliary velocities b, and the dilation has $t_{T,k} = -1$ and auxiliary velocities b_j.

Table 10. Rational seed data ($n = 13$): first two coordinates of the normals ν_i and of the directions w_i, and the points p_i. The third coordinates are determined by $\nu_i \cdot p_i = -1$ and $\nu_i \cdot w_i = 1$.

i	$\nu_{i,1}$	$\nu_{i,2}$	p_i	$w_{i,1}$	$w_{i,2}$
0	17521	5775	(28534, 25025, 26602)	8751	30617
1	6234	5692	(13687, 16771, 7786)	14540	22697
2	24365	2617	(28262, 15768, 2758)	25222	12214
3	14749	32001	(29918, 3997, 20899)	4513	12875
4	17881	2230	(14805, 20269, 15010)	31451	23560
5	29589	13848	(16491, 25687, 1222)	13330	26386
6	7165	21490	(10509, 30595, 11898)	15671	28459
7	9362	26419	(10250, 16293, 22095)	18449	25337
8	21180	6727	(931, 13174, 4843)	13326	6294
9	17687	21176	(22553, 28062, 10781)	6191	13668
10	5041	20138	(24957, 23154, 17238)	4835	8131
11	13357	23912	(19500, 17261, 3025)	7472	516
12	11928	25717	(8088, 29149, 29577)	24519	13733

Appendix D. The finite propagation

Table 11 lists the 188 transitions from degree 12 to degree 84 used in Section 9. Each residual interval is contained in the interval proved in the previous degree, each transition satisfies the hypotheses of Proposition 8.13 or 8.12, and the images cover the stated interval.

Table 11: Finite propagation. For each degree d the proved interval $[\sigma_d^-, \sigma_d^+]$ of positions and the transitions used, written $X_c[\ell, u]$: $X = \mathrm{S}$ (sparse, Proposition 8.13), L (low c) or H (high c, Proposition 8.12), with residual positions $[\ell, u]$ in degree $d - 1$ and image $[\ell + c, u + c]$.

d	$[\sigma_d^-, \sigma_d^+]$	transitions
11	[14, 14]	seed (Proposition 9.1)
12	[16, 16]	$\mathrm{S}_2[14, 14]$
13	[18, 18]	$\mathrm{S}_2[16, 16]$
14	[20, 20]	$\mathrm{S}_2[18, 18]$
15	[22, 22]	$\mathrm{S}_2[20, 20]$
16	[24, 24]	$\mathrm{S}_2[22, 22]$
17	[26, 26]	$\mathrm{S}_2[24, 24]$
18	[28, 28]	$\mathrm{S}_2[26, 26]$

d	$[\sigma_d^-, \sigma_d^+]$	transitions
19	$[30, 30]$	$S_2[28, 28]$
20	$[32, 32]$	$S_2[30, 30]$
21	$[34, 34]$	$S_2[32, 32]$
22	$[36, 37]$	$S_2[34, 34]$, $S_3[34, 34]$
23	$[38, 40]$	$S_2[36, 37]$, $S_3[36, 37]$
24	$[40, 43]$	$S_2[38, 40]$, $S_3[38, 40]$
25	$[42, 46]$	$S_2[40, 43]$, $S_3[40, 43]$
26	$[44, 49]$	$S_2[42, 46]$, $S_3[42, 46]$
27	$[46, 52]$	$S_2[44, 49]$, $S_3[44, 49]$
28	$[48, 55]$	$S_2[46, 52]$, $S_3[46, 52]$
29	$[50, 58]$	$S_2[48, 55]$, $S_3[48, 55]$
30	$[52, 61]$	$S_2[50, 58]$, $S_3[50, 58]$
31	$[54, 64]$	$S_2[52, 61]$, $S_3[52, 61]$
32	$[56, 67]$	$S_2[54, 64]$, $S_3[54, 64]$
33	$[58, 83]$	$S_2[56, 67]$, $S_3[56, 67]$, $L_{15}[56, 67]$, $L_{16}[56, 67]$
34	$[60, 100]$	$S_2[58, 83]$, $L_{17}[58, 83]$
35	$[62, 121]$	$S_2[60, 100]$, $L_{17}[60, 100]$, $H_{21}[92, 100]$
36	$[64, 142]$	$S_2[62, 121]$, $H_{21}[95, 121]$
37	$[66, 166]$	$S_2[64, 142]$, $H_{22}[97, 142]$, $H_{24}[136, 142]$
38	$[68, 191]$	$S_2[66, 166]$, $H_{25}[139, 166]$
39	$[70, 218]$	$S_2[68, 191]$, $H_{25}[143, 191]$, $H_{27}[185, 191]$
40	$[72, 246]$	$S_2[70, 212]$, $H_{26}[146, 218]$, $H_{28}[189, 218]$
41	$[74, 276]$	$S_2[72, 214]$, $H_{27}[149, 246]$, $H_{30}[239, 246]$
42	$[76, 307]$	$S_2[74, 233]$, $H_{30}[197, 276]$, $H_{31}[244, 276]$
43	$[78, 340]$	$S_2[76, 235]$, $H_{30}[202, 307]$, $H_{33}[298, 307]$
44	$[80, 374]$	$S_2[78, 255]$, $H_{31}[206, 340]$, $H_{34}[304, 340]$
45	$[82, 409]$	$S_2[80, 257]$, $H_{31}[211, 374]$, $H_{35}[310, 374]$
46	$[84, 445]$	$S_2[82, 278]$, $H_{32}[215, 409]$, $H_{36}[370, 409]$
47	$[86, 482]$	$S_2[84, 280]$, $H_{33}[219, 445]$, $H_{37}[377, 445]$
48	$[88, 520]$	$S_2[86, 302]$, $H_{34}[223, 482]$, $H_{38}[384, 482]$
49	$[90, 560]$	$S_2[88, 304]$, $H_{35}[227, 520]$, $H_{40}[507, 520]$
50	$[92, 601]$	$S_2[90, 327]$, $H_{37}[286, 560]$, $H_{41}[516, 560]$
51	$[94, 644]$	$S_2[92, 329]$, $H_{37}[292, 601]$, $H_{43}[586, 601]$
52	$[96, 688]$	$S_2[94, 353]$, $H_{38}[297, 644]$, $H_{44}[596, 644]$
53	$[98, 733]$	$S_2[96, 355]$, $H_{39}[302, 688]$, $H_{45}[671, 688]$
54	$[100, 779]$	$S_2[98, 380]$, $H_{40}[307, 733]$, $H_{46}[682, 733]$
55	$[102, 826]$	$S_2[100, 382]$, $H_{41}[312, 779]$, $H_{47}[693, 779]$
56	$[104, 874]$	$S_2[102, 408]$, $H_{42}[317, 826]$, $H_{48}[704, 826]$
57	$[106, 923]$	$S_2[104, 410]$, $H_{42}[323, 874]$, $H_{49}[785, 874]$
58	$[108, 973]$	$S_2[106, 437]$, $H_{44}[392, 923]$, $H_{50}[797, 923]$
59	$[110, 1024]$	$S_2[108, 439]$, $H_{43}[334, 973]$, $H_{51}[809, 973]$
60	$[112, 1076]$	$S_2[110, 467]$, $H_{46}[404, 1024]$, $H_{52}[821, 1024]$
61	$[114, 1129]$	$S_2[112, 469]$, $H_{47}[410, 1076]$, $H_{53}[908, 1076]$
62	$[116, 1183]$	$S_2[114, 498]$, $H_{48}[416, 1129]$, $H_{54}[921, 1129]$
63	$[118, 1238]$	$S_2[116, 500]$, $H_{49}[422, 1183]$, $H_{55}[934, 1183]$
64	$[120, 1294]$	$S_2[118, 530]$, $H_{49}[429, 1238]$, $H_{56}[947, 1238]$
65	$[122, 1351]$	$S_2[120, 532]$, $H_{50}[435, 1294]$, $H_{57}[1040, 1294]$
66	$[124, 1409]$	$S_2[122, 563]$, $H_{52}[514, 1351]$, $H_{58}[1054, 1351]$
67	$[126, 1468]$	$S_2[124, 565]$, $H_{51}[448, 1409]$, $H_{59}[1068, 1409]$
68	$[128, 1528]$	$S_2[126, 597]$, $H_{54}[528, 1468]$, $H_{60}[1082, 1468]$
69	$[130, 1589]$	$S_2[128, 599]$, $H_{55}[535, 1528]$, $H_{61}[1181, 1528]$

d	$[\sigma_d^-, \sigma_d^+]$	transitions
70	$[132, 1651]$	$S_2[130, 632]$, $H_{56}[542, 1589]$, $H_{62}[1196, 1589]$
71	$[134, 1714]$	$S_2[132, 634]$, $H_{56}[550, 1651]$, $H_{63}[1211, 1651]$
72	$[136, 1778]$	$S_2[134, 668]$, $H_{57}[557, 1714]$, $H_{64}[1226, 1714]$
73	$[138, 1843]$	$S_2[136, 670]$, $H_{57}[565, 1778]$, $H_{65}[1331, 1778]$
74	$[140, 1910]$	$S_2[138, 705]$, $H_{58}[572, 1843]$, $H_{67}[1816, 1843]$
75	$[142, 1977]$	$S_2[140, 707]$, $H_{59}[579, 1910]$, $H_{67}[1363, 1910]$
76	$[144, 2046]$	$S_2[142, 743]$, $H_{62}[669, 1977]$, $H_{69}[1957, 1977]$
77	$[146, 2116]$	$S_2[144, 745]$, $H_{63}[677, 2046]$, $H_{70}[1979, 2046]$
78	$[148, 2188]$	$S_2[146, 782]$, $H_{63}[686, 2116]$, $H_{72}[2102, 2116]$
79	$[150, 2260]$	$S_2[148, 784]$, $H_{64}[694, 2188]$, $H_{72}[2126, 2188]$
80	$[152, 2334]$	$S_2[150, 822]$, $H_{64}[614, 2260]$, $H_{73}[2149, 2260]$, $H_{74}[2253, 2260]$
81	$[154, 2409]$	$S_2[152, 824]$, $H_{65}[711, 2334]$, $H_{75}[2277, 2334]$
82	$[156, 2485]$	$S_2[154, 863]$, $H_{66}[719, 2409]$, $H_{75}[2302, 2409]$, $H_{76}[2409, 2409]$
83	$[158, 2562]$	$S_2[156, 865]$, $H_{67}[727, 2485]$, $H_{77}[2434, 2485]$
84	$[160, 2640]$	$S_2[158, 905]$, $H_{70}[827, 2562]$, $H_{78}[2459, 2562]$

Appendix E. Supplementary material

The finite computations used in this paper are collected in the supplementary archive `verification.zip`. It contains 26 files: a `README.md`, a file `manifest.json` with the SHA-256 hash of every other file, and 24 programs and data files. The programs use Python 3.12, NumPy and SymPy 1.14.0. A program `name.py` at the top level is run from the root of the unpacked archive as `python name.py`, and a program in the directory `boundary_moment_v1` as `python boundary_moment_v1/name.py`. Each program writes its JSON result to the location of the distributed result file, so that a copy of the distributed file should be kept for comparison.

The computations are of three kinds.

(i) Fixed symbolic identities and nonvanishing certificates. These are used as input to arguments valid for infinitely many degrees.

- `check_cyclic_uniform_origin_rank_v1.py` and `check_cyclic_bulk_moment_symbolic_v1.py`, with their JSON results: the identities of rational functions used in Theorem 6.7, Proposition 6.6 and Proposition 7.6.
- The directory `boundary_moment_v1`: five programs `certify_boundary_*.py` with their five JSON certificates and two shared modules. The certificates contain the data of Appendix B: the 36 and 24 leading coefficients of Tables 5 and 6, the coefficients for $\chi = -1$ and $\chi = 0$, the seven cubic obstruction functions $\mathcal{F}_\chi$ as rational functions of T, and the exact rational Horner bounds for their numerators and denominators. The program for the cubic line also checks that $\det(\mathcal{A}, \mathcal{C})$ is nonzero on each interval, but its JSON result does not store the corresponding exact bounds; these are reproduced from (7.16) as described in Appendix B.

(ii) Certificates for finitely many explicit degrees.

- `certify_cyclic_finite_composition_v1.py` with its JSON result: the computation of Appendix A for $29 \le n \le 88$ and all 3510 characters, with the chosen primes, the elements of order n and the checks of their orders, the denominator conditions, the nonzero minors, and the passage to characteristic zero. The SHA-256 hash of this certificate file is

`8d1b83ef61e872208c3034a5fb07980bf202fe0d86eb4bf91db07bdb2678b501`.

- `verify_direct_rational_seed_v1.py`, the data file `FINITE_SEED_RATIONAL_DATA_v1.json` and the record `verify_direct_rational_seed_v1_execution.json`: the rational configuration of Appendix C for $(d, s) = (11, 14)$ and the two nonzero minors of size 896.

(iii) Bases of inductions and finite transitions. `check_inductive_ranges.py` with its JSON result, and `data/compile_one_seed_bridge_v2.json`: the 188 transitions of Appendix D up to degree 84, and a check of the integer inequalities used in Theorem 9.2. Theorem 9.2 is proved in the text, and the check repeats its inequalities. The bands of Proposition 10.1, with upper endpoints $2d - 24$ and $3d - 52$, are proved only by the induction in the text. The small-band part of `check_inductive_ranges.py` concerns the weaker endpoints $2d - 26$ and $3d - 56$, and it does not verify the bands stated here.

Each certificate records the data needed to check it independently of the search that produced it. A rank certificate (Appendices A and C) records the selected nonzero minors, with their rows and columns. A symbolic or sign certificate records the exact rational functions together with the coefficients or exact interval bounds that establish their identities or their nonvanishing.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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